

Project Data Page:





Table of Contents

Cover Page	1
Vessel Design Summary:	2
Nozzle Summary:	5
Nozzle Schedule:	6
Bill of Material:	7
MDMT Summary:	8
Warnings and Errors:	10
Input Echo:	11
XY Coordinate Calculations:	21
Flg Calc [Int P]: LHS Blind Fl	22
Flg Calc [Int P]: LHS Ch Flg	25
Flg Calc [Int P]: RHS Ch Flg	34
Flg Calc [Int P]: RHS blind fl	43
Internal Pressure Calculations:	46
Element: Ch Shell	47
Element: main shell	49
Element: RHS Ch shell	51
Shop Pressure Test Results:	53
Field Pressure Test Results:	55
ASME TS Calc: Pressure Test	57
External Pressure Calculations:	70
Element and Detail Weights:	71
Nozzle Flange MAWP:	74
Wind Load Calculation:	75
Earthquake Load Calculation:	77
Center of Gravity Calculation:	79
Saddle Calcs: Road Transport:	80
Saddle Calcs: Operating Case:	94
Saddle Calcs: Test Case:	109
Saddle Calcs: Ext. Pressure:	123
Nozzle Calcs.: N1-Tube Inlet	136
Nozzle Calcs.: Noz N5	147
Nozzle Calcs.: D1	151
Nozzle Calcs.: N3-Shell Inlet	154
Nozzle Calcs.: N4-Shell Inlet	168
Nozzle Calcs.: N2 Tube outlet	182
Nozzle Calcs.: V1	193
ASME TS Calc: D	196
ASME TS Calc: CASE-2 Q-7	237
ASME TS Calc: CASE 3- START U	244
ASME TS Calc: CASE-4 Q-7	251
ASME TS Calc: CASE 5- Q-7	258



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\106-C-Clading on Both Side.pvdb

Date of Analysis : Apr 20,2026 9:47am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 2 of 264
 Vessel Design Summary: Step: 34 9:47am Apr 20,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2024.000 mm ID
 Vessel Design Length, Tangent to Tangent 19933.87 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 160 °C
 Channel Side Design Temperature 200 °C
 Shell Side Design Pressure 920.000 KPa.
 Channel Side Design Pressure 4950.000 KPa.
 Shop Shell Side Test Pressure 3478.497 KPa.
 Field Shell Side Test Pressure 1196.000 KPa.
 Shop Channel Side Test Pressure 6856.986 KPa.
 Field Channel Side Test Pressure 6435.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-266 2	...	...	K03506	
Nozzle	SA-965 F304	...	...	S30400	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg		...	...		
Nozzle Flg	SA-182 F304	...	> 130	S30400	
Tubes	SA-213 TP304	...	...	S30400	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	64 < t <= 100	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flang	4975.289	0.00	5269.000	0.0000	N/A	No
LHS Ch Flg	4969.947	0.00	6054.000	0.0000	N/A	No
Ch Shell	4969.859	0.00	5816.000	2.0000	N/A	No
main shell	939.859	103.42	2404.000	2.0000	N/A	No
RHS Ch shell	4969.859	0.00	5816.000	2.0000	N/A	No
RHS Ch Flg	4969.947	0.00	6054.000	0.0000	N/A	No
RHS blind flang	4975.289	0.00	5269.000	0.0000	N/A	No

Liquid Level: 2580.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Vessel Design Summary:

Page 3 of 264
 Step: 34 9:47am Apr 20,2026

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	268.2	265.0	273.0	265.0	256.9	...	1.00	1.00
Body Flg	707.5	433.0	355.0	355.0	346.4	...	1.00	1.00
Cylinder	3207.5	2500.0	48.0	46.0	39.3	...	1.00	1.00
Cylinder	16520.5	13108.0	20.0	20.0	8.9	...	1.00	1.00
Cylinder	19225.5	2500.0	48.0	46.0	39.3	...	1.00	1.00
Body Flg	19665.7	437.0	355.0	347.0	337.4	...	1.00	1.00
Body Flg	19933.9	265.0	273.0	265.0	256.9	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	265	174.168	...	No Calc
20	30	355	346.431	...	No Calc
30	Tsht	46	No Calc	...	3164
Tsht	40	46	No Calc	...	No Calc
40	Tsht	20	16.5501	103.419	175
Tsht	50	20	No Calc	103.419	No Calc
50	60	46	No Calc	...	2817
60	70	347	337.414	...	No Calc
70	80	265	174.168	...	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	2499.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	13108	27966.5	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	2500.02	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.993	Shell	Passed
N3-Shell Inlet	WRC-297	0.993	Shell	Passed
N4-Shell Inlet	WRC-297	0.993	Shell	Passed
N2 Tube outlet	WRC-297	0.993	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

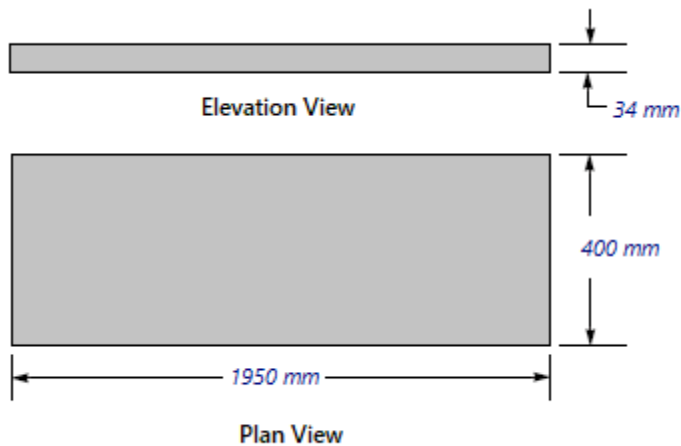
Saddle Width 375.000 mm
 Saddle Bearing Angle 120.000 deg.
 Centerline Dimension 1650.000 mm
 Wear Pad Width 675.000 mm
 Wear Pad Thickness 16.000 mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 4 of 264
 Vessel Design Summary: Step: 34 9:47am Apr 20,2026

Wear Pad Bearing Angle	132.000	deg.
Distance from Saddle to Tangent	2700.000	mm
Baseplate Length	1950.000	mm
Baseplate Thickness	34.000	mm
Baseplate Width	400.000	mm
Number of Ribs (including outside ribs)	5	
Rib Thickness	10.000	mm
Web Thickness	10.000	mm
Height of Center Web	638.000	mm
Number of Bolts in Baseplate	8	

Baseplate Sketch



Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	120026.45	Kgf
Maximum Transverse Saddle Shear Load	21150.11	Kgf
Maximum Longitudinal Saddle Shear Load	11718.26	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	100045.03	Kgf
Maximum Transverse Saddle Shear Load	597.47	Kgf
Maximum Longitudinal Saddle Shear Load	249.08	Kgf

Weights:

Fabricated - Bare w/o removable internals	139184.3	kg.
Shop Test - Fabricated + water (full)	195388.2	kg.
Shipping - Fab. + rem. intls.+ shipping app.	139184.3	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	140484.3	kg.
Empty - Fab. + intls. + details + wghts.	145504.3	kg.
Operating - Empty + operating liquid (no ca)	200752.9	kg.
Field Test - Empty weight + test medium (full)	200408.1	kg.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 5 of 264
 Nozzle Summary: Step: 32 9:47am Apr 20,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	5544	...	...	OK	16.55	OK	Passed
Noz N5	5666	...	...	OK	8.33	OK	Passed
D1	5816	...	...	OK	4.80	OK	No Calc[*]
N3-Shell Inlet	2424	OK	...	OK	8.33	OK	Passed
N4-Shell Inlet	2404	OK	...	OK	8.33	OK	Passed
N2 Tube outlet	5426	...	...	OK	17.28	OK	Passed
V1	5816	...	...	OK	4.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2404 Nozzle : N4-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 5426 Nozzle : N2 Tube outlet [Tubeside]

[*] - This was a small opening and the areas were not computed.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube Inlet	1707.525	0.000	1674.000
30	Noz N5	1707.525	270.000	658.393
30	D1	1707.525	180.000	172.328
40	N3-Shell Inlet	4912.526	0.000	597.738
40	N4-Shell Inlet	15082.525	180.000	597.738
50	N2 Tube outlet	18225.525	180.000	1750.800
50	V1	17725.527	0.000	172.328

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 6 of 264
 Nozzle Schedule: Step: 31 9:47am Apr 20,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
D1	2.000 in	Actual	LWN	3.320	16.764	...	...	346.88	600
V1	2.000 in	Actual	LWN	3.320	16.764	...	...	346.88	600
N3-Shell Inlet	12.000 in	60	WNF	12.750	14.275	543.85	20.00	283.04	300
N4-Shell Inlet	12.000 in	60	WNF	12.750	14.275	543.85	20.00	283.04	300
Noz N5	14.000 in	60	WNF	14.000	15.088	...	...	361.74	600
N1-Tube Inlet	34.488 in	Actual	WNF	34.488	19.500	...	...	645.70	600
N2 Tube outlet	36.000 in	Actual	WNF	36.000	19.500	...	...	655.16	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
D1	SA-266 2	16.000	10.000	...	...	...
V1	SA-266 2	16.000	10.000	...	...	...
N3-Shell In	SA-106 B	20.000	10.000	16.000	20.000	...
N4-Shell In	SA-106 B	20.000	10.000	16.000	20.000	...
Noz N5	SA-965 F304	20.000	20.000	...	...	...
N1-Tube Inl	SA-266 2	43.000	20.000	...	...	...
N2 Tube out	SA-266 2	44.000	20.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
D1	1707.525	180.0	300.00	0.00	Ch Shell
V1	17725.527	0.0	300.00	0.00	RHS Ch shell
N3-Shell Inlet	4912.525	0.0	250.00	0.00	main shell
N4-Shell Inlet	15082.525	180.0	250.00	0.00	main shell
Noz N5	1707.525	270.0	300.00	0.00	Ch Shell
N1-Tube Inlet	1707.525	0.0	500.00	0.00	Ch Shell
N2 Tube outlet	18225.525	180.0	500.00	0.00	RHS Ch shell

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
Bill of Material:

Step: 30 9:47am Apr 20,2026 Page 7 of 264

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	BLIND FLANGE: 273.0mm THK X 2565.0mm OD	SA-266 2
2	WELD NECK FLANGE: 355.0mm THK X 2565.0mm OD	SA-266 2
2	CYLINDER: 48.0mm THK X 2024.0mm ID X 2500.0mm	SA-516 70
1	CYLINDER: 20.0mm THK X 2024.0mm ID X 13108.0mm	SA-516 70
2	INSULATION: 320mm X 100mm THK	Hot Insulation
2	INSULATION: 416mm X 100mm THK	Hot Insulation
2	INSULATION: 2500mm X 100mm THK	Hot Insulation
2	LINING: 2500.000mm X 3.000mm THK	...
22	BAFFLES: 1976.0mm DIA X 9.5mm THK	SA-516 70
12	TIE RODS: 15mm DIA X 12280mm	SA-193 B7M
2	TUBESHEET: 2120.0mm X 205.0mm THK	SA-266 2
5285	TUBES: 13500.0mm X 19.0mm DIA X 1.6mm THK	SA-213 TP304
2	SADDLE: 375.0mm X 120 DEG	SA-516 70
1	INSULATION: 13108mm X 100mm THK	Hot Insulation
1	LINING: 13108.001mm X 3.000mm THK	...
2	GASKET: 2124.0mm OD X 2056.0mm ID	...
72	BODY FLANGE BOLTS: 88.9mm DIA	SA-193 B7
144	NUTS FOR BODY FLANGE BOLTS: 88.9mm DIA	...
1	NAMEPLATE	...



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 8 of 264
 MDMT Summary: Step: 33 9:47am Apr 20,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
main shell [8]	B	-7	-8	-29	0.992	20.000	1.00	No
Left Saddle F[24]	B	-14	-15	-29	0.992	16.000	1.00	No
Right Saddle[24]	B	-14	-15	-29	0.992	16.000	1.00	No
N3-Shell Inlet[1]	B	-7	-29	-29	1.000	20.000	1.00	No
N4-Shell Inlet[1]	B	-7	-29	-29	1.000	20.000	1.00	No
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: SS[13]	B	17	-104		0.083	51.250	1.00	No
Warmest MDMT:		17	-8					
LHS Blind Fla[11]	B	23	23		0.995	66.250	1.00	No
LHS Ch Flg [11]	B	15	7		0.865	46.000	1.00	Yes
RHS Ch Flg [11]	B	15	7		0.865	46.000	1.00	Yes
RHS blind fla[11]	B	23	23		0.995	66.250	1.00	No
Ch Shell [8]	B	15	9		0.901	46.000	1.00	Yes
RHS Ch shell [8]	B	15	9		0.901	46.000	1.00	Yes
N1-Tube Inlet [1]	B	15	9		0.905	46.000	1.00	Yes
D1 [1]	B	-13	-104	-29	0.059	16.764	1.00	No
N2 Tube outlet[1]	B	15	9		0.905	46.000	1.00	Yes
V1 [1]	B	-13	-104	-29	0.059	16.764	1.00	No
Nozl. Bolting[22]		-48			1.000	64.000	1.00	No
Tubesheet: CS[14]	B	1	-15		0.722	51.250	1.00	No
Bolting [21]		-40						
Bolting [21]		-40						
Warmest MDMT:		23	23					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
MDMT Summary:

Page 9 of 264

Step: 33 9:47am Apr 20, 2026

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side Page 10 of 264
Warnings and Errors: Step: 0 9:47am Apr 20,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 11 of 264
 Input Echo: Step: 1 9:47am Apr 20,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	920.02	KPa.
Channel Side Design Pressure	4950	KPa.
Shell Side Design Temperature	160.0	°C
Channel Side Design Temperature	200.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No



PV Elite 28 Licensee: SL User
 FileName : 106-C-Clading on Both Side Page 12 of 264
 Input Echo: Step: 1 9:47am Apr 20,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16	
Wind Load Reduction Scale Factor	1.100	
Basic Wind Speed	97.2	Km/hr
Surface Roughness Category	C: Open Terrain	
Type of Surface	Moderately Smooth	
Base Elevation	12500	mm
Percent Wind for Hydrotest	50.0	
Using User defined Wind Press. Vs Elev.	N	
Height of Hill or Escarpment H or Hh	0	mm
Distance Upwind of Crest Lh	0	mm
Distance from Crest to the Vessel x	0	mm
Type of Terrain (Hill, Escarpment)	Flat	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Site Elevation above Sea Level	0	mm

Seismic Design Code	ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor	1.000	
Importance Factor	1.250	
Table Value Fa	1.600	
Table Value Fv	2.400	
Max. Mapped Res. Acceleration [Ss]	0.147	
Max. Eff. Ground Acceleration [S]	0.065	
Force Modification Factor R	3.000	
Site Class	D	
Component Elevation Ratio z/h	0.000	
Amplification Factor Ap	1.000	
Force Factor	0.000	
Consider Vertical Acceleration	Yes	
Minimum Acceleration Multiplier	0.000	
User Value of Sds (used if > 0)	0.000	
User Value of Sd1 (used if > 0)	0.000	
Moment Reduction Factor Tau	1.000	

Overall M.A.W.P. + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 13 of 264
 Input Echo: Step: 1 9:47am Apr 20,2026

Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Blind Flang
Distance "FROM" to "TO"	265 mm
Flange Outside Diameter	2565 mm
Element Thickness	265 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	273 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	213 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2580 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Input Echo:

Step: 1 9:47am Apr 20,2026

Page 14 of 264

Height/Length of Insulation	320	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS Ch Flg
Distance "FROM" to "TO"	433 mm
Flange Inside Diameter	2035 mm
Element Thickness	355 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	355 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2035 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	416 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Ch Shell
Distance "FROM" to "TO"	2500 mm
Inside Diameter	2024 mm
Element Thickness	46 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	48 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Input Echo:

Page 15 of 264

Step: 1 9:47am Apr 20,2026

Yield Stress, Operating	230.4	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2024	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	30	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2500	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	30	
Detail Type	Lining	
Detail ID	SS Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	2500	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N1-Tube Inlet	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	34.49	in.
Nozzle Schedule	40	
Nozzle Class	600	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1567.9	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-266 2	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	Noz N5	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	14	in.
Nozzle Schedule	60	
Nozzle Class	600	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	294.93	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-965 F304	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	D1	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Input Echo:

Step: 1 9:47am Apr 20,2026

Page 16 of 264

Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	14.518	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-266 2	
Element From Node	30	
Detail Type	Weight	
Detail ID	Partion plate w	
Dist. from "FROM" Node / Offset dist	1250	mm
Miscellaneous Weight	200	Kgf
Offset from Element Centerline	0	mm

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	main shell	
Distance "FROM" to "TO"	13108	mm
Inside Diameter	2024	mm
Element Thickness	20	mm
Internal Corrosion Allowance	2	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	920	KPa.
Design Temperature Internal Pressure	160	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle Flo	
Dist. from "FROM" Node / Offset dist	2700	mm
Width of Saddle	375	mm
Height of Saddle at Bottom	1650	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	675	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Saddle	
Detail ID	Right Saddle	
Dist. from "FROM" Node / Offset dist	10420	mm
Width of Saddle	375	mm
Height of Saddle at Bottom	1650	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	675	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	132.0	



PV Elite 28

Licensee: SL User

FileName : 106-C-Cladding on Both Side

Page 17 of 264

Input Echo:

Step: 1 9:47am Apr 20,2026

Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2024	mm
Liquid Density	0.0009984	kg./cm ³
Element From Node	40	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	13108	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³
Element From Node	40	
Detail Type	Lining	
Detail ID	SS Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	13108	mm
Thickness of Lining	3	mm
Density	0.008	kg./cm ³
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3-Shell Inlet	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	12	in.
Nozzle Schedule	60	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	117.58	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N4-Shell Inlet	
Dist. from "FROM" Node / Offset dist	11670	mm
Nozzle Diameter	12	in.
Nozzle Schedule	60	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	117.58	Kgf
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Weight	
Detail ID	Fire[prrooing	
Dist. from "FROM" Node / Offset dist	2700	mm
Miscellaneous Weight	650	Kgf
Offset from Element Centerline	1651	mm
Element From Node	40	
Detail Type	Weight	



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Input Echo:

Step: 1 9:47am Apr 20,2026

Page 18 of 264

Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	10420 mm
Miscellaneous Weight	650 Kgf
Offset from Element Centerline	1651 mm

Element#	5/7
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	RHS Ch shell
Distance "FROM" to "TO"	2500 mm
Inside Diameter	2024 mm
Element Thickness	46 mm
Internal Corrosion Allowance	2 mm
Nominal Thickness	48 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2024 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	50
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm³

Element From Node	50
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	2500 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm³

Element From Node	50
Detail Type	Nozzle
Detail ID	N2 Tube outlet
Dist. from "FROM" Node / Offset dist	1500 mm
Nozzle Diameter	36 in.
Nozzle Schedule	40
Nozzle Class	600
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	1544.9 Kgf
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-266 2

Element From Node	50
-------------------	----



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Input Echo:

Step: 1 9:47am Apr 20,2026

Page 19 of 264

Detail Type	Nozzle
Detail ID	V1
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	600
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	14.518 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-266 2

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Flange
Description	RHS Ch Flg
Distance "FROM" to "TO"	437 mm
Flange Inside Diameter	2035 mm
Element Thickness	347 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	355 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	213 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2035 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	416 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Flange



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side
Input Echo:

Step: 1 9:47am Apr 20,2026 Page 20 of 264

Description	RHS blind flang
Distance "FROM" to "TO"	265 mm
Flange Outside Diameter	2565 mm
Element Thickness	265 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	273 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	4950 KPa.
Design Temperature Internal Pressure	200 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2580 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	70
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	320 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
XY Coordinate Calculations:

Step: 2 9:47am Apr 20,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flang		268.175	...	265	...
LHS Ch Flg		707.525	...	433	...
Ch Shell		3207.52	...	2500	...
main shell		16520.5	...	13108	...
RHS Ch shell		19225.5	...	2500	...
RHS Ch Flg		19665.7	...	437	...
RHS blind flang		19933.9	...	265	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 22 of 264
 Flg Calc [Int P]: LHS Blind Fl Flng: 9 9:47am Apr 20,2026

Flange Input Data Values

Description: LHS Blind Fl :

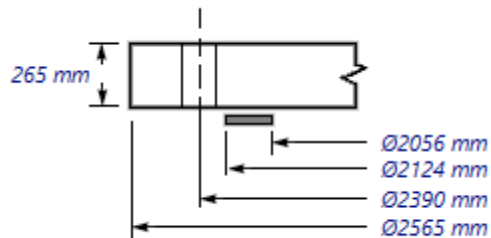
LHS Blind Flang

Flange Type		Blind
Design Pressure	P	4975.29 KPa.
Design Temperature		200 °C
Internal Corrosion Allowance	ci	0.0000 mm
External Corrosion Allowance	ce	0.0000 mm
Use Corrosion Allowance in Thickness Calcs.		Yes
Flange Outside Diameter	A	2565.000 mm
Flange Thickness	t	265.0000 mm
Flange Material		SA-266 2
Flange Material UNS number		K03506
Flange Allowable Stress At Temperature	Sfo	138.00 N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00 N./mm ²
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	7508.407 mm
Diameter of Bolt Circle	C	2390.000 mm
Nominal Bolt Diameter	a	88.9000 mm
Type of Threads	TEMA Thread Series	
Number of Bolts		36
Flange Face Outside Diameter	Fod	2135.000 mm
Flange Face Inside Diameter	Fid	2035.000 mm
Flange Facing Sketch		Sketch 1a
Gasket Outside Diameter	Go	2124.000 mm
Gasket Inside Diameter	Gi	2056.000 mm
Gasket Factor	m	3.0000
Gasket Design Seating Stress	y	68.95 N./mm ²
Gasket Thickness	tg	3.1750 mm
Raised face depth	Fd	0.0000 mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	34.000 mm
Basic Gasket Width,	$bo = N/2$	17.000 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	10.390 mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 23 of 264
 Flg Calc [Int P]: LHS Blind Fl Flng: 9 9:47am Apr 20,2026

Gasket Reaction Diameter, $G = G_o - 2 \cdot b$ 2103.220 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2103^2 \cdot 4975$$

$$= 1762683.375 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 10.4 \cdot \pi \cdot 2103 \cdot 3 \cdot 4975$$

$$= 208983.406 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(1762683 + 208983 + 0, 0)$$

$$= 1971666.750 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 10.4 \cdot \pi \cdot 2103 + 0 \cdot 0 \cdot 0$$

$$= 482671.031 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1971666 + 0 + 4 \cdot 0 / 2103) / 159, 482671 / 159)$$

$$= 1216.110 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1216.110	2032.022	
Radial Distance between Bolts and Edge:	82.550	87.500	
Circ. Spacing between the Bolts:	180.975	208.302	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 2032 \cdot 159 / (68.9 \cdot \pi (2124 + 2056))$$

$$= 35.683 \text{ mm} [\text{Note: Exceeds actual gasket width, 34.000}]$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159 \cdot 2032$$

$$= 3294495.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 1971666.75 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2390 - 2103) / 2$$

$$= 143.3899 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 1971666 \cdot 143 \cdot 1 / (265^2 \cdot 2103) + 0.3 \cdot 1 \cdot 4975 \cdot 2103^2 / 265^2$$

$$= 129.69 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 24 of 264
 Flg Calc [Int P]: LHS Blind Fl Flng: 9 9:47am Apr 20,2026

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 3294495 \cdot 143 \cdot 1 / (265^2 \cdot 2103)$$

$$= 59.60 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1971666 / 2032$$

$$= 95.16 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (482671 / 2032)$$

$$= 23.29 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	129.69	138.00	59.60	138.00
Bolting	95.16	159.00	23.29	159.00

Reqd. Blind Flange Thickness at Center 256.901 mm
 Reqd. Blind Flange Thickness at Gasket 174.147 mm

Estimated M.A.W.P. (Operating) 5294 KPa.
 Estimated Finished Weight of Flange at given Thk. 10613.1 kg.
 Estimated Unfinished Weight of Forging at given Thk 10613.1 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 1, Temperature Reduction per Fig. UCS 66.1 = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 23 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	129.69	138.00	59.60	138.00
Bolting	95.16	159.00	23.29	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 256.921 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 25 of 264
 Flg Calc [Int P]: LHS Ch Flg Flng: 10 9:47am Apr 20,2026

Flange Input Data Values

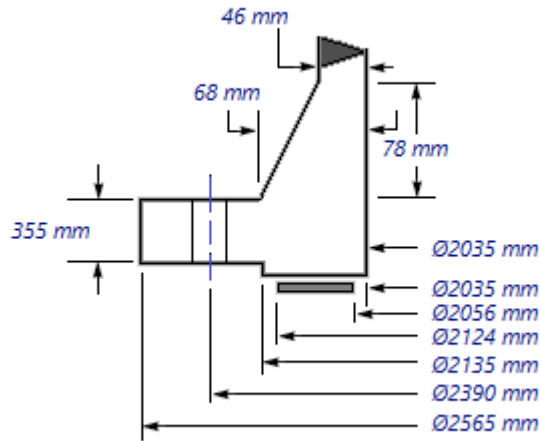
Description: LHS Ch Flg :

LHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	4969.95	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2035.000	mm
Flange Outside Diameter	A	2565.000	mm
Flange Thickness	t	355.0000	mm
Thickness of Hub at Small End	go	46.0000	mm
Thickness of Hub at Large End	gl	68.0000	mm
Length of Hub	h	78.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:
 It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2390.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2135.000	mm
Flange Face Inside Diameter	Fid	2035.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2124.000	mm
Gasket Inside Diameter	Gi	2056.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4970 \cdot (2035/2 + 0)) / (138 \cdot 1 - 0.6 \cdot 4970) + C_a$$

$$= 37.4566 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 46) / (1018 + 0.6 \cdot 46)$$

$$= 6073.598 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 46) / (1018 + 0.6 \cdot 46)$$

$$= 6073.598 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B + 2 * ci	2035.000 mm
Corroded Large Hub,	g1Cor = g1 - ci	68.000 mm
Corroded Small Hub,	g0Cor = go - ci	46.000 mm
Code R Dimension,	R = (C - B) / 2 - g1	109.500 mm

Gasket Contact Width,	N = (Go - Gi) / 2	34.000 mm
Basic Gasket Width,	bo = N / 2	17.000 mm
Effective Gasket Width,	b = Cb * sqrt(bo)	10.390 mm
Gasket Reaction Diameter,	G = Go - 2 * b	2103.220 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 2103^2 \cdot 4970$$

$$= 1760790.625 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 10.4 \cdot \pi \cdot 2103 \cdot 3 \cdot 4970$$

$$= 208759.016 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot Bcor^2 \cdot P$$

$$= \pi / 4 \cdot 2035^2 \cdot 4970$$

$$= 1648416.625 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 27 of 264
 Flg Calc [Int P]: LHS Ch Flg Flng: 10 9:47am Apr 20,2026

= H - Hd
 = 1760790 - 1648416
 = 112374.070 Kgf

Design Bolt Load for the Operating Condition [Wo]:

= max(Wo,1 + Wo,2 + H'p, 0)
 = max(1760790 + 208759 + 0, 0)
 = 1969549.750 Kgf

Design Bolt Load for the Gasket Seating Condition [Wg]:

= y • b • π • G + yPart • bPart • lp
 = 68.9 • 10.4 • π • 2103 + 0 • 0
 = 482671.031 Kgf

Required Bolt Area [Am]:

= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
 = max((1969549 + 0 + 4 • 0/2103)/159, 482671/159)
 = 1214.804 cm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

= 2a + 6t/(m + 0.5)
 = 2 • 88.9 + 6 • 355/(3 + 0.5)
 = 786.371 mm

Actual Circumferential Bolt Spacing [Bs]:

= C • sin(π / n)
 = 2390 • sin(π/36)
 = 208.302 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

= max(√(Bs/(2a + t)), 1)
 = max(√(208/(2 • 88.9 + 355)), 1)
 = 1.0000

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1214.804	2032.022	
Radial Distance between Hub and Bolts:	104.775	109.500	
Radial Distance between Bolts and Edge:	82.550	87.500	
Circ. Spacing between the Bolts:	180.975	208.302	786.371

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

= Ab • Sa/(y • π(Go + Gi))
 = 2032 • 159/(68.9 • π(2124 + 2056))
 = 35.683 mm[Note: Exceeds actual gasket width, 34.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

= Sa • Ab
 = 159 • 2032
 = 3294495.50 Kgf

Gasket Load for the Operating Condition [HG]:

= Wo - H
 = 1969549 - 1760790
 = 208759.08 Kgf

Moment Arm Calculations:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 28 of 264
 Flg Calc [Int P]: LHS Ch Flg Flng: 10 9:47am Apr 20,2026

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2390 - 2103) / 2$$

$$= 143.3899 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2390 - 2035) / 2 + 143)$$

$$= 160.4449 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2390 - 2035 - 68) / 2$$

$$= 143.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1648417.	143.5000	1.0000	236552.
Face Pressure, Mt	112374.	160.4449	1.0000	18030.
Gasket Load, Mg	208759.	143.3899	1.0000	29935.
Gasket Seating, Matm	3294496.	143.3899	1.0000	472407.

Total Moment for Operation, Mo 284517. Kg-m.
 Total Moment for Gasket seating, Mg 472407. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 305.958 mm
 Hub Ratio, $h/h_o = HL / H_o$ 0.255
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.478

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.891
 Factor V per 4.16.5 0.392
 Factor f per 4.16.5 1.278

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2565 / 2035$$

$$= 1.260$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.26^2 \cdot \log_{10}(1.26))) / (1.26^2 - 1)) / (1.26 - 1)$$

$$= 8.521$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.26^2 (1 + 8.55246 \cdot \log_{10}(1.26)) - 1) / ((1.0472 + 1.9448 \cdot 1.26^2) \cdot (1.26 - 1))$$

$$= 1.814$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.26^2 (1 + 8.55246 \cdot \log_{10}(1.26)) - 1) / (1.36136 (1.26^2 - 1) \cdot (1.26 - 1))$$

$$= 9.364$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.26^2 + 1) / (1.26^2 - 1)$$

$$= 4.397$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_o^2 \cdot h_o / V$$

$$= 9.36 \cdot 46^2 \cdot 306 / 0.39$$

$$= 15476601.000 \text{ mm}^3$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 29 of 264
 Flg Calc [Int P]: LHS Ch Flg Flng: 10 9:47am Apr 20,2026

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.89/306$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (355 \cdot 0.0029 + 1) / 1.81 + 355^3/15476601$$

$$= 4.012$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.28 \cdot 284517/2035) / (4.01 \cdot 68^2)$$

$$= 94.48 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.28 \cdot 472407/2035) / (4.01 \cdot 68^2)$$

$$= 156.87 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.38 \cdot 284517/2035) / (4.01 \cdot 355^2)$$

$$= 6.45 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.38 \cdot 472407/2035) / (4.01 \cdot 355^2)$$

$$= 10.71 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (8.52 \cdot 284517 / (355^2 \cdot 2035)) - 4.4 \cdot 6.45$$

$$= 64.35 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (8.52 \cdot 472407 / (355^2 \cdot 2035)) - 4.4 \cdot 10.7$$

$$= 106.84 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (94.5 + \max(6.45, 64.3)) / 2$$

$$= 79.41 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (157 + \max(10.7, 107)) / 2$$

$$= 131.86 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1969549/2032$$

$$= 95.05 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (482671/2032)$$

$$= 23.29 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²



PV Elite 28 Licensee: SL User
 FileName : 106-C-Clading on Both Side Page 30 of 264
 Flg Calc [Int P]: LHS Ch Flg Flng: 10 9:47am Apr 20,2026

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	94.48	207.00	156.87	207.00
Radial Flange	6.45	138.00	10.71	138.00
Tangential Flange	64.35	138.00	106.84	138.00
Maximum Average	79.41	138.00	131.86	138.00
Bolting	95.05	159.00	23.29	159.00

Minimum Required Flange Thickness 346.431 mm

Estimated M.A.W.P. (Operating) 6074 KPa.
 Estimated Finished Weight of Flange at given Thk. 5494.9 kg.
 Estimated Unfinished Weight of Forging at given Thk 6426.0 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	486.85	486.85	1.000	486.85	499.99	0.974
Tangential Flange	St	335.31	365.14	0.918	335.31	374.99	0.894
Tangential Flange	Sto	118.12	243.43	0.485	118.12	250.00	0.472
Combined	Sh/f + St	720.46	730.28	0.987	720.46	749.99	0.961
Combined	Sr + St	368.92	730.28	0.505	368.92	749.99	0.492
Combined	Sh + Sto	604.97	730.28	0.828	604.97	749.99	0.807
Radial Flange	Sr	33.61	486.85	0.069	33.61	499.99	0.067
Maximum Stress Ratio				1.000			0.974



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.48/1.17)
 = 1.2641

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)
 = max(0.25; 0.33/1.08) if $h/h0 < 0.35$, else max (0.25; 0.26/0.78)
 = 0.3067

In this case, the hub ratio $h/h0$, 0.2549 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	499.0 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	380.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 32 of 264
 Flg Calc [Int P]: LHS Ch Flg Flng: 10 9:47am Apr 20,2026

Root Area of a Single Bolt	Abs	56.4450 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	213.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.568 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2124^2 - 2056^2)$$

$$= 2232.4 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208 \cdot 2232 / (36 \cdot 56.4)$$

$$= 228.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(229; 508)$$

$$= 228.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(229; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 499)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (2232 / (36 \cdot 56.4))$$

$$:: 290.0 \geq 153.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 2232 + \pi/4 \cdot 4970 \cdot 2056^2) / (0.7 \cdot 56.4 \cdot 36)$$

$$:: 290.0 \geq 268.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$$

$$:: 290 \leq 380 (2232 / (36 \cdot 56.4))$$

$$:: 290.0 \leq 417.5 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax(thetagmax / thetafmax)$$

$$:: 290 \leq 499 (1/0.57)$$

$$:: 290.0 \leq 878.7 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.26 \cdot 56.4 \cdot 88.9$$

$$= 3858.1 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 95 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 158 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 472407 / 1 \cdot 9807 \cdot 0.39 / (3.79 \cdot 201334 \cdot 46^2 \cdot 306 \cdot 0.3)$
 $= 0.637$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 284517 / 1 \cdot 9807 \cdot 0.39 / (3.79 \cdot 191213 \cdot 46^2 \cdot 306 \cdot 0.3)$
 $= 0.404$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 472407 / 1 \cdot 9807 \cdot 0.39 / (4.01 \cdot 201334 \cdot 46^2 \cdot 306 \cdot 0.3)$
 $= 0.603$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 284517 / 1 \cdot 9807 \cdot 0.39 / (4.01 \cdot 191213 \cdot 46^2 \cdot 306 \cdot 0.3)$
 $= 0.382$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.86, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 7 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	94.48	207.00	156.87 207.00
Radial Flange	6.45	138.00	10.71 138.00
Tangential Flange	64.35	138.00	106.84 138.00
Maximum Average	79.41	138.00	131.86 138.00
Bolting	95.05	159.00	23.29 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 346.431 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 34 of 264
 Flg Calc [Int P]: RHS Ch Flg Flng: 11 9:47am Apr 20,2026

Flange Input Data Values

Description: RHS Ch Flg :

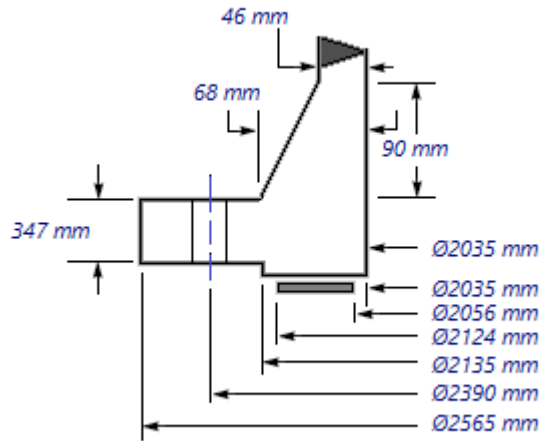
RHS Ch Flg

Flange Type	Integral Weld Neck, Integral type		
Design Pressure	P	4969.95	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2035.000	mm
Flange Outside Diameter	A	2565.000	mm
Flange Thickness	t	347.0000	mm
Thickness of Hub at Small End	go	46.0000	mm
Thickness of Hub at Large End	gl	68.0000	mm
Length of Hub	h	90.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2390.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2135.000	mm
Flange Face Inside Diameter	Fid	2035.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2124.000	mm
Gasket Inside Diameter	Gi	2056.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4970 \cdot (2035/2 + 0)) / (138 \cdot 1 - 0.6 \cdot 4970) + C_a$$

$$= 37.4566 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 46) / (1018 + 0.6 \cdot 46)$$

$$= 6073.598 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 46) / (1018 + 0.6 \cdot 46)$$

$$= 6073.598 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B + 2 * ci	2035.000 mm
Corroded Large Hub,	g1Cor = g1 - ci	68.000 mm
Corroded Small Hub,	g0Cor = go - ci	46.000 mm
Code R Dimension,	R = (C - B) / 2 - g1	109.500 mm

Gasket Contact Width,	N = (Go - Gi) / 2	34.000 mm
Basic Gasket Width,	bo = N / 2	17.000 mm
Effective Gasket Width,	b = Cb * sqrt(bo)	10.390 mm
Gasket Reaction Diameter,	G = Go - 2 * b	2103.220 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 2103^2 \cdot 4970$$

$$= 1760790.625 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 10.4 \cdot \pi \cdot 2103 \cdot 3 \cdot 4970$$

$$= 208759.016 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 2035^2 \cdot 4970$$

$$= 1648416.625 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 36 of 264
 Flg Calc [Int P]: RHS Ch Flg Flng: 11 9:47am Apr 20,2026

= H - Hd
 = 1760790 - 1648416
 = 112374.070 Kgf

Design Bolt Load for the Operating Condition [Wo]:

= max(Wo,1 + Wo,2 + H'p, 0)
 = max(1760790 + 208759 + 0, 0)
 = 1969549.750 Kgf

Design Bolt Load for the Gasket Seating Condition [Wg]:

= y • b • π • G + yPart • bPart • lp
 = 68.9 • 10.4 • π • 2103 + 0 • 0
 = 482671.031 Kgf

Required Bolt Area [Am]:

= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
 = max((1969549 + 0 + 4 • 0/2103)/159, 482671/159)
 = 1214.804 cm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

= 2a + 6t/(m + 0.5)
 = 2 • 88.9 + 6 • 347/(3 + 0.5)
 = 772.657 mm

Actual Circumferential Bolt Spacing [Bs]:

= C • sin(π / n)
 = 2390 • sin(π/36)
 = 208.302 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

= max(√(Bs/(2a + t)), 1)
 = max(√(208/(2 • 88.9 + 347)), 1)
 = 1.0000

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1214.804	2032.022	
Radial Distance between Hub and Bolts:	104.775	109.500	
Radial Distance between Bolts and Edge:	82.550	87.500	
Circ. Spacing between the Bolts:	180.975	208.302	772.657

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

= Ab • Sa/(y • π(Go + Gi))
 = 2032 • 159/(68.9 • π(2124 + 2056))
 = 35.683 mm[Note: Exceeds actual gasket width, 34.000]

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

= Sa • Ab
 = 159 • 2032
 = 3294495.50 Kgf

Gasket Load for the Operating Condition [HG]:

= Wo - H
 = 1969549 - 1760790
 = 208759.08 Kgf

Moment Arm Calculations:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 37 of 264
 Flg Calc [Int P]: RHS Ch Flg Flng: 11 9:47am Apr 20,2026

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2390 - 2103) / 2$$

$$= 143.3899 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + hg)$$

$$= 0.5 ((2390 - 2035) / 2 + 143)$$

$$= 160.4449 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2390 - 2035 - 68) / 2$$

$$= 143.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1648417.	143.5000	1.0000	236552.
Face Pressure, Mt	112374.	160.4449	1.0000	18030.
Gasket Load, Mg	208759.	143.3899	1.0000	29935.
Gasket Seating, Matm	3294496.	143.3899	1.0000	472407.

Total Moment for Operation, Mo 284517. Kg-m.
 Total Moment for Gasket seating, Mg 472407. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 305.958 mm
 Hub Ratio, h/h0 = HL / H0 0.294
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.478

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.884
 Factor V per 4.16.5 0.377
 Factor f per 4.16.5 1.156

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 2565 / 2035$$

$$= 1.260$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.26^2 \cdot \log_{10}(1.26))) / (1.26^2 - 1)) / (1.26 - 1)$$

$$= 8.521$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.26^2 (1 + 8.55246 \cdot \log_{10}(1.26)) - 1) / ((1.0472 + 1.9448 \cdot 1.26^2) \cdot (1.26 - 1))$$

$$= 1.814$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.26^2 (1 + 8.55246 \cdot \log_{10}(1.26)) - 1) / (1.36136 (1.26^2 - 1) \cdot (1.26 - 1))$$

$$= 9.364$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.26^2 + 1) / (1.26^2 - 1)$$

$$= 4.397$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 9.36 \cdot 46^2 \cdot 306 / 0.38$$

$$= 16088260.000 \text{ mm}^3$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 38 of 264
 Flg Calc [Int P]: RHS Ch Flg Flng: 11 9:47am Apr 20,2026

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.88/306$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (347 \cdot 0.0029 + 1) / 1.81 + 347^3/16088260$$

$$= 3.701$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1.16 \cdot 284517/2035) / (3.7 \cdot 68^2)$$

$$= 92.62 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1.16 \cdot 472407/2035) / (3.7 \cdot 68^2)$$

$$= 153.79 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.34 \cdot 284517/2035) / (3.7 \cdot 347^2)$$

$$= 7.19 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm/Bcor) / (L \cdot t^2)$$

$$= (2.34 \cdot 472407/2035) / (3.7 \cdot 347^2)$$

$$= 11.93 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (8.52 \cdot 284517 / (347^2 \cdot 2035)) - 4.4 \cdot 7.19$$

$$= 65.42 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (8.52 \cdot 472407 / (347^2 \cdot 2035)) - 4.4 \cdot 11.9$$

$$= 108.63 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (92.6 + \max(7.19, 65.4)) / 2$$

$$= 79.02 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (154 + \max(11.9, 109)) / 2$$

$$= 131.21 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1969549/2032$$

$$= 95.05 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (482671/2032)$$

$$= 23.29 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²



PV Elite 28 Licensee: SL User
 FileName : 106-C-Clading on Both Side Page 39 of 264
 Flg Calc [Int P]: RHS Ch Flg Flng: 11 9:47am Apr 20,2026

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	92.62	207.00	153.79	207.00
Radial Flange	7.19	138.00	11.93	138.00
Tangential Flange	65.42	138.00	108.63	138.00
Maximum Average	79.02	138.00	131.21	138.00
Bolting	95.05	159.00	23.29	159.00

Minimum Required Flange Thickness 337.414 mm

Estimated M.A.W.P. (Operating) 6074 KPa.
 Estimated Finished Weight of Flange at given Thk. 5411.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 6485.4 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	460.74	486.85	0.946	460.74	499.99	0.921
Tangential Flange	St	328.26	365.14	0.899	328.26	374.99	0.875
Tangential Flange	Sto	128.68	243.43	0.529	128.68	250.00	0.515
Combined	Sh/f + St	730.28	730.28	1.000	730.28	749.99	0.974
Combined	Sr + St	364.32	730.28	0.499	364.32	749.99	0.486
Combined	Sh + Sto	589.41	730.28	0.807	589.41	749.99	0.786
Radial Flange	Sr	36.07	486.85	0.074	36.07	499.99	0.072
Maximum Stress Ratio				1.000			0.974



Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.42/1.24)
 = 1.1460

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 0.38/1.18) if h/ho < 0.35, else max (0.25; 0.26/0.77)
 = 0.3201

In this case, the hub ratio h/ho, 0.2942 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	480.5 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sgmax	275.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.260
Fraction of Gasket Load remaining after relaxation	Phig	0.700



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 41 of 264
 Flg Calc [Int P]: RHS Ch Flg Flng: 11 9:47am Apr 20,2026

Root Area of a Single Bolt	Abs	56.4450 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	213.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.570 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (2124^2 - 2056^2)$$

$$= 2232.4 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Ab)$$

$$= 208 \cdot 2232 / (36 \cdot 56.4)$$

$$= 228.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(229; 508)$$

$$= 228.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(229; 290)$$

$$= 290.0 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(290; 480)$$

$$= 290.0 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin \cdot S (Ag / (nb \cdot Abs))$$

$$:: 290 \geq 140 (2232 / (36 \cdot 56.4))$$

$$:: 290.0 \geq 153.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin \cdot O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 290 \geq (97 \cdot 2232 + \pi/4 \cdot 4970 \cdot 2056^2) / (0.7 \cdot 56.4 \cdot 36)$$

$$:: 290.0 \geq 268.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 290 \leq 275 (2232 / (36 \cdot 56.4))$$

$$:: 290.0 \leq 302.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 290 \leq 480 (1/0.57)$$

$$:: 290.0 \leq 842.6 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 290 \cdot 0.26 \cdot 56.4 \cdot 88.9$$

$$= 3858.1 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 95 N./mm²

Bolt Allowable Stress for: SA-193 B7 Sb 158 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 472407 / 1 \cdot 9807 \cdot 0.38 / (3.48 \cdot 201334 \cdot 46^2 \cdot 306 \cdot 0.3)$
 $= 0.669$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 284517 / 1 \cdot 9807 \cdot 0.38 / (3.48 \cdot 191213 \cdot 46^2 \cdot 306 \cdot 0.3)$
 $= 0.424$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 472407 / 1 \cdot 9807 \cdot 0.38 / (3.7 \cdot 201334 \cdot 46^2 \cdot 306 \cdot 0.3)$
 $= 0.629$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 284517 / 1 \cdot 9807 \cdot 0.38 / (3.7 \cdot 191213 \cdot 46^2 \cdot 306 \cdot 0.3)$
 $= 0.398$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.86, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 7 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	92.62	207.00	153.79 207.00
Radial Flange	7.19	138.00	11.93 138.00
Tangential Flange	65.42	138.00	108.63 138.00
Maximum Average	79.02	138.00	131.21 138.00
Bolting	95.05	159.00	23.29 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 337.414 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 43 of 264
 Flg Calc [Int P]: RHS blind fl Flng: 12 9:47am Apr 20,2026

Flange Input Data Values

Description: RHS blind fl :

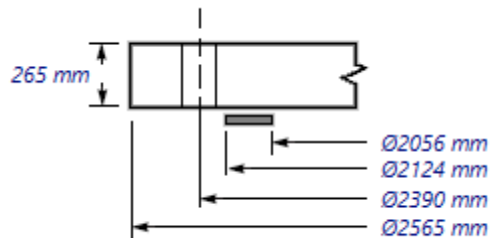
RHS blind flang

Flange Type		Blind	
Design Pressure	P	4975.29	KPa.
Design Temperature		200	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2565.000	mm
Flange Thickness	t	265.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Perimeter along the Center of the Bolts	L	7508.407	mm
Diameter of Bolt Circle	C	2390.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		36	
Flange Face Outside Diameter	Fod	2135.000	mm
Flange Face Inside Diameter	Fid	2035.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2124.000	mm
Gasket Inside Diameter	Gi	2056.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Go - Gi)/2$	34.000	mm
Basic Gasket Width,	$bo = N/2$	17.000	mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	10.390	mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 44 of 264
 Flg Calc [Int P]: RHS blind fl Flng: 12 9:47am Apr 20,2026

Gasket Reaction Diameter, $G = G_o - 2 \cdot b$ 2103.220 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2103^2 \cdot 4975$$

$$= 1762683.375 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 10.4 \cdot \pi \cdot 2103 \cdot 3 \cdot 4975$$

$$= 208983.406 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(1762683 + 208983 + 0, 0)$$

$$= 1971666.750 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.9 \cdot 10.4 \cdot \pi \cdot 2103 + 0 \cdot 0 \cdot 0$$

$$= 482671.031 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1971666 + 0 + 4 \cdot 0 / 2103) / 159, 482671 / 159)$$

$$= 1216.110 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1216.110	2032.022	
Radial Distance between Bolts and Edge:	82.550	87.500	
Circ. Spacing between the Bolts:	180.975	208.302	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 2032 \cdot 159 / (68.9 \cdot \pi (2124 + 2056))$$

$$= 35.683 \text{ mm} [\text{Note: Exceeds actual gasket width, 34.000}]$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159 \cdot 2032$$

$$= 3294495.50 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 1971666.75 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2390 - 2103) / 2$$

$$= 143.3899 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 1971666 \cdot 143 \cdot 1 / (265^2 \cdot 2103) + 0.3 \cdot 1 \cdot 4975 \cdot 2103^2 / 265^2$$

$$= 129.69 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 45 of 264
 Flg Calc [Int P]: RHS blind fl Flng: 12 9:47am Apr 20,2026

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 3294495 \cdot 143 \cdot 1 / (265^2 \cdot 2103)$$

$$= 59.60 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1971666 / 2032$$

$$= 95.16 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (482671 / 2032)$$

$$= 23.29 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	129.69	138.00	59.60	138.00
Bolting	95.16	159.00	23.29	159.00

Reqd. Blind Flange Thickness at Center 256.901 mm
 Reqd. Blind Flange Thickness at Gasket 174.147 mm

Estimated M.A.W.P. (Operating) 5294 KPa.
 Estimated Finished Weight of Flange at given Thk. 10613.1 kg.
 Estimated Unfinished Weight of Forging at given Thk 10613.1 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 1, Temperature Reduction per Fig. UCS 66.1 = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 23 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 23 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	129.69	138.00	59.60	138.00
Bolting	95.16	159.00	23.29	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 256.921 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 46 of 264
 Internal Pressure Calculations: Step: 7 9:47am Apr 20,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flang		4975.3	273	...	2565	138	1
LHS Ch Flg		4969.9	355	...	2035	138	2
Ch Shell		4969.9	48	2	2024	138	3
main shell		939.86	20	2	2024	138	4
RHS Ch shell		4969.9	48	2	2024	138	5
RHS Ch Flg		4969.9	355	...	2035	138	6
RHS blind flang		4975.3	273	...	2565	138	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flang		4950	5269	5294	265	256.901	1
LHS Ch Flg		4950	6054	6074	355	346.431	2
Ch Shell		4950	5816	6106	46	39.3271	3
main shell		920	2404	2695	20	8.93479	4
RHS Ch shell		4950	5816	6106	46	39.3271	5
RHS Ch Flg		4950	6054	6074	347	337.414	6
RHS blind flang		4950	5269	5294	265	256.901	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 2404 KPa.
 Shell Side MAPnc = 2695 KPa.
 Channel Side MAWP = 5269 KPa.
 Channel Side MAPnc = 5294 KPa.

Elements Suitable for Design Internal Pressure.

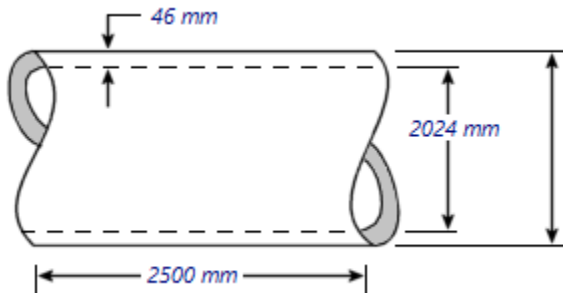
PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 47 of 264
 Element: Ch Shell Step: 8 9:47am Apr 20,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	4969.859 KPa.	Int. Design Temperature	200 °C
Material	SA-516 70	Ext. Design Temperature	0 °C
Length	2500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2024.000 mm	Actual Stress	117.5 N./mm ²
Spec. Min. Thk t,e, or sa	46.0000 mm	Spec. Nominal Thk.	48.000 mm
Surface Area	166504.4 cm ²	SG of Contents	1.000
Element Volume	8045050.5 cm ³	Weight of Contents	8040.1 Kg
Empty Weight	6054.1 Kg	Operating Weight	14094.3 Kg
		Comp. All. Ope. Stress	129.0 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4970 \cdot 1014) / (138 \cdot 1 - 0.6 \cdot 4970) \\
 &= 37.3271 + 2.0000 = 39.3271 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 19.859 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 44) / (1014 + 0.6 \cdot 44) \\
 &= 5836 - 19.9 = 5816 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 46) / (1012 + 0.6 \cdot 46) \\
 &= 6106 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4970 \cdot (1014 + 0.6 \cdot 44)) / (1 \cdot 44) \\
 &= 117.521 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side Page 48 of 264
Element: Ch Shell Step: 8 9:47am Apr 20,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f (1 - R_f / R_o)$) 2.317 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 46$, $t_r = 39.6$, $c = 2$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C
Min Metal Temp. at Required thickness (UCS 66.1) 9 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was
specified as being heat treated.

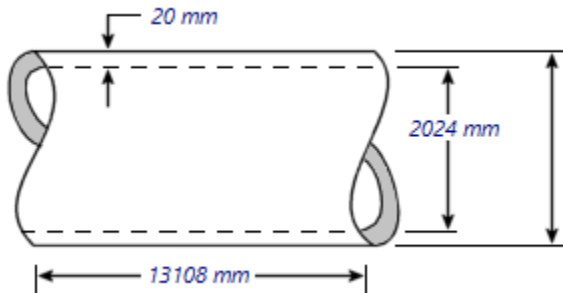
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5268.591 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	46.000 mm	Basic MDMT	15 °C
Stress/Thk. Ratio	0.901	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	9 °C
		Joint Efficiency E^*	100 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: main shell

Cylindrical Shell Design Information:					
Design Pressure	P	939.859 KPa.	Int. Design Temperature		160 °C
Material		SA-516 70	Ext. Design Temperature		160 °C
Length		13108.001 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	2.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2024.000 mm	Actual Stress		53.5 N./mm ²
Spec. Min. Thk t,e, or sa		20.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		849955.4 cm ²	SG of Contents		1.000
Element Volume		22433082.0 cm ³	Weight of Contents		22419.4 Kg
Empty Weight		13047.5 Kg	Operating Weight		35466.9 Kg
			Comp. All. Ope. Stress		123.7 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (940 \cdot 1014) / (138 \cdot 1 - 0.6 \cdot 940) \\
 &= 6.9348 + 2.0000 = 8.9348 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 19.859 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 18) / (1014 + 0.6 \cdot 18) \\
 &= 2424 - 19.9 = 2404 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 20) / (1012 + 0.6 \cdot 20) \\
 &= 2695 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (940 \cdot (1014 + 0.6 \cdot 18)) / (1 \cdot 18) \\
 &= 53.512 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 50 of 264
 Element: main shell Step: 9 9:47am Apr 20,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.978 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 20$, $t_r = 17.9$, $c = 2$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -8 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2403.844 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	20.000 mm	Basic MDMT	-7 °C
Stress/Thk. Ratio	0.992	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-8 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: main shell

Cylindrical Shell From 40 to Tsht Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 198566240 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.000	2064.00	13108.02	114.67	6.3508	0.0001518	15.07

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 15.1) / (3 \cdot 115) = 175 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.550	2064.00	13108.02	141.85	6.3508	0.0001108	11.00

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 11) / (3 \cdot 142) = 103 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
18.000	2064.00	27966.46	114.67	13.5496	0.0000897	8.91

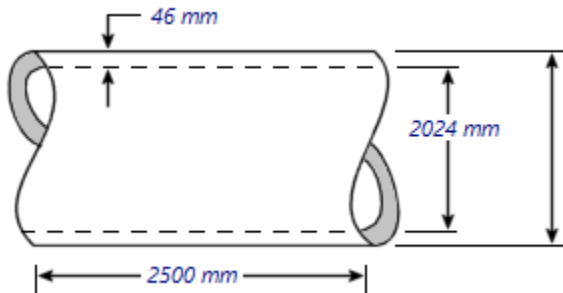
$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 8.91) / (3 \cdot 115) = 104 \text{ KPa.}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 51 of 264
 Element: RHS Ch shell Step: 10 9:47am Apr 20,2026

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: RHS Ch shell

Cylindrical Shell Design Information:			
Design Pressure P	4969.859 KPa.	Int. Design Temperature	200 °C
Material	SA-516 70	Ext. Design Temperature	200 °C
Length	2500.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	2.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2024.000 mm	Actual Stress	117.5 N./mm ²
Spec. Min. Thk t _e , or sa	46.0000 mm	Spec. Nominal Thk.	48.000 mm
Surface Area	166504.4 cm ²	SG of Contents	1.000
Element Volume	8045050.5 cm ³	Weight of Contents	8040.1 Kg
Empty Weight	6054.1 Kg	Operating Weight	14094.3 Kg
		Comp. All. Ope. Stress	123.7 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (4970 \cdot 1014) / (138 \cdot 1 - 0.6 \cdot 4970) \\
 &= 37.3271 + 2.0000 = 39.3271 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 19.859 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 44) / (1014 + 0.6 \cdot 44) \\
 &= 5836 - 19.9 = 5816 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138 \cdot 1 \cdot 46) / (1012 + 0.6 \cdot 46) \\
 &= 6106 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (4970 \cdot (1014 + 0.6 \cdot 44)) / (1 \cdot 44) \\
 &= 117.521 \text{ N./mm}^2
 \end{aligned}$$



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side Page 52 of 264
Element: RHS Ch shell Step: 10 9:47am Apr 20,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.317 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 46$, $t_r = 39.6$, $c = 2$ mm, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 15 °C
Min Metal Temp. at Required thickness (UCS 66.1) 9 °C

Note:
Post Weld Heat Treatment is required for this Element/Joint and it was
specified as being heat treated.

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	5268.591 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	46.000 mm	Basic MDMT	15 °C
Stress/Thk. Ratio	0.901	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	9 °C
		Joint Efficiency E^*	100 %

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 53 of 264
 Shop Pressure Test Results: Step: 11 9:47am Apr 20,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3124.997 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1196.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3478.497 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2644.228 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1315.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1315.600 KPa.
App 27-4	= M.A.W.P.	2403.844 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	6849.168 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	6435.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6856.986 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	5795.450 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	7078.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7078.500 KPa.
App 27-4	= M.A.W.P.	5268.591 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch Shell	155.4	234.0	0.664	6876.83
main shell	179.1	234.0	0.766	3498.34
RHS Ch shell	155.4	234.0	0.664	6876.83

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	138.00	138.00	1.000
Noz N5	Nozzle	138.00	121.40	1.137
D1	Nozzle	138.00	138.00	1.000
N3-Shell Inlet	Nozzle	118.00	118.00	1.000
N3-Shell Inlet	Pad	138.00	138.00	1.000
N4-Shell Inlet	Nozzle	118.00	118.00	1.000
N4-Shell Inlet	Pad	138.00	138.00	1.000
N2 Tube outlet	Nozzle	138.00	138.00	1.000
V1	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flang	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
Ch Shell	138.00	138.00	1.000
main shell	138.00	138.00	1.000
RHS Ch shell	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS blind flang	138.00	138.00	1.000



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side Page 54 of 264
Shop Pressure Test Results: Step: 11 9:47am Apr 20,2026

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	138.00	126.00	1.095
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	151.72	225.00	0.674
Noz N5	89.87	184.50	0.487
D1	14.55	225.00	0.065
N3-Shell Inlet	43.96	216.00	0.204
N4-Shell Inlet	43.96	216.00	0.204
N2 Tube outlet	158.49	225.00	0.704
V1	14.55	225.00	0.065

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 55 of 264
 Field Pressure Test Results: Step: 12 9:47am Apr 20,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 3124.997 KPa.
 UG99b(b)(35) = 1.30 * Design Pres * Sa/S 1196.000 KPa.
 UG99(c) = 1.30 * M.A.P. - Head(Hyd) 3478.497 KPa.
 UG100 = 1.10 * M.A.W.P. * Sa/S 2644.228 KPa.
 PED = max(1.43*DP, 1.25*DP*ratio) 1315.600 KPa.
 PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 1315.600 KPa.
 App 27-4 = M.A.W.P. 2403.844 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b) = 1.30 * M.A.W.P. * Sa/S 6849.168 KPa.
 UG99b(b)(35) = 1.30 * Design Pres * Sa/S 6435.000 KPa.
 UG99(c) = 1.30 * M.A.P. - Head(Hyd) 6856.986 KPa.
 UG100 = 1.10 * M.A.W.P. * Sa/S 5795.450 KPa.
 PED = max(1.43*DP, 1.25*DP*ratio) 7078.500 KPa.
 PED&UG99(b)(35) = max(1.43*DP, 1.3*DP*ratio) 7078.500 KPa.
 App 27-4 = M.A.W.P. 5268.591 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 920 • 1
 = 1196.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 4950 • 1
 = 6435.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch Shell	152.6	234.0	0.652	6454.84
main shell	69.2	234.0	0.296	1215.84
RHS Ch shell	152.6	234.0	0.652	6454.84

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	138.00	138.00	1.000
Noz N5	Nozzle	138.00	121.40	1.137
D1	Nozzle	138.00	138.00	1.000
N3-Shell Inlet	Nozzle	118.00	118.00	1.000
N3-Shell Inlet	Pad	138.00	138.00	1.000
N4-Shell Inlet	Nozzle	118.00	118.00	1.000
N4-Shell Inlet	Pad	138.00	138.00	1.000
N2 Tube outlet	Nozzle	138.00	138.00	1.000
V1	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
Field Pressure Test Results:

Step: 12 9:47am Apr 20,2026 Page 56 of 264

Description	Ambient	Operating	Ratio
LHS Blind Flang	138.00	138.00	1.000
LHS Ch Flg	138.00	138.00	1.000
Ch Shell	138.00	138.00	1.000
main shell	138.00	138.00	1.000
RHS Ch shell	138.00	138.00	1.000
RHS Ch Flg	138.00	138.00	1.000
RHS blind flang	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	138.00	126.00	1.095
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	142.41	225.00	0.633
Noz N5	84.36	184.50	0.457
D1	13.65	225.00	0.061
N3-Shell Inlet	15.28	216.00	0.071
N4-Shell Inlet	15.28	216.00	0.071
N2 Tube outlet	148.77	225.00	0.661
V1	13.65	225.00	0.061

Elements Suitable for Test Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell

Shell Test Pressure		1196.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2024.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		6435.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2024.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5285
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13090.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		138.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	205.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	986.713 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	28995.0 cm ²



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 58 of 264
 ASME TS Calc: Pressure Test Case: 1 9:47a Apr 20,2026

Perimeter of Tube Layout Cp 6328.5 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 205.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
 Tubesheet Outside Diameter A 2120.000 mm

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 610.000 mm
 Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes
 Shell Side Min. Design Pressure Psd,min -102.0000 KPa.

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
 Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
 Junction Stress Reduction option Perform Plastic Calculation

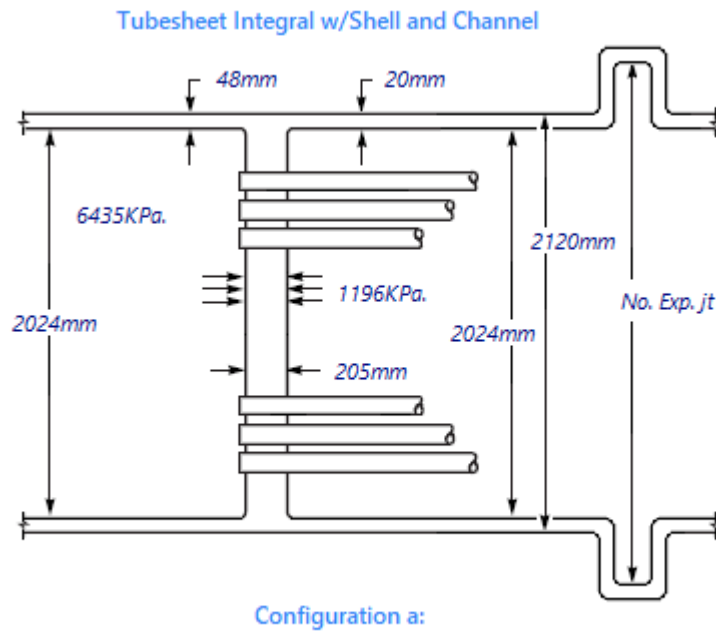
Tubesheet Integral with Both
 Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: Pressure Test

Page 59 of 264
 Case: 1 9:47a Apr 20,2026



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
 Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: Pressure Test

Page 60 of 264
 Case: 1 9:47a Apr 20,2026

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (6537 \cdot (19/2 - 0)) / (138 \cdot 1 + 0.4 \cdot 6537)$
 $= 0.4428 + 0.0000 = 0.4428 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-1 at 21.11 °C
 Elastic Modulus for Material 193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	13090.00	11.55	50.0000	0.0082522	98.41

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 98.4) / (3 \cdot 11.5) = 11365 \text{ KPa.}$

Results for Required Thickness: No Conversion

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3857	19.05	13090.00	49.39	50.0000	0.0004509	43.53

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 43.5) / (3 \cdot 49.4) = 1175 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4428 mm
Allowable Internal Pressure at Corroded thickness	25683.32 KPa.
Required Internal Design Pressure	6537.00 KPa.
Allowable External Pressure at Corroded thickness	11364.53 KPa.
Required External Design Pressure	1196.00 KPa.
Required Thickness due to Shell Side pressure	0.3857 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$= l_{tx} / h$ (modified for corrosion if present)
 $= 0/205$
 $= 0.0000$ (must be $0 \leq p \leq 1$)

Effective tube hole diameter [d*]:

$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 1.65 (195117184/201322320) \cdot (138/138) \cdot (0); 19 - 2 \cdot 1.65)$
 $= 19.0500 \text{ mm}$

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 987 + 19$
 $= 1992.476 \text{ mm}$

Determine the basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (25.4 - 19) / 25.4$
 $= 0.250$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: Pressure Test

Page 61 of 264
 Case: 1 9:47a Apr 20,2026

Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 1992 / 2$$

$$= 996.2380 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$$

$$= 25.4 / \sqrt{1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1992 \cdot 25.4) / (\pi \cdot 1992^2) }$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1012 / 996$$

$$= 1.0158$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1012 / 996$$

$$= 1.0158$$

Compute parameter [xt]:

$$= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 5285 / ((19 - 2 \cdot 1.65) / (2 \cdot 996))^2$$

$$= 0.6698$$

Determine parameter [xs]:

$$= 1 - Nt / (dt / (2 \cdot a_o))^2$$

$$= 1 - 5285 / (19 / (2 \cdot 996))^2$$

$$= 0.5169$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 20 (2024 + 20) 202701232 / 13090$$

$$= 202802.8594 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 1.65 (19 - 1.65) 195117184 / 13090$$

$$= 137.0993 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 202803 / (5285 \cdot 137)$$

$$= 0.27989$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 202803 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2024 + 20) 20)$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: Pressure Test

Page 62 of 264
 Case: 1 9:47a Apr 20, 2026

= 0.0090 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.009 \cdot 202701232 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 2670262016.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2024 + 20) \cdot 2.67E+09 / 205^3 \cdot (1 + 205 \cdot 0.009 + (205 \cdot 0.009)^2 / 2)$$

$$= 17263948.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2024^2 / (4 \cdot 202701232 \cdot 20) \cdot (1 - 2024 \cdot 0.3 / (2(2024 + 20)))$$

$$= 0.21509E+00 \text{ mm/N./mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2024 + 48)48}$$

$$= 0.0058 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0058 \cdot 201322320 \cdot 48^3 / (6(1 - 0.3^2))$$

$$= 23505158144.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2024 + 48 / \cos(0)) \cdot 23.51E+09 / 205^3 \cdot (1 + 205 \cdot 0.0058 \cdot (1 + 1) / 2 + (205 \cdot 0.0058)^2 / 2 \cdot 1)$$

$$= 97680824.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2024^2 / (4 \cdot 201322320 \cdot 48) \cdot (1 - 2024 \cdot 0.3 / (2(2024 + 48)))$$

$$= 0.90447E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

h/p = 8.070867 ; μ* = 0.250000 ; E* = 43608536 KPa.
 E*/E = 0.216611 ; ν* = 0.396457

Note: As h/p (8.071) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t_t(dt - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2)5285 \cdot 195117184 \cdot 1.65(19 - 1.65)996^2) / (43608536 \cdot 13090 \cdot 205^3))^{1/4}$$

$$= 3.3156$$

Values from Table 4.18.3

$$Z_d = 0.041184 ; Z_v = 0.088477 ; Z_m = 0.444934$$

$$Z_a = 0.311063E+01 ; Z_w = 0.088477$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: Pressure Test

Page 63 of 264
 Case: 1 9:47a Apr 20, 2026

$$= A / D_o$$

$$= 2120/1992$$

$$= 1.0640$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 43608536 \cdot (17263948 + 97680824 + 201322320 \cdot \ln(1.06))$$

$$= 1.7637$$

Compute parameter [ϕ]:

$$= (1 + v^*) F$$

$$= (1 + 0.4) 1.76$$

$$= 2.4629$$

Compute parameter [Q1]:

$$= (\psi s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.02 - 1 - 2.46 \cdot 0.088) / (1 + 2.46 \cdot 0.44)$$

$$= -0.096424$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.041 + -0.096 \cdot 0.088) / 2 \cdot 3.32^4$$

$$= 1.9732$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.088 + -0.096 \cdot 0.44) / 2 \cdot 3.32^4$$

$$= 2.7540$$

Compute parameter [U]:

$$= (Z_w + (\psi s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.088 + (1.02 - 1) 0.44) 3.32^4 / (1 + 2.46 \cdot 0.44)$$

$$= 5.5080$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ωs]:

$$= \psi s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 2.67E+09 \cdot 0.009 \cdot 0.21508874 (1 + 205 \cdot 0.009)$$

$$= 14914.5400 \text{ mm}^2$$

Calculate parameter [$\omega^* s$]:

$$= a o^2 (\psi s^2 - 1) \cdot (\psi s - 1) / 4 - \omega s$$

$$= 996^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 14915$$

$$= -14789.3359 \text{ mm}^2$$

Parameter [ωc]:

$$= \psi c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta \cdot h \cdot \beta C / 2)]$$

$$= 1.02 \cdot 23.51E+09 [0.090446845 (1 + 0.0058 \cdot 1 \cdot 205) - 0.000 (1/205 + 1 \cdot 0.0058/2)]$$

$$= 27159.8477 \text{ mm}^2$$

Parameter [$\omega^* c$]:

$$= a o^2 [(\psi c^2 + 1) \cdot (\psi c - 1) / 4 - (\psi s - 1) / 2] - \omega c$$

$$= 996^2 [(1.02^2 + 1) \cdot (1.02 - 1) / 4 - (1.02 - 1) / 2] - 27160$$

$$= -27034.6445 \text{ mm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: Pressure Test

Page 64 of 264
 Case: 1 9:47a Apr 20, 2026

Pressure [P*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= -102 \{ 0.52 + 2(1 - 0.52) 0.31 + [2/0.28 \cdot (2024/1992)^2] 0.3 - [(1.02^2 - 1)/(1 \cdot 0.28)] - [(1 - 1)/(2 \cdot 1 \cdot 0.28)] [(0^2 - (2024)^2)/1992^2] \}$$

$$= -297.2796 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.67 + 2(1 - 0.67) 0.31 + 1/(1 \cdot 0.28)] 6435$$

$$= 28618.3223 \text{ KPa.}$$

Pressure [P_γ]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 5285 \cdot 137 \cdot 0 / (3.14 \cdot 996^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W^* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 5.51 \cdot 0 / (2 \cdot \pi \cdot 996^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega^*s \cdot Ps - \omega^*c \cdot Pt)$$

$$= -(5.51/996^2) \cdot (-14789 \cdot -102 - 27035 \cdot 6435)$$

$$= -973.8358 \text{ KPa.}$$

Calculate the pressure [P_ω]:

$$= (U/ao^2) \cdot (\omega s \cdot P^*s - \omega c \cdot P^*c)$$

$$= (5.51/996^2) \cdot (14915 \cdot 0 - 27160 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + P_{\gamma} + P_{\omega} + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.28 / (1 + 1 \cdot 0.28(1.97 + (1.02 - 1)2.75)) \cdot (-297 - 28618 + 0 + 0 + -0 + -974)$$

$$= -5347.4004 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega^*s \cdot Ps - \omega^*c \cdot Pt) - (\omega s \cdot P^*s - \omega c \cdot P^*c)) + W^* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-14789 \cdot -102 - 27035 \cdot 6435) - (14915 \cdot 0 - 27160 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 2.46 \cdot 0.44)$$

$$= 8538.0244 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.096 + 2 \cdot 8538 / (-5347 \cdot 996^2)$$

$$= -0.12798$$

Fm Value from Table 4.18.3 = 0.063988



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 65 of 264
 ASME TS Calc: Pressure Test Case: 1 9:47a Apr 20, 2026

Tubesheet Bending Stress - original thickness [σ]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.064 / 0.25) \cdot (2 \cdot 996 / (205 - 0))^2 \cdot -5347$
 $= -193.9526 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 250 = 356 \text{ N./mm}^2, 4.18.233$

Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.058 / 0.25) \cdot (2 \cdot 996 / (138 - 0))^2 \cdot -4935$
 $= -356.2252 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + C_{ats} + C_{atc}$
 $= 138 + 0 + 0$
 $= 137.8058 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$
 $= (2 \cdot 0.25 \cdot 205 / 996) \cdot 190$
 $= 19.5482 \text{ N./mm}^2$

Shear Stress is not required to be computed; [P_e] $\leq$ τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:
 $= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$
 $= (28995 \cdot -3494) / (0.25 \cdot 34.4 \cdot 6328)$
 $= -185.9677 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:
 $= H + C_{ats} + C_{atc}$
 $= 34.4 + 0 + 0$
 $= 34.4380 \text{ mm}$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(H_{reqB}; H_{reqS})$
 $= \max(138; 34.4)$
 $= 137.8058 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{tmin} = -0.022, F_{tmax} = 1.8$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$
 $= ((-102 \cdot 0.52 - 6435 \cdot 0.67) - (-5347) \cdot -0.022) / (0.67 - 0.52)$
 $= -29.3092 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$
 $= ((-102 \cdot 0.52 - 6435 \cdot 0.67) - (-5347) \cdot 1.8) / (0.67 - 0.52)$
 $= 34.5820 \text{ N./mm}^2$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 34.6 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= \beta T \cdot S_{y,t} = 194.7465 \text{ N./mm}^2$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 66 of 264
 ASME TS Calc: Pressure Test Case: 1 9:47a Apr 20,2026

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$
 $= \max((3.25 - 0.25(0.041 + -0.13 \cdot 0.088)3.32^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2 \text{)}$

Determine the factor [rt]:
 $= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$
 $= (\sqrt{ (19^2 + (19 - 2 \cdot 1.65)^2) }) / 4$
 $= 6.1794 \text{ mm}$

Determine the factor [Ct]:
 $= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt)}$
 $= \sqrt{ (2 \cdot \pi^2 \cdot 195117184 / 205)}$
 $= 151.4439$

Determine the factor [Ft]:
 $= k \cdot L / r$
 $= 1 \cdot 610 / 6.18$
 $= 98.7147$

Buckling Allowable Stress [Stb]:
 $= (Sy_t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$
 $= (205 / 2) \cdot (1 - 98.7 / (2 \cdot 151))$
 $= 194.746 \text{ N./mm}^2 \text{ (Never greater than } Sot \text{)}$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
 $= \sigma_{t,max} \cdot \text{Tube Area}$
 $= 34.6 \cdot 0.9$
 $= 318.06 \text{ Kg}$

Tube Metal Area [At]:
 $= \pi (do - t) t$
 $= \pi (19 - 1.65) 1.65$
 $= 0.902 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (do - t) S$
 $= \pi \cdot 1.65 (19 - 1.65) 195$
 $= 1791.117 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft \text{)}$
 $= 0.85 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 195$
 $= 1763.5480 \text{ Kg}$

Weld Strength Factor [fw]:
 $= S / Sw \text{ where } Sw = \min(S \text{ and } St)$
 $= 195 / 195$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft \text{)}$
 $= 0.55 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 195$
 $= 1141.1193 \text{ Kg}$

Minimum Length of Weld [ar]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 67 of 264
 ASME TS Calc: Pressure Test Case: 1 9:47a Apr 20,2026

$$= 2(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do})$$

$$= 2(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 1.65 \cdot (19 - 1.65) \cdot 1 \cdot 1) - 0.75 \cdot 19})$$

$$= 2.075 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0374 mm
 Minimum Required Groove Weld Leg [agr]: 1.0374 mm

Tube-Tubesheet Jt allowable, 1791 is >= tube strength 1791 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= \beta T \cdot Sy,s = 0.95 \cdot 260 = 370 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= ao^2 / ((Ds + ts)ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts)ts)$
 $= 996^2 / ((2024 + 20)20) \cdot [-5347 + (1.02^2 - 1)(-102 - 6435)] + 1012^2 \cdot 6435 / ((2024 + 20)20)$
 $= 26.3274 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 20 mm, the minimum Shell length adjacent to the tubesheet should be 362.154 mm

Stress/Force Summary for Loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	194.0 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	47.8 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	34.6 <=	194.7 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-29.3 <=	-194.7 N./mm ²	Pass
Maximum Force on any one Tube	318.1 <=	1791.1 Kgf	Pass
Axial Membrane Stress in Shell	26.3 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D1 un-corr. (Psd,min + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	137.806	205.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.037	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.037	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses				Case	Pass/
		Bend	Allwd	Shear	Allwd		Type	Fail
D1uc	137.806 ...	194	356	48	190	Ps+Pt-Th	D1	Ok
D2uc	21.880 ...	22	356	6	190	Ps+Pt-Th	D2	Ok
D3uc	125.670 ...	170	356	42	190	Ps+Pt-Th	D3	Ok
Max:	137.8058 ... mm		0.544		0.252	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 68 of 264
 ASME TS Calc: Pressure Test Case: 1 9:47a Apr 20,2026

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	26	370	...	...	:	...	...	...	...	Ok
D2uc	16	370	...	...	:	...	...	...	...	Ok
D3uc	44	370	...	...	:	...	...	...	...	Ok
Max RATIO	0.119				:	...				:

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	35	194	-29	-194	318	1791	...	370	...	370	Ok
D2uc	4	194	-4	-194	39	1791	...	370	...	370	Ok
D3uc	31	194	-25	-194	282	1791	...	370	...	370	Ok
Max RATIO	0.178		0.150		0.178		...		...		:

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	137.806	205.000	Ok
Tube thickness :	0.443	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 362.154 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
 Due to Pressure Shell Extends by : +.259 mm
 Due to Pressure + Thermal Shell Extends by : +.259 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
ASME TS Calc: Pressure Test

Page 69 of 264
Case: 1 9:47a Apr 20,2026

1	0.2594	...	...	...	...	...
2	-0.1146	...	...	...	...	...
3	0.1350	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	265	No Calc	No Calc
20	30	No Calc	...	355	No Calc	No Calc
30	Tsht	2499.97	2116	44	0.0034236	114.139
Tsht	40	No Calc	...	44	No Calc	No Calc
40	Tsht	13108	2064	18	0.0001518	15.0716
Tsht	50	No Calc	...	18	No Calc	No Calc
50	60	2500.02	2116	44	0.0034236	101.623
60	70	No Calc	...	347	No Calc	No Calc
70	80	No Calc	...	265	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	265	174.168	...	No Calc
20	30	355	346.431	...	No Calc
30	Tsht	46	No Calc	...	3164
Tsht	40	46	No Calc	...	No Calc
40	Tsht	20	16.5501	103.419	175
Tsht	50	20	No Calc	103.419	No Calc
50	60	46	No Calc	...	2817
60	70	347	337.414	...	No Calc
70	80	265	174.168	...	No Calc

Minimum

175

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	2499.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	13108	27966.5	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	2500.02	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	10933.5	...	10933.5	...	1093.35
20	30	5494.91	1408589	5494.91	1408589	549.491
30	40	6054.11	8045050	5807.47	8076879	605.411
40	50	13047.5	22433082	11754.3	22599970	1304.75
50	60	6054.11	8045050	5807.47	8076879	605.411
60	70	5529.75	1447626	5529.75	1447626	552.975
70	80	10933.5	...	10933.5	...	1093.35
Total		58047	41379400.00	56260	41609944.00	5804



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	...	132.5	-7.49998	...	Liquid: 10
10	Insl	232.975	160	...	...	Insulation
20	Liqd	1407.73	216.5	...	...	Liquid: 20
20	Insl	92.1574	208	...	...	Insulation
30	Liqd	8040.14	1250	...	...	Liquid: 30
30	Insl	613.353	1250	...	...	Insulation
30	Lini	381.403	1250	...	...	SS Clad
30	Nozl	1724.71	1000	1450	...	N1-Tube Inlet
30	Nozl	324.427	1000	...	1189.8	Noz N5
30	Nozl	15.9702	1000	-1037.4	...	D1
30	Wght	200	1250	...	...	Partition plate w
40	Sadl	731.346	2700	1313	...	Left Saddle Flo
40	Sadl	731.346	10420	1313	...	Right Saddle
40	Liqd	22419.4	6554	...	...	Liquid: 40
40	Insl	3140.47	6554	...	...	Insulation
40	Lini	1999.77	6554	...	...	SS Clad
40	Nozl	129.339	1500	1161.43	...	N3-Shell Inlet
40	Nozl	129.339	11670	-1161.43	...	N4-Shell Inlet
40	Wght	650	2700	-1651	...	Fire[proofing
40	Wght	650	10420	-1651	...	Fire proofing
50	Liqd	8040.14	1250	...	...	Liquid: 50
50	Insl	613.353	1250	...	...	Insulation
50	Lini	381.403	1250	...	...	Clad
50	Nozl	1699.38	1500	-1469.2	...	N2 Tube outlet
50	Nozl	15.9702	1000	1037.4	...	V1
60	Liqd	1446.74	218.5	...	...	Liquid: 60
60	Insl	94.6631	208	...	...	Insulation
70	Liqd	...	132.5	-7.49998	...	Liquid: 70
70	Insl	232.975	160	...	...	Insulation
30	FTsh	3536.63	2602.5	...	...	Fixed Tubesheet
30	Tube	56823	9250	...	...	Tubes
30	RTsh	3536.63	15897.5	...	...	Tubesheet
40	Bafl	135.042	610	...	...	Baffles
40	Bafl	135.042	1220	...	...	Baffles
40	Bafl	135.042	1830	...	...	Baffles
40	Bafl	135.042	2440	...	...	Baffles
40	Bafl	135.042	3050	...	...	Baffles



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Element and Detail Weights: Step: 14 9:47am Apr 20,2026 Page 72 of 264

40 Baf1	135.042	3660	...	...	Baffles
40 Baf1	135.042	4270	...	...	Baffles
40 Baf1	135.042	4880	...	...	Baffles
40 Baf1	135.042	5490	...	...	Baffles
40 Baf1	135.042	6100	...	...	Baffles
40 Baf1	135.042	6710	...	...	Baffles
40 Baf1	135.042	7320	...	...	Baffles
40 Baf1	135.042	7930	...	...	Baffles
40 Baf1	135.042	8540	...	...	Baffles
40 Baf1	135.042	9150	...	...	Baffles
40 Baf1	135.042	9760	...	...	Baffles
40 Baf1	135.042	10370	...	...	Baffles
40 Baf1	135.042	10980	...	...	Baffles
40 Baf1	135.042	11590	...	...	Baffles
40 Baf1	135.042	12200	...	...	Baffles
40 Baf1	135.042	12810	...	...	Baffles
40 Baf1	135.679	13420	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	1462.7
Liquid	41354.2
Insulation	5019.9
Lining	2762.6
Nozzles	4039.1
Weights	1500.0
Exchanger Components	66867.9
Liquid in Tubes	13894.4

Sum of the Detail Weights 136900.8 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	63852.1	63852.1	63852.1	63852.1	63852.1	63852.1
Saddles	1462.7	1462.7	1462.7	1462.7	1462.7	1462.7
Nozzles	4039.1	4039.1	4039.1	4039.1	4039.1	4039.1
Wld Weights	200.0	200.0	200.0	200.0	200.0	200.0
Exchanger	66867.9	66867.9	66867.9	66867.9	66867.9	66867.9
Insulation	...	...	...	...	5019.9	5019.9 *
Lining	2762.6	2762.6	2762.6	2762.6	2762.6	2762.6
Empty Weights	...	...	...	1300.0	1300.0	...
Test Weights	...	1300.0	...	...	...	...
Ope Weights	...	...	...	...	...	1300.0
Ope. Liquid	...	...	...	...	...	41354.2
Tube Ope Lqd	...	...	...	...	...	13894.4
Test Liquid	...	41009.4	...	...	...	...
Tube Tst Lqd	...	13894.5	...	...	...	...
Totals	139184.3	195388.2	139184.3	140484.3	145504.3	200752.9

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt. - Bare weight without removable internals 139184.3 kg.



PV Elite 28 Licensee: SL User

FileName : 106-C-Cladding on Both Side

Page 73 of 264

Element and Detail Weights: Step: 14 9:47am Apr 20,2026

Shop Test Wt.	- Fabricated weight + water (full)	195388.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	139184.3 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	140484.3 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	145504.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	200752.9 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	198787.7 kg.
Field Test Wt.	- Empty weight + test medium (full)	200408.1 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	13902970.0 cm ³
Weight of Ope Liq in tubes :	13894.4 kg.
Weight of Water in tubes :	13894.5 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	73672	...
20	30	53074.5	27682.3
30	40	166504	158965
40	50	849955	833483
50	60	166504	158965
60	70	53892.9	28449.5
70	80	73672	...

Total	1437275.5	1207544.2 cm ²
Total	1547.1	1299.8 ft ²

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side
Nozzle Flange MAWP:

Page 74 of 264
Step: 15 9:47am Apr 20,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1-Tube Inlet	6860.1	9940.0	200	600	GR 2.1	...	...	...	...
Noz N5	6860.1	9940.0	200	600	GR 2.1	...	...	...	...
D1	6860.1	9940.0	200	600	GR 2.1	...	...	...	...
N3-Shell Inlet	3646.1	4970.0	160	300	GR 2.1	...	...	...	...
N4-Shell Inlet	3646.1	4970.0	160	300	GR 2.1	...	...	...	...
N2 Tube outlet	6860.1	9940.0	200	600	GR 2.1	...	...	...	...
V1	6860.1	9940.0	200	600	GR 2.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 3646.059 KPa.
Lowest Flange Pressure Rating was (Amb)[ShellSide]: 4970.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 6860.113 KPa.
Lowest Flange Pressure Rating was (Amb)[TubeSide]: 9940.003 KPa.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 75 of 264
 Wind Load Calculation: Step: 16 9:47am Apr 20,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	12500 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.21 \cdot 0.93) / (1 + 1.7 \cdot 3.4 \cdot 0.21)))$
 $= \min(0.85, 0.89)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.611
Structure Height to Diameter ratio	8.898

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5 : Z_g = 274320 \text{ mm}$

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1650 + 12500 = 14150 \text{ mm}$
 $= 46.4 \text{ ft. Imperial Units}$

Velocity Pressure coefficient evaluated at height z $[K_z]$:
 Because z (46.4 ft.) $> 15 \text{ ft.}$
 $= 2.01(z / Z_g)^{2/\alpha}$
 $= 2.01(46.4/900)^{2/9.5}$
 $= 1.08$

Type of Hill: No Hill



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 76 of 264
 Wind Load Calculation: Step: 16 9:47am Apr 20,2026

Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 = (1 + K1 • K2 • K3)²
 = (1 + 0 • 0 • 0)²
 = 1

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 = max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)
 = max(16, 0.00256 • 1.08 • 1 • 1 • 1 • 60.4²)
 = 16 psf [78.1] Kgs/m²

Force on the first element [F]:
 = qz • G • Cf • WindArea
 = 16 • 0.85 • 0.61 • 9.46
 = 78.6 lbs. [35.6] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flang	14150.0	0.000	0.000	0.000	1.077	1.000	78.120
LHS Ch Flg	14150.0	0.000	0.000	0.000	1.077	1.000	78.120
Ch Shell	14150.0	0.000	0.000	0.000	1.077	1.000	78.120
main shell	14150.0	0.000	0.000	0.000	1.077	1.000	78.120
RHS Ch shell	14150.0	0.000	0.000	0.000	1.077	1.000	78.120
RHS Ch Flg	14150.0	0.000	0.000	0.000	1.077	1.000	78.120
RHS blind flang	14150.0	0.000	0.000	0.000	1.077	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Partion plate	0.00	14150.00	78.12	0.00
Fire[prrofig	0.00	12500.00	78.12	0.00
Fire proofing	0.00	12500.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	14150	3318	8792.7	78.12	39.2115
20	30	14150	2672.58	11572.3	78.12	51.607
30	40	14150	2779.2	69480	78.12	309.849
40	50	14150	2716.8	356118	78.12	1588.13
50	60	14150	2779.2	69480	78.12	309.849
60	70	14150	2670.47	11669.9	78.12	52.0426
70	80	14150	3318	8792.7	78.12	39.2115

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	1.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.15 = 0.24 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.16 \\ SDS &= 2/3 * Sms = 2/3 * 0.24 = 0.16 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.16 = 0.1 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(9.620.75)$
 = 0.109 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.11$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.16 \cdot 199290 \cdot 1.25$
 = 11718.263 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.16 \cdot 199290$
 = 6249.7 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1282.5	22143.4	1555.76



PV Elite 28 Licensee: SL User

FileName : 106-C-Clading on Both Side

Page 78 of 264

Earthquake Load Calculation:

Step: 17

9:47am Apr 20,2026

20	30	1017.5	22143.4	1234.3
30	40	1012	22143.4	1227.63
40	Sadl	1012	22143.4	1227.63
Sadl	50	1012	22143.4	1227.63
40	50	1012	22143.4	1227.63
50	60	1012	22143.4	1227.63
60	70	1017.5	22143.4	1234.3
70	80	1282.5	22143.4	1555.76

The Earthquake Weight does not include the weight of the saddles.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side Page 79 of 264
Center of Gravity Calculation: Step: 18 9:47am Apr 20,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	9972.525 mm
Center of Gravity of the Liquid	9975.646 mm
Center of Gravity of the Insulation	9973.589 mm
Center of Gravity of the Lining	9966.526 mm
Center of Gravity of the Nozzles	9251.363 mm
Center of Gravity of the Added Weights (Operating)	8903.857 mm
Center of Gravity of the Added Weights (Empty)	8903.857 mm
Center of Gravity of the Tubesheet(s)	9957.525 mm
Center of Gravity of the Tubes	9957.525 mm
Center of Gravity of the Baffles	10428.896 mm
Center of Gravity of Bare Shell New and Cold	9972.260 mm
Center of Gravity of Bare Shell Corroded	9972.442 mm
Vessel CG in the Operating Condition	9951.327 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	9942.972 mm
Vessel CG in the Test Condition	9950.862 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



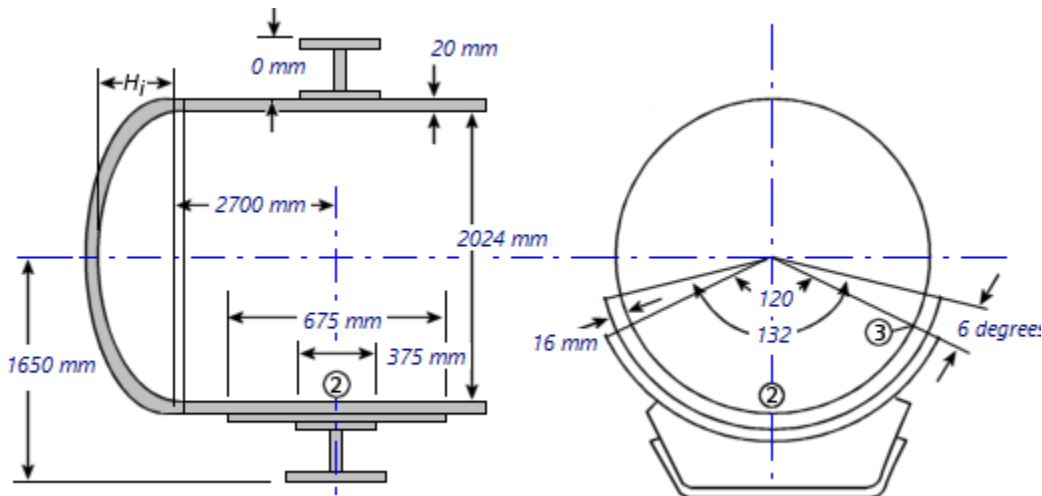
PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 80 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1022.00	mm
Shell Thickness used in this Case	t	20.000	mm
Stiffened Vessel Length per 4.15.6	L	13108.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	675.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1650.00	mm
Coefficient of Friction	mu	0.30	

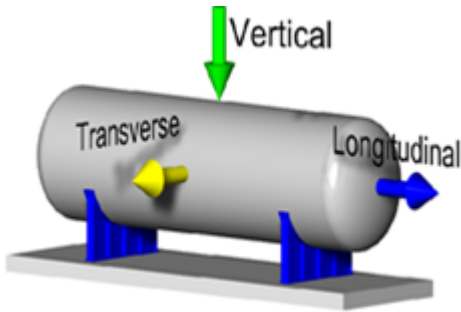


Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		92776.42	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 81 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026



Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-8.55	-234.00
Long. Stress at Top of Midspan	8.55	138.00
Long. Stress at Bottom of Midspan	8.55	138.00
Long. Stress at Top of Saddles	67.14	138.00
Long. Stress at Bottom of Saddles	-37.21	-234.00
Long. Stress at Bottom of Saddles	37.21	138.00
Tangential Shear in Shell	30.64	110.40
Circ. Stress at Horn of Saddle	66.22	207.00
Circ. Compressive Stress in Shell	3.21	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 577 • 0.00256 • 207² • 1
 = 63501.6 lbf
 = 28804.320 Kgf

Longitudinal Force due to Wind during Transport [F_{lw}]:
 = π • D²/4 • 0.00256 • wv² • wlsf • Wind Dia. Mult
 = π • 7.43²/4 • 0.00256 • 207² • 1 • 1.2
 = 5724.3 lbf
 = 2596.5 Kgf

Transverse Force due to Lateral Acceleration [F_t]:
 = g_z • Shipping Weight • g_w + F_{tw}
 = 0.1 • 134959 • 1 + 28804
 = 42300.2 Kgf

Longitudinal Force due to Axial Acceleration [F_l]:
 = g_x • Shipping Weight • g_w + F_{lw}
 = 0.1 • 134959 • 1 + 2597
 = 16092.4 Kgf

Saddle Reaction Force due to Transverse Load [F_{wt}]:
 = F_t/Num of Saddles • B / E
 = 42300/2 • 1650/1815
 = 19225.4 Kgf

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:
 = F_l • B / L_s
 = 16092 • 1650/7720
 = 3439.4 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:
 = Saddle Load • g_y
 = 70049 • 0.05



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 82 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

= 3502.431 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 70049 + 3502 + max(3439, 19225)
 = 92776.4 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	93507.77	Kgf
Transverse Shear Load	21150.11	Kgf
Longitudinal Shear Load	16092.43	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin(θ/2), 2•Rm)
 = min(2(2024/2 + 20 + 16)sin(120/2), 2•1022)
 = 1815.189 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -92776•2700[1-(1-2700/13108+(1022²-0²)/
 (2•2700•13108))/(1+(4•0)/(3•13108))]
 = -47902.1 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 92776•13108/4((1+2(1022²-0²)/(13108²))/(1+(4•0)/
 (3•13108))-4•2700/13108)
 = 57229.5 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P • Rm/(2t) - M2/(π•Rm²t)
 = 0 • 1022/(2•20) - 57229/(π•1022²•20)
 = -8.55 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = 0 • 1022/(2 • 20) + 57229/(π • 1022² • 20)
 = 8.55 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P • Rm/(2t) - M1/(K1•π•Rm²t)
 = 0•1022/(2•20)--47902/(0.11•π•1022²•20)
 = 67.14 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 0•1022/(2•20)+-47902/(0.19•π•1022²•20)
 = -37.21 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 92776 (13108 - 2 • 2700)/(13108 + (4 • 0/3))
 = 54556.0 Kgf



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 83 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 54556 / (1022 \cdot 20)$$

$$= 30.64 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1022 \cdot 20)}$$

$$= 111.515 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(1022 \cdot 20)}, 2 \cdot 2700)$$

$$= 598.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 92776 \cdot 0.1 / (598(20 + 1 \cdot 16))$$

$$= -3.21 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -92776 / (4(20 + 1 \cdot 16)598) -$$

$$3 \cdot 0.053 \cdot 92776 / (2(20 + 1 \cdot 16)^2)$$

$$= -66.22 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1949.9999	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	638.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1012 \cdot 20)} + 675)20^3) / 12$$

$$= 59.796 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 59.8 + 179(224 - 10)^2$$

$$= 82032.812 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	896.9	20.0	10.0	179.4	179387.4	0.820E+05
Wearplate	675.0	16.0	28.0	108.0	302400.0	0.414E+05
Web	10.0	568.0	320.0	56.8	1817599.6	0.205E+05
BasePlate	400.0	34.0	621.0	136.0	8445599.0	0.215E+06
Totals	...	...	...	480.2	10744986.0	0.359E+06

Distance to Centroid [C1]:

$$= AY / A$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 84 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

$$= 10744986/480$$

$$= 223.767 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 92776$$

$$= 18882.0137 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 6.1560 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1650 - 1012 \cdot \sin(120/2) / 1.05$$

$$= 813.083 \text{ mm}$$

Bending Moment, M = Fh * d = 15352.9434 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 9.3921 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(92776 + 731)400 / (4 \cdot 1950 \cdot 173)}$$

$$= 28.530 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1012 + 20 + 16)$$

$$= 1815.189 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1815 / (5 - 1)$$

$$= 453.797 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 454 \cdot 400/2$$

$$= 907.595 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 93508 / (1950 \cdot 400)$$

$$= 11.988 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 908 \cdot 12$$

$$= 10880.403 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 85 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

$$= 36.5 + 22.7$$

$$= 59.190 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 10880/59.2$$

$$= 18.027 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375 - 10$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/59.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4395/59.2)}$$

$$= 86.165 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1650 - \cos(120/2) (1012 + 20 + 16) - 34$$

$$= 1092.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/260)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1092/86.2$$

$$= 12.673$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 16092 / (2 \cdot 1950) \cdot 454 \cdot 1092/2$$

$$= 1022.399 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (12.7) / (8 \cdot 123) - (12.7^3) / (8 \cdot 123^3))$$

$$= 151.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 18/152 + (1022 \cdot 188/43945308) / 173$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 86 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

= 0.366

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	453.8	10.0	0.0	45.4	0.0	3.78
Totals	...	...	...	81.9	...	0.440E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/81.9
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 375/2
 = 187.500 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 1650 - cos(1048/2)(34 + 0 + 0) - 0
 = 568.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4398/81.9}$
 = 73.291 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 568/73.3
 = 7.750

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 16092/(2 • 1950)
 = 4.126 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 4.13 • 454 • 568
 = 1063.594 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 1064/234572
 = 44.465 N./mm²

Compressive Allowable, KL/r < Cc (7.75 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (7.75)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (7.75) / (8 \cdot 123) - (7.75^3) / (8 \cdot 123^3))$
 = 153.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 26.1/154 + (1064 • 188/4398)/173
 = 0.426

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 87 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

Vessel Mean Radius	Rm	1022.00	mm
Shell Thickness used in this Case	t	20.000	mm
Stiffened Vessel Length per 4.15.6	L	13108.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	675.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1650.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		91647.04	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	333.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-8.45	-234.00	
Long. Stress at Top of Midspan	8.45	138.00	
Long. Stress at Bottom of Midspan	8.45	138.00	
Long. Stress at Top of Saddles	66.32	138.00	
Long. Stress at Bottom of Saddles	-36.76	-234.00	
Long. Stress at Bottom of Saddles	36.76	138.00	
Tangential Shear in Shell	30.27	110.40	
Circ. Stress at Horn of Saddle	65.42	207.00	
Circ. Compressive Stress in Shell	3.17	138.00	

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 577 \cdot 0.00256 \cdot 207^2 \cdot 1 \\
 &= 63501.6 \text{ lbf} \\
 &= 28804.320 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 7.43^2/4 \cdot 0.00256 \cdot 207^2 \cdot 1 \cdot 1.2 \\
 &= 5724.3 \text{ lbf} \\
 &= 2596.5 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 134959 \cdot 1 + 28804
 \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 88 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

= 42300.2 Kgf

Longitudinal Force due to Axial Acceleration [FI]:

= $g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$
 = $0.1 \cdot 134959 \cdot 1 + 2597$
 = 16092.4 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:

= $F_t / \text{Num of Saddles} \cdot B / E$
 = $42300/2 \cdot 1650/1815$
 = 19225.4 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:

= $F_l \cdot B / L_s$
 = $16092 \cdot 1650/7720$
 = 3439.4 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

= Saddle Load $\cdot g_y$
 = $68973 \cdot 0.05$
 = 3448.651 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = $68973 + 3449 + \max(3439, 19225)$
 = 91647.0 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	92378.39	Kgf
Transverse Shear Load	21150.11	Kgf
Longitudinal Shear Load	16092.43	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 = $\min(2(2024/2 + 20 + 16) \sin(120/2), 2 \cdot 1022)$
 = 1815.189 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 = $-91647 \cdot 2700 [1 - (1 - 2700/13108 + (1022^2 - 0^2) / (2 \cdot 2700 \cdot 13108)) / (1 + (4 \cdot 0) / (3 \cdot 13108))]$
 = -47319.0 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 = $91647 \cdot 13108 / 4 ((1 + 2(1022^2 - 0^2)/(13108^2)) / (1 + (4 \cdot 0) / (3 \cdot 13108)) - 4 \cdot 2700/13108)$
 = 56532.8 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 = $0 \cdot 1022 / (2 \cdot 20) - 56533 / (\pi \cdot 1022^2 \cdot 20)$
 = -8.45 N./mm²



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 89 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1022 / (2 \cdot 20) + 56533 / (\pi \cdot 1022^2 \cdot 20)$$

$$= 8.45 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0 \cdot 1022 / (2 \cdot 20) - 47319 / (0.11 \cdot \pi \cdot 1022^2 \cdot 20)$$

$$= 66.32 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0 \cdot 1022 / (2 \cdot 20) + 47319 / (0.19 \cdot \pi \cdot 1022^2 \cdot 20)$$

$$= -36.76 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 91647 \cdot (13108 - 2 \cdot 2700) / (13108 + (4 \cdot 0 / 3))$$

$$= 53891.9 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 53892 / (1022 \cdot 20)$$

$$= 30.27 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1022 \cdot 20)}$$

$$= 111.515 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(1022 \cdot 20)}, 2 \cdot 2700)$$

$$= 598.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 91647 \cdot 0.1 / (598(20 + 1 \cdot 16))$$

$$= -3.17 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -91647 / (4(20 + 1 \cdot 16)598) - 3 \cdot 0.053 \cdot 91647 / (2(20 + 1 \cdot 16)^2)$$

$$= -65.42 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1949.9999	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	638.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 90 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1012 \cdot 20)} + 675)^3) / 12$$

$$= 59.796 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 59.8 + 179 (224 - 10)^2$$

$$= 82032.812 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	896.9	20.0	10.0	179.4	179387.4	0.820E+05
Wearplate	675.0	16.0	28.0	108.0	302400.0	0.414E+05
Web	10.0	568.0	320.0	56.8	1817599.6	0.205E+05
BasePlate	400.0	34.0	621.0	136.0	8445599.0	0.215E+06
Totals	...	...	...	480.2	10744986.0	0.359E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10744986 / 480$$

$$= 223.767 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 91647$$

$$= 18652.1602 \text{ Kg}$$

Tension Stress, $St = (Fh / As) = 6.0811 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1650 - 1012 \cdot \sin(120/2) / 1.05$$

$$= 813.083 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 15166.0498 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 9.2778 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(91647 + 731) \cdot 400) / (4 \cdot 1950 \cdot 173)}$$

$$= 28.357 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1012 + 20 + 16)$$

$$= 1815.189 \text{ mm}$$

Distance between Ribs [e]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 91 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

= Web Length / (Nribs - 1)
 = 1815/(5 - 1)
 = 453.797 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 454 • 400/2
 = 907.595 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 92378/(1950 • 400)
 = 11.843 Kgf/cm²

Axial Load [P]:
 = Ap • Bp
 = 908 • 11.8
 = 10748.992 Kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 36.5 + 22.7
 = 59.190 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 10749/59.2
 = 17.809 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 375 - 10
 = 365.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/59.2
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 375/2
 = 187.500 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4395/59.2}$
 = 86.165 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1650 - cos(120/2)(1012 + 20 + 16) - 34
 = 1092.000 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 92 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20,2026

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1092/86.2$$

$$= 12.673$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 16092 / (2 \cdot 1950) \cdot 454 \cdot 1092/2$$

$$= 1022.399 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.7)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (12.7) / (8 \cdot 123) - (12.7^3) / (8 \cdot 123^3))$$

$$= 151.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 17.8/152 + (1022 \cdot 188/43945308) / 173$$

$$= 0.364$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	453.8	10.0	0.0	45.4	0.0	3.78
Totals	...	...	...	81.9	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/81.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1650 - \cos(1048/2) (34 + 0 + 0) - 0$$

$$= 568.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4398/81.9) }$$

$$= 73.291 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 568/73.3$$

$$= 7.750$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 16092 / (2 \cdot 1950)$$

$$= 4.126 \text{ Kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.13 \cdot 454 \cdot 568$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 93 of 264
 Saddle Calcs: Road Transport: Step: 19 9:47am Apr 20, 2026

= 1063.594 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 1064/234572

= 44.465 N./mm²

Compressive Allowable, $KL/r < C_c$ (7.75 < 123) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(7.75)²/(2 • 123²))260/

(5/3+3•(7.75)/(8• 123)-(7.75³)/(8•123³)

= 153.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C1/I)/S_{ba}$

= 25.8/154 + (1064 • 188/4398)/173

= 0.424

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



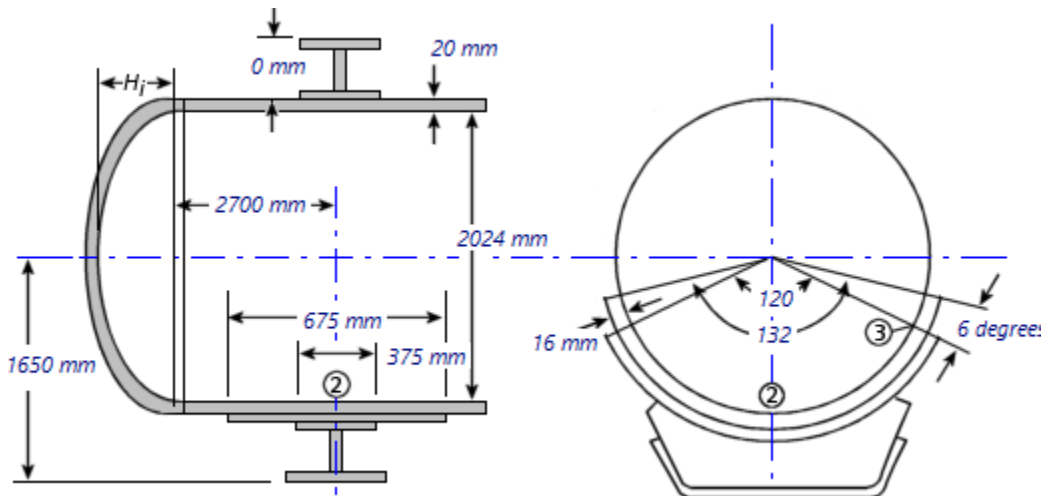
PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 94 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1023.00	mm
Shell Thickness used in this Case	t	20.000	mm
Stiffened Vessel Length per 4.15.6	L	13108.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	675.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1650.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	119295.11	Kgf
Pressure used in Analysis	P 929.920	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	14.23	138.00
Long. Stress at Bottom of Midspan	38.62	138.00
Long. Stress at Top of Saddles	122.15	138.00
Long. Stress at Bottom of Saddles	-26.63	-235.80
Long. Stress at Bottom of Saddles	26.63	138.00
Tangential Shear in Shell	43.74	110.40
Circ. Stress at Horn of Saddle	94.89	172.50



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 95 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Circ. Compressive Stress in Shell	4.46	138.00
-----------------------------------	------	--------

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (2390/2 + 0) \cdot 1650/1815$
 $= 3258.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(498, 0) \cdot 1650/7720$
 $= 106.5 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(11718, 0) \cdot 1650/7720$
 $= 2504.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (11718/2 + 0) \cdot 1650/1815$
 $= 15977.8 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 103317 + \max(106, 3259, 2505, 15978)$
 $= 119295.1 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (2.26 \cdot 1.2)^2)$
 $= 4885.121 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	120026.45	Kgf
Transverse Shear Load	5859.13	Kgf
Longitudinal Shear Load	11718.26	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2 (2024/2 + 20 + 16) \sin(120/2), 2 \cdot 1023)$
 $= 1815.189 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -119295 \cdot 2700 [1 - (1 - 2700/13108 + (1023^2 - 0^2) / (2 \cdot 2700 \cdot 13108)) / (1 + (4 \cdot 0) / (3 \cdot 13108))]$
 $= -61584.9 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$



PV Elite 28 Licensee: SL User

FileName : 106-C-Cladding on Both Side

Page 96 of 264

Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20, 2026

$$= 119295 \cdot 13108 / 4 \cdot ((1 + 2(1023^2 - 0^2) / (13108^2)) / (1 + (4 \cdot 0) / (3 \cdot 13108))) - 4 \cdot 2700 / 13108$$

$$= 73596.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 930 \cdot 1023 / (2 \cdot 18) - 73597 / (\pi \cdot 1023^2 \cdot 18)$$

$$= 14.23 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 930 \cdot 1023 / (2 \cdot 18) + 73597 / (\pi \cdot 1023^2 \cdot 18)$$

$$= 38.62 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 930 \cdot 1023 / (2 \cdot 18) - 61585 / (0.11 \cdot \pi \cdot 1023^2 \cdot 18)$$

$$= 122.15 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 930 \cdot 1023 / (2 \cdot 18) + 61585 / (0.19 \cdot \pi \cdot 1023^2 \cdot 18)$$

$$= -26.63 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 119295 (13108 - 2 \cdot 2700) / (13108 + (4 \cdot 0 / 3))$$

$$= 70150.0 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 70150 / (1023 \cdot 18)$$

$$= 43.74 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1023 \cdot 18)}$$

$$= 105.845 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(1023 \cdot 18)}, 2 \cdot 2700)$$

$$= 586.69 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(138 / 138, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 119295 \cdot 0.1 / (587(18 + 1 \cdot 16))$$

$$= -4.46 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -119295 / (4(18 + 1 \cdot 16)587) -$$

$$3 \cdot 0.053 \cdot 119295 / (2(18 + 1 \cdot 16)^2)$$

$$= -94.89 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 7720.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature})$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 97 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

$$= 0.12491\text{E-}04 \cdot 7720 (160 - 21.1)$$

$$= 13.393 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1949.9999	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	638.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1014 \cdot 18)} + 675)18^3)/12$$

$$= 43.048 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 43 + 159 (231 - 9)^2$$

$$= 78888.531 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	885.8	18.0	9.0	159.4	143492.4	0.789E+05
Wearplate	675.0	16.0	26.0	108.0	280800.0	0.456E+05
Web	10.0	568.0	318.0	56.8	1806240.1	0.195E+05
BasePlate	400.0	34.0	619.0	136.0	8418400.0	0.204E+06
Totals	...	...	...	460.2	10648933.0	0.348E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10648933/460$$

$$= 231.380 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 119295$$

$$= 24279.1406 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 7.9156 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1650 - 1014 \cdot \sin(120/2) / 1.05$$

$$= 811.429 \text{ mm}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Clading on Both Side Page 98 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Bending Moment, $M = F_h \cdot d = 19701.1816 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 12.8284 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(119295 + 731) \cdot 400) / (4 \cdot 1950 \cdot 175)}$
 $= 32.199 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120/2) (1012 + 20 + 16)$
 $= 1815.189 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1815 / (5 - 1)$
 $= 453.797 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 454 \cdot 400 / 2$
 $= 907.595 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 120026 / (1950 \cdot 400)$
 $= 15.388 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 908 \cdot 15.4$
 $= 13966.073 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 22.7$
 $= 59.190 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 13966 / 59.2$
 $= 23.140 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 375 - 10$
 $= 365.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 59.2$
 $= 0.000 \text{ mm}$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 99 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{4395/59.2}$$

$$= 86.165 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1650 - \cos(120/2) (1012 + 20 + 16) - 34$$

$$= 1092.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199943008}{262}}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1092/86.2$$

$$= 12.673$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 11718 / (2 \cdot 1950) \cdot 454 \cdot 1092/2$$

$$= 744.495 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($12.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (12.7)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (12.7) / (8 \cdot 123) - (12.7^3) / (8 \cdot 123^3))$$

$$= 152.8 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 23.1/153 + (744 \cdot 188/43945308) / 175$$

$$= 0.330$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	453.8	10.0	0.0	45.4	0.0	3.78
Totals	...	...	...	81.9	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/81.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1650 - \cos(1048/2) (34 + 0 + 0) - 0$$

$$= 568.000 \text{ mm}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 100 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{(4398/81.9)}$
 $= 73.291 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 568/73.3$
 $= 7.750$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 11718 / (2 \cdot 1950)$
 $= 3.005 \text{ Kgf/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 3 \cdot 454 \cdot 568$
 $= 774.493 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 774/234572$
 $= 32.379 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($7.75 < 123$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (7.75)^2 / (2 \cdot 123^2)) 262 /$
 $(5/3 + 3 \cdot (7.75) / (8 \cdot 123) - (7.75^3) / (8 \cdot 123^3))$
 $= 154.7 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 33.5/155 + (774 \cdot 188/4398) / 175$
 $= 0.402$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	240.0000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	231.8406 cm ²
Saddle Load Q0 (Weight)	Q0	104048.7 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	2504.6 Kgf
Maximum Transverse Force	Ft	5859.1 Kgf
Maximum Longitudinal Force	F1	11718.3 Kgf
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 11718 / (232 \cdot 8)$
 $= 0.6 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 5859 / (232 \cdot 8)$
 $= 0.3 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss



PV Elite 28 Licensee: SL User
 FileName : 106-C-Clading on Both Side Page 101 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (BoltShearAllowable \cdot Nbolts)$
 = $11718 / (82.8 \cdot 8)$
 = 1.7349 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1650 \cdot 5859 + 0$
 = 9667.76 Kg-m.

Eccentricity (e):
 = $Rmom / QO$
 = $9668 / 104049$
 = 92.91 mm < $B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0, 1.73, 0]$
 = 1.7349 cm²

MDMT Calculations for: Left Saddle Flo, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 16$, $c = 2$ mm, $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = 0 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-15 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1023.00	mm
Shell Thickness used in this Case	t	20.000	mm
Stiffened Vessel Length per 4.15.6	L	13108.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	675.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1650.00	mm
Coefficient of Friction	mu	0.00	



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 102 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 118200.38 Kgf
 Pressure used in Analysis P 929.920 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	14.34	138.00
Long. Stress at Bottom of Midspan	38.51	138.00
Long. Stress at Top of Saddles	121.27	138.00
Long. Stress at Bottom of Saddles	-26.14	-235.80
Long. Stress at Bottom of Saddles	26.14	138.00
Tangential Shear in Shell	43.34	110.40
Circ. Stress at Horn of Saddle	94.02	172.50
Circ. Compressive Stress in Shell	4.42	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2390/2 + 0) \cdot 1650/1815$
 $= 3258.6 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(498, 0) \cdot 1650/7720$
 $= 106.5 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(11718, 0) \cdot 1650/7720$
 $= 2504.6 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (11718/2 + 0) \cdot 1650/1815$
 $= 15977.8 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 102223 + \max(106, 3259, 2505, 15978)$
 $= 118200.4 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (2.26 \cdot 1.2)^2)$
 $= 4885.121 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	118931.72	Kgf
Transverse Shear Load	5859.13	Kgf
Longitudinal Shear Load	11718.26	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2 (2024/2 + 20 + 16) \sin(120/2), 2 \cdot 1023)$
 $= 1815.189 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 103 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -118200 \cdot 2700 [1 - (1 - 2700/13108 + (1023^2 - 0^2)/(2 \cdot 2700 \cdot 13108))/(1 + (4 \cdot 0)/(3 \cdot 13108))]$$

$$= -61019.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L]$$

$$= 118200 \cdot 13108/4 [(1 + 2(1023^2 - 0^2)/(13108^2))/(1 + (4 \cdot 0)/(3 \cdot 13108)) - 4 \cdot 2700/13108]$$

$$= 72921.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$$

$$= 930 \cdot 1023/(2 \cdot 18) - 72922/(\pi \cdot 1023^2 \cdot 18)$$

$$= 14.34 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 930 \cdot 1023/(2 \cdot 18) + 72922/(\pi \cdot 1023^2 \cdot 18)$$

$$= 38.51 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 t)$$

$$= 930 \cdot 1023/(2 \cdot 18) - 61020/(0.11 \cdot \pi \cdot 1023^2 \cdot 18)$$

$$= 121.27 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm/(2t) + M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 930 \cdot 1023/(2 \cdot 18) + 61020/(0.19 \cdot \pi \cdot 1023^2 \cdot 18)$$

$$= -26.14 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 118200 (13108 - 2 \cdot 2700)/(13108 + (4 \cdot 0/3))$$

$$= 69506.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 69506/(1023 \cdot 18)$$

$$= 43.34 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1023 \cdot 18)}$$

$$= 105.845 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{(1023 \cdot 18)}, 2 \cdot 2700)$$

$$= 586.69 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(138/138, 1)$$

$$= 1.0000$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 104 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 118200 \cdot 0.1 / (587(18 + 1 \cdot 16))$$

$$= -4.42 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -118200 / (4(18 + 1 \cdot 16)587) -$$

$$3 \cdot 0.053 \cdot 118200 / (2(18 + 1 \cdot 16)^2)$$

$$= -94.02 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1949.9999	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	638.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1014 \cdot 18)} + 675)18^3) / 12$$

$$= 43.048 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 43 + 159(231 - 9)^2$$

$$= 78888.531 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	885.8	18.0	9.0	159.4	143492.4	0.789E+05
Wearplate	675.0	16.0	26.0	108.0	280800.0	0.456E+05
Web	10.0	568.0	318.0	56.8	1806240.1	0.195E+05
BasePlate	400.0	34.0	619.0	136.0	8418400.0	0.204E+06
Totals	...	...	...	460.2	10648933.0	0.348E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10648933 / 460$$

$$= 231.380 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 118200$$

$$= 24056.3379 \text{ Kgf}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 105 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Tension Stress, $St = (Fh/As)$ = 7.8430 N./mm²
 Allowed Stress, $Sa = 0.6 * Yield Str$ = 157.1972 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 1650 - 1014 \cdot \sin(120/2) / 1.05$
 $= 811.429 \text{ mm}$

Bending Moment, $M = Fh * d$ = 19520.3906 Kg-m.

Bending Stress, $Sb = (M * C1 / I)$ = 12.7107 N./mm²
 Allowed Stress, $Sa = 2/3 * Yield Str$ = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + SaddleWt)BasePlateWidth / (4 * BasePlateLength * AllStress))}$
 $= \sqrt{(3(118200 + 731)400 / (4 * 1950 * 175))}$
 $= 32.052 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)$
 $= 2 * \cos(90 - 120/2) (1012 + 20 + 16)$
 $= 1815.189 \text{ mm}$

Distance between Ribs [e]:
 $= Web Length / (Nribs - 1)$
 $= 1815 / (5 - 1)$
 $= 453.797 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * Bpwid / 2$
 $= 454 * 400 / 2$
 $= 907.595 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + SaddleWt) / (BasePlateLength * BasePlateWidth)$
 $= 118932 / (1950 * 400)$
 $= 15.248 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 908 * 15.2$
 $= 13838.691 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= Rib Area + Web Area$
 $= 36.5 + 22.7$
 $= 59.190 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 13839 / 59.2$
 $= 22.929 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 106 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

= Saddle Width - Web Thickness
 = 375 - 10
 = 365.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/59.2
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 375/2
 = 187.500 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{4395/59.2}$
 = 86.165 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1650 - cos(120/2)(1012 + 20 + 16) - 34
 = 1092.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199943008/262}$
 = 122.739

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1092/86.2
 = 12.673

Bending Moment [Rm]:
 $= FI / (2 \cdot Bpln) \cdot e \cdot L / 2$
 $= 11718 / (2 \cdot 1950) \cdot 454 \cdot 1092/2$
 = 744.495 Kg-m.

Compressive Allowable, $KL/r < Cc$ (12.7 < 123) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (12.7)^2 / (2 \cdot 123^2)) 262 /$
 $(5/3 + 3 \cdot (12.7) / (8 \cdot 123) - (12.7^3) / (8 \cdot 123^3))$
 = 152.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 22.9/153 + (744 \cdot 188/43945308) / 175$
 = 0.328

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	453.8	10.0	0.0	45.4	0.0	3.78
Totals	...	...	...	81.9	...	0.440E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/81.9
 = 0.000 mm

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 107 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

= Saddle Width / 2
 = 375/2
 = 187.500 mm

Length of Inner Rib [L]:
 = Saddle Height - Outside Radius - Bpthk
 = 1650 - cos(1048/2)(34 + 0 + 0) - 0
 = 568.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{4398/81.9}$
 = 73.291 mm

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 568/73.3
 = 7.750

Unit Force [Force,u]:
 = FI / (2 • Baseplate Length)
 = 11718/(2 • 1950)
 = 3.005 Kg/mm

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 = 3 • 454 • 568
 = 774.493 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:
 = Bending Moment / Section Modulus
 = 774/234572
 = 32.379 N./mm²

Compressive Allowable, KL/r < Cc (7.75 < 123) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (7.75)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (7.75) / (8 \cdot 123) - (7.75^3) / (8 \cdot 123^3))$
 = 154.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 = Sc/Sca + (Mbase,c • C1/I)/Sba
 = 33.2/155 + (774 • 188/4398)/175
 = 0.400

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	240.0000 cm²
BasePlate Allowable Stress	S	138.00 N./mm²
Area Available in a Single Bolt	BltArea	231.8406 cm²
Saddle Load Q0 (Weight)	Q0	102953.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2504.6 Kg
Maximum Transverse Force	Ft	5859.1 Kg
Maximum Longitudinal Force	F1	11718.3 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 31214.6 Kg



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 108 of 264
 Saddle Calcs: Operating Case: Step: 20 9:47am Apr 20,2026

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:
 $= \max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 31215 / (232 \cdot 8)$
 $= 1.7 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:
 $= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 5859 / (232 \cdot 8)$
 $= 0.3 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 11718 / (82.8 \cdot 8)$
 $= 1.7349 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot F_t + \text{Sum of X Moments}$
 $= 1650 \cdot 5859 + 0$
 $= 9667.76 \text{ Kg-m}$.

Eccentricity (e):
 $= R_{mom} / QO$
 $= 9668 / 102954$
 $= 93.90 \text{ mm} < B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[Bltarearl, Bltarears, Bltareart]$
 $= \max[0, 1.73, 0]$
 $= 1.7349 \text{ cm}^2$

MDMT Calculations for: Right Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, $t_g = 16$, $c = 2 \text{ mm}$, $E^* = 1$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = $0 \text{ } ^\circ\text{C}$
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	-15 $^\circ\text{C}$
Min Metal Temp. w/o impact per UG-20(f)	-29 $^\circ\text{C}$

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



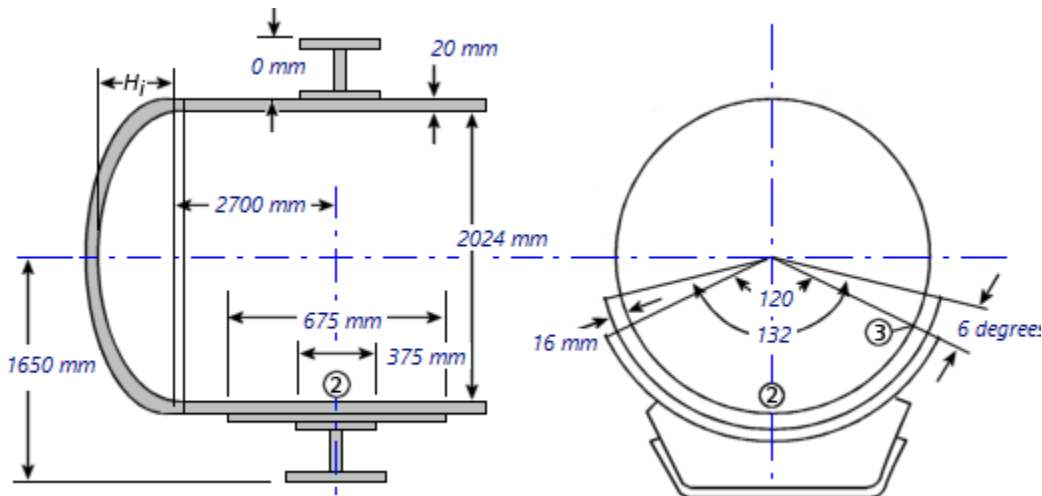
PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 109 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1022.00	mm
Shell Thickness used in this Case	t	20.000	mm
Stiffened Vessel Length per 4.15.6	L	13108.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	675.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1650.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	99313.68	Kgf
Pressure used in Analysis	P	1206.017 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	21.66	234.00
Long. Stress at Bottom of Midspan	39.97	234.00
Long. Stress at Top of Saddles	102.69	234.00
Long. Stress at Bottom of Saddles	-9.02	-234.00
Long. Stress at Bottom of Saddles	9.02	234.00
Tangential Shear in Shell	32.80	138.58
Circ. Stress at Horn of Saddle	70.89	350.99



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 110 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

Circ. Compressive Stress in Shell	3.44	234.00
-----------------------------------	------	--------

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (1195/2 + 0) \cdot 1650/1815$
 $= 1629.3 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(249, 0) \cdot 1650/7720$
 $= 53.2 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 97684 + \max(53.2, 1629, 0, 0)$
 $= 99313.7 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4 (2.26 \cdot 1.2)^2)$
 $= 4885.121 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	100045.03	Kgf
Transverse Shear Load	597.47	Kgf
Longitudinal Shear Load	249.08	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2024/2 + 20 + 16) \sin(120/2), 2 \cdot 1022)$
 $= 1815.189 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -99314 \cdot 2700 [1 - (1 - 2700/13108 + (1022^2 - 0^2)/(2 \cdot 2700 \cdot 13108)) / (1 + (4 \cdot 0)/(3 \cdot 13108))]$
 $= -51277.4 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 99314 \cdot 13108/4 ((1 + 2(1022^2 - 0^2)/(13108^2)) / (1 + (4 \cdot 0)/(3 \cdot 13108)) - 4 \cdot 2700/13108)$
 $= 61262.0 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 1206 \cdot 1022/(2 \cdot 20) - 61262/(\pi \cdot 1022^2 \cdot 20)$
 $= 21.66 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Saddle Calcs: Test Case:

Step: 21 9:47am Apr 20,2026

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1206 \cdot 1022 / (2 \cdot 20) + 61262 / (\pi \cdot 1022^2 \cdot 20)$$

$$= 39.97 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1206 \cdot 1022 / (2 \cdot 20) - 51277 / (0.11 \cdot \pi \cdot 1022^2 \cdot 20)$$

$$= 102.69 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 1206 \cdot 1022 / (2 \cdot 20) + 51277 / (0.19 \cdot \pi \cdot 1022^2 \cdot 20)$$

$$= -9.02 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 99314 (13108 - 2 \cdot 2700) / (13108 + (4 \cdot 0 / 3))$$

$$= 58400.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 58400 / (1022 \cdot 20)$$

$$= 32.80 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1022 \cdot 20}$$

$$= 111.515 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{1022 \cdot 20}, 2 \cdot 2700)$$

$$= 598.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.76 \cdot 99314 \cdot 0.1 / (598(20 + 1 \cdot 16))$$

$$= -3.44 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot t_r)^2)$$

$$= -99314 / (4(20 + 1 \cdot 16) 598) -$$

$$3 \cdot 0.053 \cdot 99314 / (2(20 + 1 \cdot 16)^2)$$

$$= -70.89 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1949.9999	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	638.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{1012 \cdot 20} + 675) 20^3) / 12$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 112 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

$$= 59.796 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 59.8 + 179 (224 - 10)^2$$

$$= 82032.812 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	896.9	20.0	10.0	179.4	179387.4	0.820E+05
Wearplate	675.0	16.0	28.0	108.0	302400.0	0.414E+05
Web	10.0	568.0	320.0	56.8	1817599.6	0.205E+05
BasePlate	400.0	34.0	621.0	136.0	8445599.0	0.215E+06
Totals	...	...	...	480.2	10744986.0	0.359E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10744986/480$$

$$= 223.767 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 99314$$

$$= 20212.4883 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 6.5898 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1650 - 1012 \cdot \sin(120/2) / 1.05$$

$$= 813.083 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 16434.7500 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 10.0539 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(99314 + 731)400) / (4 \cdot 1950 \cdot 173)}$$

$$= 29.510 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1012 + 20 + 16)$$

$$= 1815.189 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 1815 / (5 - 1)$$

$$= 453.797 \text{ mm}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 113 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 454 \cdot 400 / 2$$

$$= 907.595 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 100045 / (1950 \cdot 400)$$

$$= 12.826 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 908 \cdot 12.8$$

$$= 11641.068 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 36.5 + 22.7$$

$$= 59.190 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 11641 / 59.2$$

$$= 19.287 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 375 - 10$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 59.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 375 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (4395 / 59.2) }$$

$$= 86.165 \text{ mm}$$

Length of Outer Rib [L]:

$$= Saddle \text{ Height} - \cos(\theta / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1650 - \cos(120 / 2) (1012 + 20 + 16) - 34$$

$$= 1092.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Elastic \text{ Modulus} / Yield \text{ Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 199943008 / 260) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 114 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

$$= KL/r$$

$$= 1 \cdot 1092/86.2$$

$$= 12.673$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 249 / (2 \cdot 1950) \cdot 454 \cdot 1092/2$$

$$= 15.824 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (12.7) / (8 \cdot 123) - (12.7^3) / (8 \cdot 123^3))$$

$$= 151.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 19.3/152 + (15.8 \cdot 188/43945308) / 173$$

$$= 0.131$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	453.8	10.0	0.0	45.4	0.0	3.78
Totals	...	...	...	81.9	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/81.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 1650 - \cos(1048/2) (34 + 0 + 0) - 0$$

$$= 568.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4398/81.9) }$$

$$= 73.291 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 568/73.3$$

$$= 7.750$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 249 / (2 \cdot 1950)$$

$$= 0.064 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.064 \cdot 454 \cdot 568$$

$$= 16.462 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 115 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

= Bending Moment / Section Modulus
 = 16.5/234572
 = 0.688 N./mm²

Compressive Allowable, $KL/r < C_c$ (7.75 < 123) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (7.75)^2 / (2 \cdot 123^2)) 260 /$
 $(5/3 + 3 \cdot (7.75) / (8 \cdot 123) - (7.75^3) / (8 \cdot 123^3))$
 = 153.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
 = 27.9/154 + (16.5 • 188/4398)/173
 = 0.186

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	240.0000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	231.8406 cm ²
Saddle Load QO (Weight)	QO	98415.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	53.2 Kg
Maximum Transverse Force	Ft	597.5 Kg
Maximum Longitudinal Force	Fl	249.1 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 249/(232 • 8)
 = 0.0 N./mm². Must be less than 29.0 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = 597/(232 • 8)
 = 0.0 N./mm². Must be less than 29.0 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = 249/(29 • 8)
 = 0.1054 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = 1650 • 597 + 0
 = 985.85 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = 986/98416
 = 10.02 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 116 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0, 0.11, 0]
 = 0.1054 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1022.00	mm
Shell Thickness used in this Case	t	20.000	mm
Stiffened Vessel Length per 4.15.6	L	13108.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	675.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1650.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		98215.20	Kgf
Pressure used in Analysis	P	1206.017	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	21.76	234.00
Long. Stress at Bottom of Midspan	39.87	234.00
Long. Stress at Top of Saddles	101.89	234.00
Long. Stress at Bottom of Saddles	-8.58	-234.00
Long. Stress at Bottom of Saddles	8.58	234.00
Tangential Shear in Shell	32.44	138.58
Circ. Stress at Horn of Saddle	70.10	350.99
Circ. Compressive Stress in Shell	3.40	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = Ftr(Ft/Num of Saddles + Z Force Load) • B / E
 = 3 (1195/2 + 0) • 1650/1815
 = 1629.3 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
 = max(Fl,wind; Sum of X Forces) • B / Ls
 = max(249, 0) • 1650/7720
 = 53.2 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 117 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20, 2026

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 96586 + max(53.2, 1629, 0, 0)
 = 98215.2 Kgf

Longitudinal Wind Force [F_L]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766 (0 + $\pi/4$ (2.26 • 1.2)²)
 = 4885.121 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	98946.54	Kgf
Transverse Shear Load	597.47	Kgf
Longitudinal Shear Load	249.08	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(2024/2 + 20 + 16)sin(120/2), 2•1022)
 = 1815.189 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -98215•2700 [1 - (1 - 2700/13108 + (1022²-0²)/(2•2700•13108))/(1+(4•0)/(3•13108))]
 = -50710.2 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = Q•L/4 ((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 98215•13108/4 ((1+2(1022²-0²)/(13108²))/(1+(4•0)/(3•13108)) - 4•2700/13108)
 = 60584.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = 1206 • 1022/(2•20) - 60584/(π •1022²•20)
 = 21.76 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 1206 • 1022/(2 • 20) + 60584/(π • 1022² • 20)
 = 39.87 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 = P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 1206•1022/(2•20)-50710/(0.11• π •1022²•20)
 = 101.89 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 1206•1022/(2•20)+50710/(0.19• π •1022²•20)
 = -8.58 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = Q(L-2a)/(L+(4•hm/3))



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 118 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

$$= 98215 (13108 - 2 \cdot 2700) / (13108 + (4 \cdot 0/3))$$

$$= 57754.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 57754 / (1022 \cdot 20)$$

$$= 32.44 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{ (R_m \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (1022 \cdot 20) }$$

$$= 111.515 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (R_m \cdot t) }, 2a)$$

$$= \min(375 + 1.56 \cdot \sqrt{ (1022 \cdot 20) }, 2 \cdot 2700)$$

$$= 598.03 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1 (t + \eta \cdot t_r))$$

$$= -0.76 \cdot 98215 \cdot 0.1 / (598 (20 + 1 \cdot 16))$$

$$= -3.40 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot t_r) b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot t_r)^2)$$

$$= -98215 / (4(20 + 1 \cdot 16) 598) -$$

$$3 \cdot 0.053 \cdot 98215 / (2(20 + 1 \cdot 16)^2)$$

$$= -70.10 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1949.9999	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	638.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (1012 \cdot 20) } + 675) 20^3) / 12$$

$$= 59.796 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 59.8 + 179 (224 - 10)^2$$

$$= 82032.812 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	896.9	20.0	10.0	179.4	179387.4	0.820E+05
Wearplate	675.0	16.0	28.0	108.0	302400.0	0.414E+05
Web	10.0	568.0	320.0	56.8	1817599.6	0.205E+05
BasePlate	400.0	34.0	621.0	136.0	8445599.0	0.215E+06
Totals	...	...	...	480.2	10744986.0	0.359E+06



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 119 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20, 2026

Distance to Centroid [C1]:

$$= AY / A$$

$$= 10744986/480$$

$$= 223.767 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 98215$$

$$= 19988.9219 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 6.5169 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1650 - 1012 \cdot \sin(120/2) / 1.05$$

$$= 813.083 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 16252.9678 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 9.9427 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) }$$

$$= \sqrt{ (3(98215 + 731) 400 / (4 \cdot 1950 \cdot 173)) }$$

$$= 29.347 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1012 + 20 + 16)$$

$$= 1815.189 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1815 / (5 - 1)$$

$$= 453.797 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 454 \cdot 400 / 2$$

$$= 907.595 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 98947 / (1950 \cdot 400)$$

$$= 12.685 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 908 \cdot 12.7$$

$$= 11513.250 \text{ Kgf}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 120 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 36.5 + 22.7
 = 59.190 cm²

Compressive Stress [Sc]:

= P/Ar
 = 11513/59.2
 = 19.076 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 375 - 10
 = 365.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0/59.2
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 375/2
 = 187.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{4395/59.2}$
 = 86.165 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1650 - cos(120/2)(1012 + 20 + 16) - 34
 = 1092.000 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199943008/260}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1092/86.2
 = 12.673

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 249 / (2 • 1950) • 454 • 1092/2
 = 15.824 Kg-m.

Compressive Allowable, KL/r < Cc (12.7 < 123) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (12.7)^2 / (2 \cdot 123^2)) 260 / (5/3 + 3 \cdot (12.7) / (8 \cdot 123) - (12.7^3) / (8 \cdot 123^3))$
 = 151.7 N./mm²



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 121 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 19.1/152 + (15.8 \cdot 188/43945308) / 173$$

$$= 0.130$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	453.8	10.0	0.0	45.4	0.0	3.78
Totals	...	...	...	81.9	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/81.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1650 - \cos(1048/2)(34 + 0 + 0) - 0$$

$$= 568.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (4398/81.9) }$$

$$= 73.291 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 568/73.3$$

$$= 7.750$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 249 / (2 \cdot 1950)$$

$$= 0.064 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.064 \cdot 454 \cdot 568$$

$$= 16.462 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 16.5/234572$$

$$= 0.688 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($7.75 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.75)^2 / (2 \cdot 123^2)) 260 /$$

$$(5/3 + 3 \cdot (7.75) / (8 \cdot 123) - (7.75^3) / (8 \cdot 123^3))$$

$$= 153.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$

$$= 27.6/154 + (16.5 \cdot 188/4398) / 173$$

$$= 0.184$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 122 of 264
 Saddle Calcs: Test Case: Step: 21 9:47am Apr 20,2026

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	8
Total Number of Bolts in Tension/Baseplate	Nbt	4
Bolt Material Specification	SA-307 A	
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	240.0000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	231.8406 cm ²
Saddle Load Q0 (Weight)	Q0	97317.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	53.2 Kg
Maximum Transverse Force	Ft	597.5 Kg
Maximum Longitudinal Force	Fl	249.1 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 $= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 249 / (232 \cdot 8)$
 $= 0.0 \text{ N./mm}^2$. Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 597 / (232 \cdot 8)$
 $= 0.0 \text{ N./mm}^2$. Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 $= 0.0$ (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 $= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 249 / (82.8 \cdot 8)$
 $= 0.0369 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 1650 \cdot 597 + 0$
 $= 985.85 \text{ Kg-m}$.

Eccentricity (e):
 $= Rmom / Q0$
 $= 986 / 97317$
 $= 10.13 \text{ mm} < Bplen/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 $= 0$ (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0, 0.037, 0]$
 $= 0.0369 \text{ cm}^2$

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



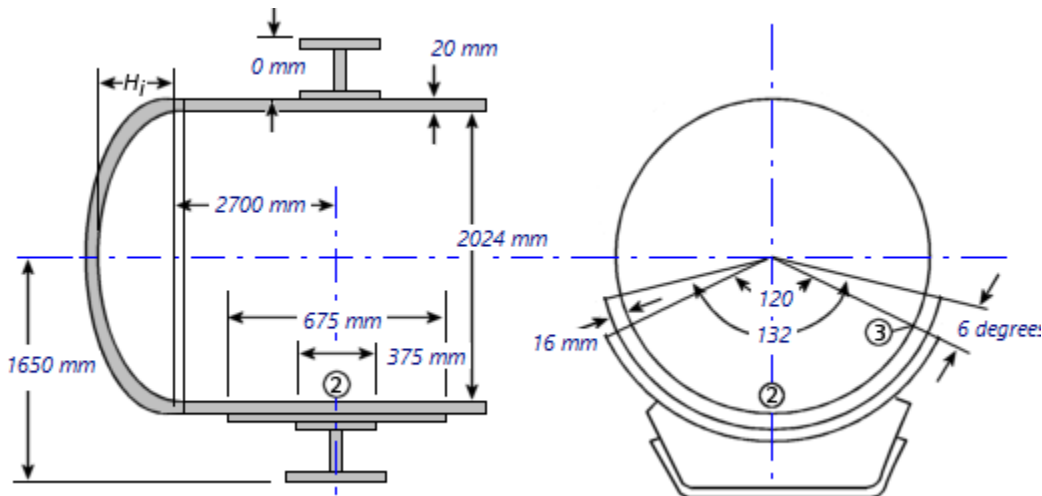
PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 123 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	1023.00	mm
Shell Thickness used in this Case	t	20.000	mm
Stiffened Vessel Length per 4.15.6	L	13108.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	675.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1650.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	119295.11	Kgf
Pressure used in Analysis	P	-103.419 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-15.13	-235.80
Long. Stress at Top of Midspan	15.13	138.00
Long. Stress at Bottom of Midspan	9.26	138.00
Long. Stress at Top of Saddles	92.78	138.00
Long. Stress at Bottom of Saddles	-55.99	-235.80
Long. Stress at Bottom of Saddles	55.99	138.00
Tangential Shear in Shell	43.74	110.40



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 124 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

Circ. Stress at Horn of Saddle	94.89	172.50
Circ. Compressive Stress in Shell	4.46	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (2390/2 + 0) \cdot 1650/1815$
 $= 3258.6\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(498, 0) \cdot 1650/7720$
 $= 106.5\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(11718, 0) \cdot 1650/7720$
 $= 2504.6\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (11718/2 + 0) \cdot 1650/1815$
 $= 15977.8\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 103317 + \max(106, 3259, 2505, 15978)$
 $= 119295.1\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.26 \cdot 1.2)^2)$
 $= 4885.121\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	120026.45	Kgf
Transverse Shear Load	5859.13	Kgf
Longitudinal Shear Load	11718.26	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2024/2 + 20 + 16) \sin(120/2), 2 \cdot 1023)$
 $= 1815.189\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -119295 \cdot 2700 [1 - (1 - 2700/13108 + (1023^2 - 0^2) / (2 \cdot 2700 \cdot 13108)) / (1 + (4 \cdot 0) / (3 \cdot 13108))]$
 $= -61584.9\ Kg \cdot m$

Moment per Equation 4.15.2 [M2]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 125 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20, 2026

$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\
 &= 119295 \cdot 13108 / 4 \cdot ((1 + 2(1023^2 - 0^2) / (13108^2)) / (1 + (4 \cdot 0) / (3 \cdot 13108)) - 4 \cdot 2700 / 13108) \\
 &= 73596.9 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 1023 / (2 \cdot 18) - 73597 / (\pi \cdot 1023^2 \cdot 18) \\
 &= -15.13 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 1023 / (2 \cdot 18) + 73597 / (\pi \cdot 1023^2 \cdot 18) \\
 &= 9.26 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 1023 / (2 \cdot 18) - 61585 / (0.11 \cdot \pi \cdot 1023^2 \cdot 18) \\
 &= 92.78 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= -103 \cdot 1023 / (2 \cdot 18) + 61585 / (0.19 \cdot \pi \cdot 1023^2 \cdot 18) \\
 &= -55.99 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 119295 \cdot (13108 - 2 \cdot 2700) / (13108 + (4 \cdot 0 / 3)) \\
 &= 70150.0 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.17 \cdot 70150 / (1023 \cdot 18) \\
 &= 43.74 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1023 \cdot 18)} \\
 &= 105.845 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(375 + 1.56 \cdot \sqrt{(1023 \cdot 18)}, 2 \cdot 2700) \\
 &= 586.69 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$\begin{aligned}
 &= \min(Sr/S, 1) \\
 &= \min(138/138, 1) \\
 &= 1.0000
 \end{aligned}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.76 \cdot 119295 \cdot 0.1 / (587(18 + 1 \cdot 16)) \\
 &= -4.46 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\
 &= -119295 / (4(18 + 1 \cdot 16)587) - 3 \cdot 0.053 \cdot 119295 / (2(18 + 1 \cdot 16)^2) \\
 &= -94.89 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1949.9999	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	400.0000	mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 126 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

Number of Ribs (inc. outside ribs)	Nribs	5
Rib Thickness	Ribtk	10.0000 mm
Web Thickness	Webtk	10.0000 mm
Web Location	Webloc	Center
Saddle Design Temperature		21.1 °C
Saddle Yield Stress	Sy	262.0 N./mm ²
Height of Web at Center	Hw,c	638.0 mm
Friction Coefficient	mu	0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(1014 \cdot 18)} + 675)18^3)/12$
 $= 43.048 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 43 + 159(231 - 9)^2$
 $= 78888.531 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	885.8	18.0	9.0	159.4	143492.4	0.789E+05
Wearplate	675.0	16.0	26.0	108.0	280800.0	0.456E+05
Web	10.0	568.0	318.0	56.8	1806240.1	0.195E+05
BasePlate	400.0	34.0	619.0	136.0	8418400.0	0.204E+06
Totals	...	...	...	460.2	10648933.0	0.348E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 10648933/460$
 $= 231.380 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120)\cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 119295$
 $= 24279.1406 \text{ Kg}$

Tension Stress, St = (Fh/As) = 7.9156 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1650 - 1014 \cdot \sin(120/2) / 1.05$
 $= 811.429 \text{ mm}$

Bending Moment, M = Fh * d = 19701.1816 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 12.8284 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt{3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 127 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

$$= \sqrt{3(119295 + 731)400 / (4 \cdot 1950 \cdot 175)}$$

$$= 32.199 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (1012 + 20 + 16)$$

$$= 1815.189 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1815 / (5 - 1)$$

$$= 453.797 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 454 \cdot 400 / 2$$

$$= 907.595 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 120026 / (1950 \cdot 400)$$

$$= 15.388 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 908 \cdot 15.4$$

$$= 13966.073 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 36.5 + 22.7$$

$$= 59.190 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 13966 / 59.2$$

$$= 23.140 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 375 - 10$$

$$= 365.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0 / 59.2$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375 / 2$$

$$= 187.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 128 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

$$= \sqrt{(4395/59.2)}$$

$$= 86.165 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1650 - \cos(120/2) (1012 + 20 + 16) - 34$$

$$= 1092.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/262)}$$

$$= 122.739$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1092/86.2$$

$$= 12.673$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{\text{plen}}) \cdot e \cdot L / 2$$

$$= 11718 / (2 \cdot 1950) \cdot 454 \cdot 1092/2$$

$$= 744.495 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($12.7 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.7)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (12.7) / (8 \cdot 123) - (12.7)^3 / (8 \cdot 123^3))$$

$$= 152.8 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 23.1/153 + (744 \cdot 188/43945308) / 175$$

$$= 0.330$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	453.8	10.0	0.0	45.4	0.0	3.78
Totals	...	...	...	81.9	...	0.440E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/81.9$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 375/2$$

$$= 187.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 1650 - \cos(1048/2) (34 + 0 + 0) - 0$$

$$= 568.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(4398/81.9)}$$

$$= 73.291 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 129 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

$$= 1 \cdot 568/73.3$$

$$= 7.750$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 11718 / (2 \cdot 1950)$$

$$= 3.005 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3 \cdot 454 \cdot 568$$

$$= 774.493 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 774/234572$$

$$= 32.379 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($7.75 < 123$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.75)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (7.75) / (8 \cdot 123) - (7.75^3) / (8 \cdot 123^3))$$

$$= 154.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 33.5/155 + (774 \cdot 188/4398) / 175$$

$$= 0.402$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1023.00	mm
Shell Thickness used in this Case	t	20.000	mm
Stiffened Vessel Length per 4.15.6	L	13108.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2700.00	mm
Saddle Width	b1 or b	375.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	675.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1650.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		118200.38	Kgf
Pressure used in Analysis	P	-103.419	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-15.02	-235.80
Long. Stress at Top	of Midspan	15.02	138.00
Long. Stress at Bottom	of Midspan	9.14	138.00



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 130 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

Long. Stress at Top of Saddles	91.91	138.00
Long. Stress at Bottom of Saddles	-55.51	-235.80
Long. Stress at Bottom of Saddles	55.51	138.00
Tangential Shear in Shell	43.34	110.40
Circ. Stress at Horn of Saddle	94.02	172.50
Circ. Compressive Stress in Shell	4.42	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2390/2 + 0) \cdot 1650/1815$
 $= 3258.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(498, 0) \cdot 1650/7720$
 $= 106.5 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(11718, 0) \cdot 1650/7720$
 $= 2504.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (11718/2 + 0) \cdot 1650/1815$
 $= 15977.8 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 102223 + \max(106, 3259, 2505, 15978)$
 $= 118200.4 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766 (0 + \pi/4(2.26 \cdot 1.2)^2)$
 $= 4885.121 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	118931.72	Kgf
Transverse Shear Load	5859.13	Kgf
Longitudinal Shear Load	11718.26	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(2024/2 + 20 + 16) \sin(120/2), 2 \cdot 1023)$
 $= 1815.189 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$



PV Elite 28

Licensee: SL User

FileName : 106-C-Cladding on Both Side

Page 131 of 264

Saddle Calcs: Ext. Pressure:

Step: 22

9:47am Apr 20, 2026

$$= -118200 \cdot 2700 [1 - (1 - 2700/13108 + (1023^2 - 0^2) / (2 \cdot 2700 \cdot 13108)) / (1 + (4 \cdot 0) / (3 \cdot 13108))] \\ = -61019.7 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a/L) \\ = 118200 \cdot 13108 / 4 ((1 + 2(1023^2 - 0^2) / (13108^2)) / (1 + (4 \cdot 0) / (3 \cdot 13108)) - 4 \cdot 2700 / 13108) \\ = 72921.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t) \\ = -103 \cdot 1023 / (2 \cdot 18) - 72922 / (\pi \cdot 1023^2 \cdot 18) \\ = -15.02 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\ = -103 \cdot 1023 / (2 \cdot 18) + 72922 / (\pi \cdot 1023^2 \cdot 18) \\ = 9.14 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t) \\ = -103 \cdot 1023 / (2 \cdot 18) - 61020 / (0.11 \cdot \pi \cdot 1023^2 \cdot 18) \\ = 91.91 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\ = -103 \cdot 1023 / (2 \cdot 18) + 61020 / (0.19 \cdot \pi \cdot 1023^2 \cdot 18) \\ = -55.51 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\ = 118200 (13108 - 2 \cdot 2700) / (13108 + (4 \cdot 0 / 3)) \\ = 69506.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t) \\ = 1.17 \cdot 69506 / (1023 \cdot 18) \\ = 43.34 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)} \\ = 0.78 \cdot \sqrt{(1023 \cdot 18)} \\ = 105.845 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\ = \min(375 + 1.56 \cdot \sqrt{(1023 \cdot 18)}, 2 \cdot 2700) \\ = 586.69 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1) \\ = \min(138/138, 1) \\ = 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\ = -0.76 \cdot 118200 \cdot 0.1 / (587(18 + 1 \cdot 16)) \\ = -4.42 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2) \\ = -118200 / (4(18 + 1 \cdot 16)587) - 3 \cdot 0.053 \cdot 118200 / (2(18 + 1 \cdot 16)^2) \\ = -94.02 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 132 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1949.9999	mm
Baseplate Thickness	Bpthk	34.0000	mm
Baseplate Width	Bpwid	400.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	5	
Rib Thickness	Ribtk	10.0000	mm
Web Thickness	Webtk	10.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	638.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1014 \cdot 18)} + 675)^3) / 12$
 $= 43.048 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 43 + 159 (231 - 9)^2$
 $= 78888.531 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	885.8	18.0	9.0	159.4	143492.4	0.789E+05
Wearplate	675.0	16.0	26.0	108.0	280800.0	0.456E+05
Web	10.0	568.0	318.0	56.8	1806240.1	0.195E+05
BasePlate	400.0	34.0	619.0	136.0	8418400.0	0.204E+06
Totals	...	...	...	460.2	10648933.0	0.348E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 10648933 / 460$
 $= 231.380 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.2 \cdot 118200$
 $= 24056.3379 \text{ Kg}$

Tension Stress, St = (Fh/As) = 7.8430 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$
 $= 1650 - 1014 \cdot \sin(120 / 2) / 1.05$
 $= 811.429 \text{ mm}$

Bending Moment, M = Fh * d = 19520.3906 Kg-m.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 133 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

Bending Stress, $S_b = (M * C_1 / I) = 12.7107 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 174.6635 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(118200 + 731)400 / (4 * 1950 * 175))}$
 $= 32.052 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (1012 + 20 + 16)$
 $= 1815.189 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 1815 / (5 - 1)$
 $= 453.797 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 454 * 400 / 2$
 $= 907.595 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 118932 / (1950 * 400)$
 $= 15.248 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 908 * 15.2$
 $= 13838.691 \text{ Kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 36.5 + 22.7$
 $= 59.190 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 13839 / 59.2$
 $= 22.929 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	375.0	...	37.5	...	0.439E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 375 - 10$
 $= 365.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0 / 59.2$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 134 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

= Saddle Width / 2
 = 375/2
 = 187.500 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{4395/59.2}$
 = 86.165 mm

Length of Outer Rib [L]:
 $= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 1650 - \cos(120/2) (1012 + 20 + 16) - 34$
 = 1092.000 mm

Intermediate Term [Cc]:
 $= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 $= \sqrt{2 \cdot \pi^2 \cdot 199943008/262}$
 = 122.739

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 1092/86.2$
 = 12.673

Bending Moment [Rm]:
 $= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$
 $= 11718 / (2 \cdot 1950) \cdot 454 \cdot 1092/2$
 = 744.495 Kg-m.

Compressive Allowable, $KL/r < Cc$ (12.7 < 123) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (12.7)^2 / (2 \cdot 123^2)) 262 / (5/3 + 3 \cdot (12.7) / (8 \cdot 123) - (12.7)^3 / (8 \cdot 123^3))$
 = 152.8 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 $= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 22.9/153 + (744 \cdot 188/43945308) / 175$
 = 0.328

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	365.0	0.0	36.5	0.0	0.439E+04
Web	453.8	10.0	0.0	45.4	0.0	3.78
Totals	...	...	...	81.9	...	0.440E+04

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0/81.9$
 = 0.000 mm

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 375/2$
 = 187.500 mm

Length of Inner Rib [L]:
 $= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$
 $= 1650 - \cos(1048/2) (34 + 0 + 0) - 0$
 = 568.000 mm

Radius of Gyration [r]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 135 of 264
 Saddle Calcs: Ext. Pressure: Step: 22 9:47am Apr 20,2026

$$\begin{aligned}
 &= \sqrt{\text{Total Inertia} / \text{Total Area}} \\
 &= \sqrt{4398/81.9} \\
 &= 73.291 \text{ mm}
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 568/73.3 \\
 &= 7.750
 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Baseplate Length}) \\
 &= 11718 / (2 \cdot 1950) \\
 &= 3.005 \text{ Kg/mm}
 \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned}
 &= \text{Unit Force} \cdot e \cdot L \\
 &= 3 \cdot 454 \cdot 568 \\
 &= 774.493 \text{ Kg-m.}
 \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned}
 &= \text{Bending Moment} / \text{Section Modulus} \\
 &= 774/234572 \\
 &= 32.379 \text{ N./mm}^2
 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($7.75 < 123$) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\
 &= (1 - (7.75)^2 / (2 \cdot 123^2)) 262 / \\
 &\quad (5/3 + 3 \cdot (7.75) / (8 \cdot 123) - (7.75^3) / (8 \cdot 123^3)) \\
 &= 154.7 \text{ N./mm}^2
 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned}
 &= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba} \\
 &= 33.2/155 + (774 \cdot 188/4398) / 175 \\
 &= 0.400
 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 136 of 264
 Nozzle Calcs.: N1-Tube Inlet Nozl: 15 9:47am Apr 20,2026

Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	5293.880	KPa.
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2024.00	mm
Shell Finished (Minimum) Thickness	t	46.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1707.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

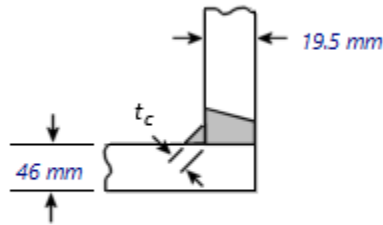
Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		34.4882	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	19.5000	mm
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	82.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		43.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		a	
Flange Class		600	
Flange Grade		GR 2.1	
Flange Series		B	

The Pressure Design option was Overall MAWP.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 137 of 264
 Nozzle Calcs.: N1-Tube Inlet Nozl: 15 9:47am Apr 20,2026



UW-16 (a)

Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 34.488 in.
 Actual Thickness Used in Calculation 0.768 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5294 \cdot 1014 / (138 \cdot 1 - 0.6 \cdot 5294)$
 $= 39.8180 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5294 \cdot 438 / (138 \cdot 1 + 0.4 \cdot 5294)$
 $= 16.5496 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 160 / 0.5773$
 $= 277.9942 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1674.0005	mm
Parallel to Vessel Wall, opening length	d	837.0002	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	110.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$



PV Elite 28

Licensee: SL User

FileName : 106-C-Cladding on Both Side

Page 138 of 264

Nozzle Calcs.: N1-Tube Inlet

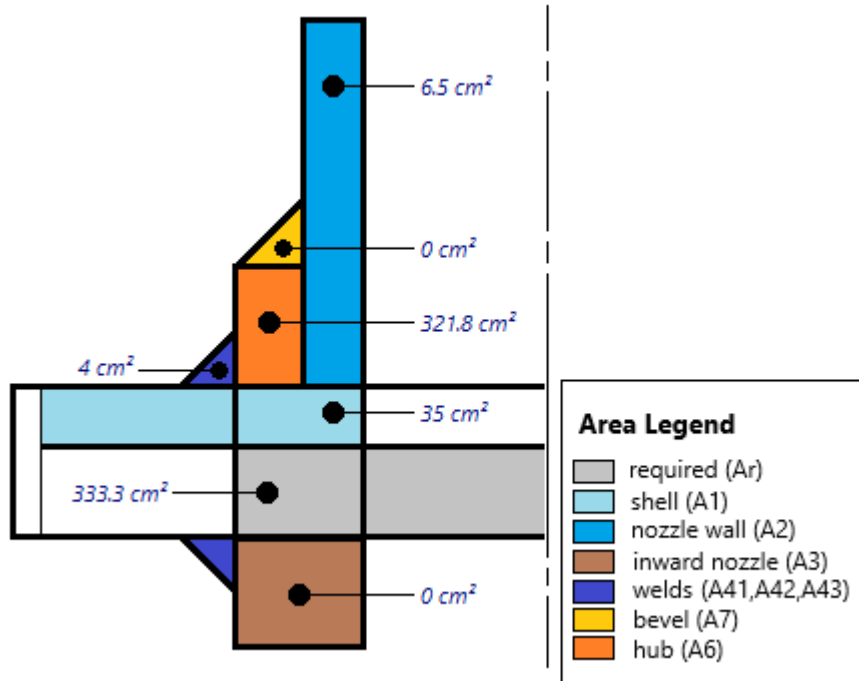
Nozl: 15 9:47am Apr 20,2026

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1, 1)

= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	333.277	NA	NA
Area in Shell	A1	35.003	NA	NA
Area in Nozzle Wall	A2	6.491	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		4.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	321.783	NA	NA
TOTAL AREA AVAILABLE	Atot	367.277	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)

= (837 • 39.8 • 1 + 2 • 180 • 39.8 • 1 • (1-1))

= 333.277 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1 • t - F • tr) - 2 • tn(E1 • t - F • tr) • (1 - fr1)

= 837 (1 • 44 - 1 • 39.8) - 2 • 180

(1 • 44 - 1 • 39.8) • (1 - 1)

= 35.003 cm²



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 139 of 264
 Nozzle Calcs.: N1-Tube Inlet Nozl: 15 9:47am Apr 20,2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 110) (19.5 - 16.5) 1$$

$$= 6.491 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 20^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{np}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(110, 500, 70)) \cdot (180 - 19.5) \cdot 1$$

$$= 321.783 \text{ cm}^2 \text{ Includes Bevel Area of } 97.083 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 16.5496 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 41.8180 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 41.8180 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 3.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.33, \max(41.8, 3.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(16.5, 8.33)$$

$$= 16.5496 \text{ mm}$$

Available Nozzle Neck Thickness = 19.5000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 74.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 270.2 N./mm ²	Passed
Occasional	: 55.5,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 11.9,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 19.5$, $t_r = 16.5$, $c = 0 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-17 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 46$, $t_r = 39.8$, $c = 2 \text{ mm}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.9$, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	9 °C



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 140 of 264
 Nozzle Calcs.: N1-Tube Inlet Nozl: 15 9:47am Apr 20,2026

Gov. MDMT of the nozzle to shell joint welded assembly : 9 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 160 / 0.5773$
 $= 277.9942 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1255.5004 mm
 Parallel to Vessel Wall, opening length d 627.7502 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 110.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		222.196	222.196
Area Available in Shell A1		17.502	17.502
Area Available in Nozzle Wall A2		6.491	6.491
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		4.000	4.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		321.783	321.783
TOTAL AREA AVAILABLE Atot		349.775	349.775

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 14.1400 = $0.7 * W_o$ mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5544 KPa.

The Drop for this Nozzle is : 99.6953 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 645.6953 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2024.000	mm
Shell Thickness	46.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 141 of 264
 Nozzle Calcs.: N1-Tube Inlet Nozl: 15 9:47am Apr 20,2026

Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	1197.000	mm	
Nozzle Thickness	180.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-266 2		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be **SUS**tained loads.

Design Internal Pressure	Dp	5293.88	KPa.
Radial Load	P	-8260.24	Kgf
Circumferential Shear	Vc	5965.75	Kgf
Longitudinal Shear	Vl	5965.75	Kgf
Circumferential Moment	Mc	16514.19	Kg-m.
Longitudinal Moment	Ml	16514.19	Kg-m.
Torsional Moment	Mt	18991.26	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.566) is > 0.5
 check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.650) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 1197.0 + 2 \cdot 1.65 \cdot \sqrt{1036.0(46.0 - 2.0)}$
 $= 1901.564 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2072.000 mm
Nozzle Outside Diameter	(d) =	1197.000 mm
Vessel Thickness used	(T) =	44.000 mm
Nozzle Thickness used	(t) =	180.000 mm
T / t	=	0.244
d / t	=	10.000

$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.964$

Nr/P	=	0.051
Mr/P	=	0.082
M0/P	=	0.026
N0/P	=	0.058
MrD/Mc	=	0.256
NrDL/Mc	=	0.091
M0d/Mc	=	0.077
N0DL/Mc	=	0.085



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N1-Tube Inlet

Nozl: 15 9:47am Apr 20,2026 Page 142 of 264

MrD/M1 = 0.070
 NrDL/M1 = 0.048
 MØD/M1 = 0.021
 NØDL/M1 = 0.061

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	119	124	119	124
Outplane Stress Summary	115	125	141	115

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-29	29	29	-29
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	49	67	115	15

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	115	125	141	115

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-32	32	32	-32
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N1-Tube Inlet

Nozl: 15 9:47am Apr 20,2026

Page 143 of 264

Outplane Stress Summary	30	81	106	29
-------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-107	107	107	-107
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	119	124	119	124

Inplane Stress Summary	28	207	255	5
------------------------	----	-----	-----	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	31	207	255	29
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	18	12	18

Hoop Stress Summary	10	16	18	25
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	4	5	9	5
----------------------	---	---	---	---



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N1-Tube Inlet

Nozl: 15 9:47am Apr 20,2026

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	10	16	18	25

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
hoop direction)				
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	18	12	18
Hoop Stress Summary	8	15	20	26

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
pipe cross-section)				
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	0	9	13	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	8	15	20	26



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 145 of 264
 Nozzle Calcs.: N1-Tube Inlet Nozl: 15 9:47am Apr 20,2026

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	119	124	119	124	119	124	119	124
Circ. Pl (SUS)	-2	-2	6	6	-4	-4	8	8
Circ. Q (SUS)	-2	2	14	-14	-87	87	127	-127
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	-1	-1	5	5	-3	-3	7	7
Long. Q (SUS)	-9	9	49	-49	-26	26	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	119.0	124.0	119.0	124.0	119.0	124.0	119.0	124.0
Pm+Pl (SUS)	117.0	122.0	125.0	130.0	115.0	120.0	127.0	132.0
Pm+Pl+Q (Total)	115.0	124.0	139.0	116.0	30.4	207.0	254.0	28.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.00	138.00	Passed
Pm+Pl (SUS)	132.00	207.00	Passed
Pm+Pl+Q (TOTAL)	254.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	18	12	18	12	18	12	18
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-1	1	1	-1	-5	5	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	12.0	18.0	12.0	18.0	12.0	18.0	12.0	18.0
Pm+Pl (SUS)	10.0	16.0	18.0	24.0	9.0	15.0	19.0	25.0
Pm+Pl+Q (Total)	10.0	16.0	18.0	24.0	9.0	15.0	19.0	25.0

Nozzle Stress Summation Comparison (N./mm²):

Type of	Max. S.I.	S.I. Allowable	Result
---------	-----------	----------------	--------



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side Page 146 of 264
Nozzle Calcs.: N1-Tube Inlet Nozl: 15 9:47am Apr 20,2026

Stress Int.			
Pm (SUS)	18.00	138.00	Passed
Pm+Pl (SUS)	25.00	207.00	Passed
Pm+Pl+Q (TOTAL)	25.00	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 147 of 264
 Nozzle Calcs.: Noz N5 Nozl: 16 9:47am Apr 20,2026

Input, Nozzle Desc: Noz N5

From: 30

Pressure for Reinforcement Calculations	P	5293.880	KPa.
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2024.00	mm
Shell Finished (Minimum) Thickness	t	46.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1707.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

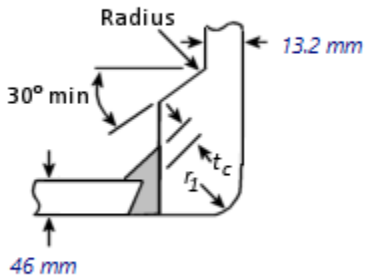
Type of Element Connected to the Parent : Nozzle

Material		SA-965 F304	
Material UNS Number		S30400	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	121.40	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		60	
Flange Material		SA-182 F304	
Hub Height of Integral Nozzle	h	80.0000	mm
Height of Beveled Transition	L'	37.4772	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		g	
Flange Class		600	
Flange Grade		GR 2.1	

The Pressure Design option was Overall MAWP.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Clading on Both Side Page 148 of 264
 Nozzle Calcs.: Noz N5 Nozl: 16 9:47am Apr 20,2026



UW-16.1 [g]

Reinforcement CALCULATION, Description: Noz N5

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	14.000 in.
Actual Thickness Used in Calculation	0.520 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5294 \cdot 1014 / (138 \cdot 1 - 0.6 \cdot 5294)$
 $= 39.8180 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5294 \cdot 178 / (121 \cdot 1 + 0.4 \cdot 5294)$
 $= 7.6209 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 76.8 / 0.5773$
 $= 133.0187 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	658.3934	mm
Parallel to Vessel Wall, opening length	d	329.1967	mm
Normal to Vessel Wall (Thickness Limit), no pad	T1np	110.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 121 / 138)$
 $= 0.880$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 121 / 138)$
 $= 0.880$

Weld Strength Reduction Factor [fr3]:



PV Elite 28

Licensee: SL User

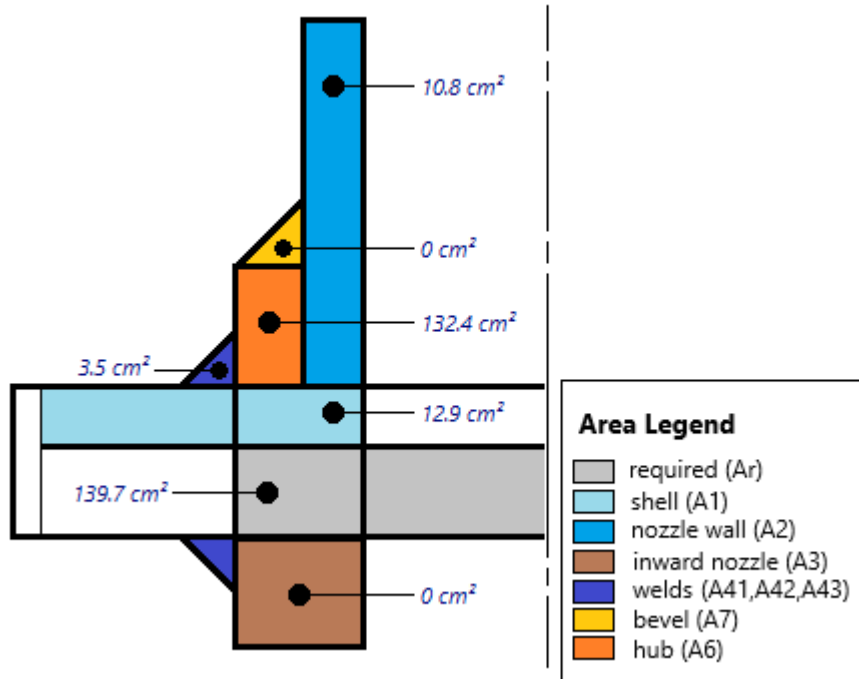
FileName : 106-C-Cladding on Both Side

Page 149 of 264

Nozzle Calcs.: Noz N5

Nozl: 16 9:47am Apr 20,2026

= min(fr2, fr4)
= min(0.88, 1)
= 0.880



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	139.701	NA	NA
Area in Shell	A1	12.861	NA	NA
Area in Nozzle Wall	A2	10.801	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		3.519	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	132.409	NA	NA
TOTAL AREA AVAILABLE	Atot	159.590	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)
= (329 • 39.8 • 1 + 2 • 90 • 39.8 • 1 • (1-0.88))
= 139.701 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d (E1 • t - F • tr) - 2 • tn (E1 • t - F • tr) • (1 - fr1)
= 329 (1 • 44 - 1 • 39.8) - 2 • 90
(1 • 44 - 1 • 39.8) • (1 - 0.88)
= 12.861 cm²

Area Available in Nozzle Projecting Outward [A2]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 150 of 264
 Nozzle Calcs.: Noz N5 Nozl: 16 9:47am Apr 20,2026

$$= (2 \cdot t_{np}) (t_n - t_{rn}) \cdot fr^2$$

$$= (2 \cdot 110) (13.2 - 7.62) \cdot 0.88$$

$$= 10.801 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr^2 + (W_i - c_{an}/0.707)^2 \cdot fr^2$$

$$= 20^2 \cdot 0.88 + (0)^2 \cdot 0.88$$

$$= 3.519 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot fr^2$$

$$= (2 \cdot \min(110, 300, 80)) \cdot (90 - 13.2) \cdot 0.88$$

$$= 132.409 \text{ cm}^2 \text{ Includes Bevel Area of } 24.312 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 7.6209 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 3.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 41.8180 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 41.8180 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 3.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 8.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[8.33, \max(41.8, 3.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(7.62, 8.33)$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Weld Size Calculations, Description: Noz N5

Intermediate Calc. for nozzle/shell Welds $T_{min} = 19.0000 \text{ mm}$

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 \cdot W_o \text{ mm}$

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5666 KPa.

The Drop for this Nozzle is : 15.7414 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 361.7414 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 151 of 264
 Nozzle Calcs.: D1 Nozl: 17 9:47am Apr 20,2026

Input, Nozzle Desc: D1

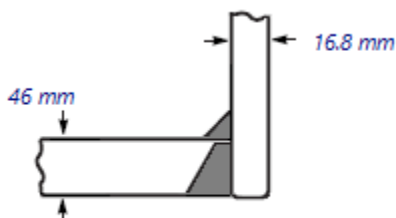
From: 30

Pressure for Reinforcement Calculations	P	5293.880	KPa.
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2024.00	mm
Shell Finished (Minimum) Thickness	t	46.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1707.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-182 F304	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.1	

The Pressure Design option was Overall MAWP.





PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 152 of 264
 Nozzle Calcs.: D1 Nozl: 17 9:47am Apr 20,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5294 \cdot 1014 / (138 \cdot 1 - 0.6 \cdot 5294)$
 $= 39.8180 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \cdot (\exp([P / (S_n \cdot E)] - 1))$ per Appendix 1-2 (a)(1)
 $= 25.4 \cdot (\exp([5294 / (138 \cdot 1)] - 1))$
 $= 0.9934 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.3280	mm
Parallel to Vessel Wall	Rn+tn+t	86.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: D1.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.9934	mm
Wall Thickness per UG16(b),	tr16b = 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 41.8180	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 41.8180	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000	mm
Wall Thickness per table UG-45	tb3 = 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(41.8, 3.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.99, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.99, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.059$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: D1



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side Page 153 of 264
Nozzle Calcs.: D1 Nozl: 17 9:47am Apr 20,2026

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8787 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.8787 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 154 of 264
 Nozzle Calcs.: N3-Shell Inlet Nozl: 18 9:47am Apr 20,2026

Input, Nozzle Desc: N3-Shell Inlet From: 40

Pressure for Reinforcement Calculations	P	2423.703	KPa.
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2024.00	mm
Design Length of Section	L	13108.0156	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4912.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

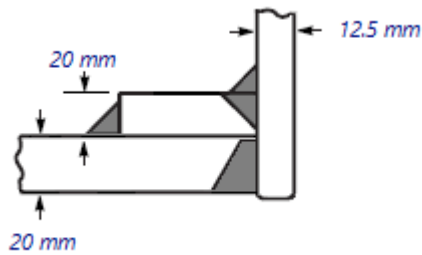
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		60	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	543.8500	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		110.0000	mm
Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was Overall MAWP.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N3-Shell Inlet Nozl: 18 9:47am Apr 20,2026



Reinforcement CALCULATION, Description: N3-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 11.767 in.
 Actual Thickness Used in Calculation 0.492 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2424 \cdot 1014 / (138 \cdot 1 - 0.6 \cdot 2424)$
 $= 18.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2424 \cdot 149 / (118 \cdot 1 - 0.6 \cdot 2424)$
 $= 3.1079 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9318 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	597.7382	mm
Parallel to Vessel Wall, opening length	d	298.8691	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		45.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

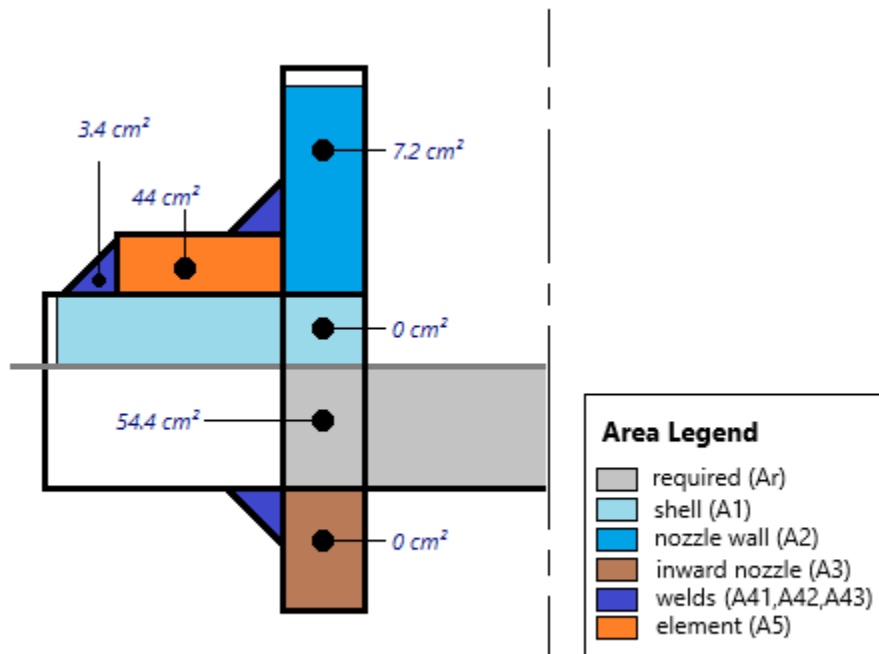


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	54.448	22.006	NA
Area in Shell	A1	0.000	10.186	NA
Area in Nozzle Wall	A2	7.220	8.895	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		3.415	3.415	NA
Area in Element	A5	44.000	44.000	NA
TOTAL AREA AVAILABLE	Atot	54.636	66.496	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 542.9128 20.0000 mm
 Based on given Pad Diameter: 543.8500 19.9148 mm
 Based on the Estimated Diameter Limit: 596.1507 16.0898 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (299 \cdot 18 \cdot 1 + 2 \cdot 12.5 \cdot 18 \cdot 1 \cdot (1 - 0.86))$
 $= 54.448 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 299(1 \cdot 18 - 1 \cdot 18) - 2 \cdot 12.5$
 $(1 \cdot 18 - 1 \cdot 18) \cdot (1 - 0.86)$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 157 of 264
 Nozzle Calcs.: N3-Shell Inlet Nozl: 18 9:47am Apr 20,2026

= 0.000 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 45) \cdot (12.5 - 3.11) \cdot 0.86$$

$$= 7.220 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 16^2 \cdot 1$$

$$= 3.415 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (544 - 324) \cdot 20 \cdot 1$$

$$= 44.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1079 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 20.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 20.0000 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7603 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.33, \max(20, 3.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.11, 8.33)$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 12.4904 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.5,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 263.5 N./mm ²	Passed
Occasional	: 13.9,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 13.9,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 12.5, tr = 3.11, c = 0 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.25, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.5, tr = 3.11, c = 0 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.25, Temp. Reduction = 78 °C



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 158 of 264
 Nozzle Calcs.: N3-Shell Inlet Nozl: 18 9:47am Apr 20,2026

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.5, tr = 3.11, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.25$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20, tr = 18, c = 2 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 12.5, tr = 3.11, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.25$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Weld Size Calculations, Description: N3-Shell Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	12.4904 mm
Intermediate Calc. for pad/shell Welds	TminPad	18.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.0000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (54.4 - 0.44699E-05 + 2 \cdot 12.5 \cdot 0.86 \cdot (1 \cdot 18 - 18)) \cdot 138)$
 $= 76617.03 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (7.22 + 44 + 3.42 - 0 \cdot 0.86) \cdot 138$
 $= 76880.79 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (7.22 + 0 + 0.86 + (3.84)) \cdot 138$
 $= 16773.93 \text{ Kg}$

Weld Load [W3]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 159 of 264
 Nozzle Calcs.: N3-Shell Inlet Nozl: 18 9:47am Apr 20,2026

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (7.22 + 0 + 3.42 + 44 + (3.84)) \cdot 138$$

$$= 82291.16 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 29992. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 544 \cdot 16 \cdot 0.49 \cdot 138$$

$$= 94245. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 156) \cdot (12.5 - 0) \cdot 0.7 \cdot 118$$

$$= 51452. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 324 \cdot 20 \cdot 0.74 \cdot 138$$

$$= 105942. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 324 \cdot (20 - 2) \cdot 0.74 \cdot 138$$

$$= 95348. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (94245 + 51452) = 145697 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (29992 + 105942 + 95348 + 0) = 231282 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (94245 + 95348 + 0) = 189593 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 145697 Kgf, must exceed W = 76617 Kgf	or W1 = 76880 Kgf
Path 2-2 = 231281 Kgf, must exceed W = 76617 Kgf	or W2 = 16773 Kgf
Path 3-3 = 189592 Kgf, must exceed W = 76617 Kgf	or W3 = 82291 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2424 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 13.0384 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 283.0384 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	2.0000 mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 160 of 264
 Nozzle Calcs.: N3-Shell Inlet Nozl: 18 9:47am Apr 20,2026

Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2024.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	298.869	mm
Nozzle Thickness	12.4904	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	543.850	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2423.70	KPa.
Radial Load	P	-2294.50	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(1033.0(20.0 - 2.0))}$$

$$= 773.837 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2066.000 mm
Nozzle Outside Diameter	(d) =	323.850 mm
Vessel Thickness used	(T) =	38.000 mm
Nozzle Thickness used	(t) =	12.490 mm
T / t	=	3.042
d / t	=	25.928

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.156$$

Nr/P	=	0.085
Mr/P	=	0.095
M0/P	=	0.090
N0/P	=	0.216



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N3-Shell Inlet

Nozl: 18 9:47am Apr 20,2026

MrD/Mc = 0.169
 NrDL/Mc = 0.124
 MØd/Mc = 0.190
 NØDL/Mc = 0.145
 MrD/Ml = 0.132
 NrDL/Ml = 0.103
 MØD/Ml = 0.140
 NØDL/Ml = 0.301

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	63	65	63	65
Outplane Stress Summary	44	75	106	46

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	31	31	31	31
Inplane Stress Summary	17	42	66	5

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	44	75	106	46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-3	-3	3	3



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N3-Shell Inlet

Page 162 of 264

Nozl: 18 9:47am Apr 20,2026

Outplane Bending (Mc)	-30	30	30	-30
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31

Outplane Stress Summary	8	53	78	0
-------------------------	---	----	----	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-27	27	27	-27
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	65	63	65

Inplane Stress Summary	42	82	104	34
------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	43	82	104	35
------------------------	----	----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36

Hoop Stress Summary	28	31	44	47
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N3-Shell Inlet

Page 163 of 264

Nozl: 18 9:47am Apr 20,2026

Axial Bending (ML)	-120	120	120	-120
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	-70	79	198	-134

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	11	11	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	101	82	199	182

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36
Hoop Stress Summary	32	35	40	43

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-160	160	160	-160
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	-110	119	238	-174

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	11	11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 164 of 264
 Nozzle Calcs.: N3-Shell Inlet Nozl: 18 9:47am Apr 20,2026

Two * Max Shear Stress 143 120 239 218

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{1023.0(20.0 - 2.0)}$
 $= 771.654 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2046.000 mm
 Nozzle Outside Diameter (d) = 543.850 mm
 Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 122.490 mm
 $T / t = 0.147$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.834$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.063
 Mr/P = 0.112
 M0/P = 0.033
 N0/P = 0.053
 MrD/Mc = 0.272
 NrDL/Mc = 0.102
 M0d/Mc = 0.082
 N0DL/Mc = 0.074
 MrD/Ml = 0.085
 NrDL/Ml = 0.079
 M0D/Ml = 0.025
 N0DL/Ml = 0.068

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	135	137	135	137
Outplane Stress Summary	137	133	168	121

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	46	-46	46	-46
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-36	36	36	-36
Inplane Pressure Stress	67	67	67	67
Inplane Stress Summary	76	56	160	-5



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N3-Shell Inlet

Nozl: 18 9:47am Apr 20,2026

Page 165 of 264

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			2	2	2	2
Shear Stress Summary			3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			137	133	168	127

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			3	3	3	3
Outplane Bending (P)			13	-13	13	-13
Outplane Membrane (Mc)			-5	-5	5	5
Outplane Bending (Mc)			-34	34	34	-34
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			67	67	67	67
Outplane Stress Summary			44	87	125	27

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			4	4	4	4
Inplane Bending (P)			46	-46	46	-46
Inplane Membrane (Mc)			-7	-7	7	7
Inplane Bending (Mc)			-116	116	116	-116
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			135	137	135	137
Inplane Stress Summary			62	204	310	-13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			2	2	2	2
Shear Stress Summary			1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 166 of 264
 Nozzle Calcs.: N3-Shell Inlet Nozl: 18 9:47am Apr 20,2026

Two * Max Shear Stress 62 204 310 42

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	65	63	65	63	65	63	65
Circ. Pl (SUS)	-5	-5	11	11	-2	-2	4	4
Circ. Q (SUS)	-14	14	30	-30	-19	19	35	-35
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-1	-1	3	3	0	0	6	6
Long. Q (SUS)	-13	13	29	-29	-22	22	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	63.0	65.0	63.0	65.0	63.0	65.0	63.0	65.0
Pm+Pl (SUS)	58.0	60.0	74.0	76.0	61.0	63.0	67.0	69.0
Pm+Pl+Q (Total)	44.1	74.1	104.1	46.1	42.1	82.1	102.1	35.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	138.00	Passed
Pm+Pl (SUS)	76.00	207.00	Passed
Pm+Pl+Q (TOTAL)	104.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135	137	135	137	135	137	135	137
Circ. Pl (SUS)	-1	-1	7	7	-3	-3	11	11
Circ. Q (SUS)	3	-3	23	-23	-70	70	162	-162
Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	-1	-1	9	9	-2	-2	8	8
Long. Q (SUS)	10	-10	82	-82	-21	21	47	-47
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	135.0	137.0	135.0	137.0	135.0	137.0	135.0	137.0
Pm+Pl (SUS)	134.0	136.0	142.0	144.0	132.0	134.0	146.0	148.0
Pm+Pl+Q (Total)	137.1	133.1	165.5	127.1	62.2	204.0	308.0	42.2



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	137.00	138.00	Passed
Pm+Pl (SUS)	148.00	207.00	Passed
Pm+Pl+Q (TOTAL)	308.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	36	33	36	33	36	33	36
Circ. Pl (SUS)	-5	-5	11	11	0	0	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-12	-12	14	14	-12	-12	14	14
Long. Q (SUS)	-75	75	165	-165	-115	115	205	-205
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	33.0	36.0	33.0	36.0	33.0	36.0	33.0	36.0
Pm+Pl (SUS)	28.2	31.1	44.3	47.2	33.1	36.1	39.4	42.3
Pm+Pl+Q (Total)	101.0	81.0	195.2	182.4	144.5	119.4	235.5	217.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	118.00	Passed
Pm+Pl (SUS)	47.23	177.00	Passed
Pm+Pl+Q (TOTAL)	235.51	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 168 of 264
 Nozzle Calcs.: N4-Shell Inlet Nozl: 19 9:47am Apr 20,2026

Input, Nozzle Desc: N4-Shell Inlet From: 40

Pressure for Reinforcement Calculations	P	2423.703	KPa.
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	103.42	KPa.
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2024.00	mm
Design Length of Section	L	13108.0156	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		15082.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

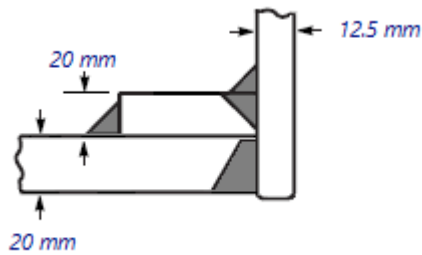
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		60	
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	543.8500	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		110.0000	mm
Flange Class		300	
Flange Grade		GR 2.1	

The Pressure Design option was Overall MAWP.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 169 of 264
 Nozzle Calcs.: N4-Shell Inlet Nozl: 19 9:47am Apr 20,2026



Reinforcement CALCULATION, Description: N4-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	11.767 in.
Actual Thickness Used in Calculation	0.492 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2424 \cdot 1014 / (138 \cdot 1 - 0.6 \cdot 2424)$
 $= 18.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2424 \cdot 149 / (118 \cdot 1 - 0.6 \cdot 2424)$
 $= 3.1079 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.9318 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	597.7382 mm
Parallel to Vessel Wall, opening length	d	298.8691 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		45.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$



PV Elite 28

Licensee: SL User

FileName : 106-C-Cladding on Both Side

Page 170 of 264

Nozzle Calcs.: N4-Shell Inlet

Nozl: 19 9:47am Apr 20,2026

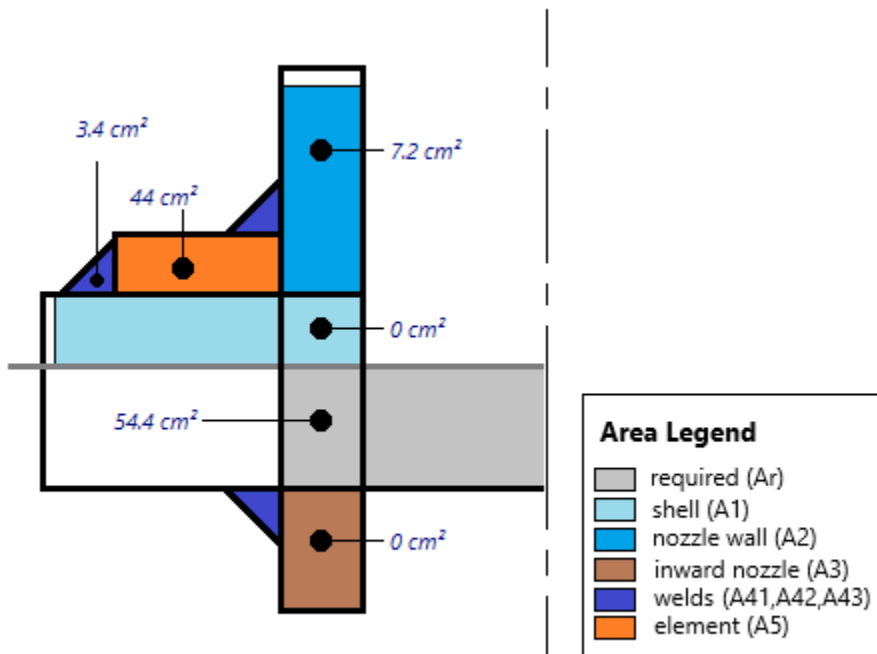


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	54.448	22.006	NA
Area in Shell	A1	0.000	10.186	NA
Area in Nozzle Wall	A2	7.220	8.895	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	3.415	3.415	NA
Area in Element	A5	44.000	44.000	NA
TOTAL AREA AVAILABLE	Atot	54.636	66.496	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
Based on given Pad Thickness: 542.9128 20.0000 mm
Based on given Pad Diameter: 543.8500 19.9148 mm
Based on the Estimated Diameter Limit: 596.1507 16.0898 mm

Area Required [A]:
= (d • tr • F + 2 • tn • tr • F • (1-fr1)) UG-37(c)
= (299•18•1+2•12.5•18•1•(1-0.86))
= 54.448 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
= d(E1•t - F•tr) - 2 • tn (E1•t - F•tr) • (1 - fr1)
= 299 (1 • 18 - 1 • 18) - 2 • 12.5
(1 • 18 - 1 • 18) • (1 - 0.86)



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 171 of 264
 Nozzle Calcs.: N4-Shell Inlet Nozl: 19 9:47am Apr 20,2026

= 0.000 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 45) \cdot (12.5 - 3.11) \cdot 0.86$$

$$= 7.220 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 16^2 \cdot 1$$

$$= 3.415 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (544 - 324) \cdot 20 \cdot 1$$

$$= 44.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.1079 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 20.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 20.0000 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.7603 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.33, \max(20, 3.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(3.11, 8.33)$$

$$= 8.3312 \text{ mm}$$

Available Nozzle Neck Thickness = 12.4904 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 31.5,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 263.5 N./mm ²	Passed
Occasional	: 13.9,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 13.9,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 12.5, tr = 3.11, c = 0 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.25, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 12.5, tr = 3.11, c = 0 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.25, Temp. Reduction = 78 °C



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 172 of 264
 Nozzle Calcs.: N4-Shell Inlet Nozl: 19 9:47am Apr 20,2026

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 12.5, tr = 3.11, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.25$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 20, tr = 18, c = 2 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 12.5, tr = 3.11, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.25$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-22 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Weld Size Calculations, Description: N4-Shell Inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	12.4904 mm
Intermediate Calc. for pad/shell Welds	TminPad	18.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.0000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (54.4 - 0.44699E-05 + 2 \cdot 12.5 \cdot 0.86 \cdot (1 \cdot 18 - 18)) \cdot 138)$
 $= 76617.03 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (7.22 + 44 + 3.42 - 0 \cdot 0.86) \cdot 138$
 $= 76880.79 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (7.22 + 0 + 0.86 + (3.84)) \cdot 138$
 $= 16773.93 \text{ Kg}$

Weld Load [W3]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 173 of 264
 Nozzle Calcs.: N4-Shell Inlet Nozl: 19 9:47am Apr 20,2026

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t_{fr1})) \cdot S$$

$$= (7.22 + 0 + 3.42 + 44 + (3.84)) \cdot 138$$

$$= 82291.16 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 324 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 29992. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 544 \cdot 16 \cdot 0.49 \cdot 138$$

$$= 94245. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 156) \cdot (12.5 - 0) \cdot 0.7 \cdot 118$$

$$= 51452. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 324 \cdot 20 \cdot 0.74 \cdot 138$$

$$= 105942. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 324 \cdot (20 - 2) \cdot 0.74 \cdot 138$$

$$= 95348. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (94245 + 51452) = 145697 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (29992 + 105942 + 95348 + 0) = 231282 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (94245 + 95348 + 0) = 189593 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 145697 Kgf, must exceed W = 76617 Kgf	or W1 = 76880 Kgf
Path 2-2 = 231281 Kgf, must exceed W = 76617 Kgf	or W2 = 16773 Kgf
Path 3-3 = 189592 Kgf, must exceed W = 76617 Kgf	or W3 = 82291 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2404 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 103 KPa.

The Drop for this Nozzle is : 13.0384 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 283.0384 mm

Input Echo, WRC297 Item 1, Description: N4-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	2.0000 mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N4-Shell Inlet

Page 174 of 264
 9:47am Apr 20, 2026

Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2024.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	298.869	mm
Nozzle Thickness	12.4904	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	543.850	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2423.70	KPa.
Radial Load	P	-2294.00	Kgf
Circumferential Shear	Vc	1529.68	Kgf
Longitudinal Shear	Vl	1529.68	Kgf
Circumferential Moment	Mc	1284.43	Kg-m.
Longitudinal Moment	Ml	1284.43	Kg-m.
Torsional Moment	Mt	1834.90	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(1033.0(20.0 - 2.0))}$$

$$= 773.837 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2066.000	mm
Nozzle Outside Diameter	(d) =	323.850	mm
Vessel Thickness used	(T) =	38.000	mm
Nozzle Thickness used	(t) =	12.490	mm
T / t	=	3.042	
d / t	=	25.928	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.156$$

Nr/P	=	0.085
Mr/P	=	0.095
M0/P	=	0.090
N0/P	=	0.216



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N4-Shell Inlet

Nozl: 19 9:47am Apr 20,2026

MrD/Mc = 0.169
 NrDL/Mc = 0.124
 MØd/Mc = 0.190
 NØDL/Mc = 0.145
 MrD/Ml = 0.132
 NrDL/Ml = 0.103
 MØD/Ml = 0.140
 NØDL/Ml = 0.301

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-22	22	22	-22
Normal Pressure Stress	63	65	63	65
Outplane Stress Summary	44	75	106	46

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-21	21	21	-21
Inplane Pressure Stress	31	31	31	31
Inplane Stress Summary	17	42	66	5

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	44	75	106	46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	-3	-3	3	3



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N4-Shell Inlet

Page 176 of 264

Nozl: 19 9:47am Apr 20,2026

Outplane Bending (Mc)	-30	30	30	-30
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31

Outplane Stress Summary	8	53	78	0
-------------------------	---	----	----	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-27	27	27	-27
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	65	63	65

Inplane Stress Summary	42	82	104	34
------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	43	82	104	35
------------------------	----	----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-8	-8	8	8
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36

Hoop Stress Summary	28	31	44	47
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-13	-13	13	13



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N4-Shell Inlet

Page 177 of 264

Nozl: 19 9:47am Apr 20,2026

Axial Bending (ML)	-120	120	120	-120
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	-70	79	198	-134

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	11	11	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	101	82	199	182

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	33	36	33	36
Hoop Stress Summary	32	35	40	43

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	45	-45	45	-45
Axial Membrane (Mc)	-13	-13	13	13
Axial Bending (Mc)	-160	160	160	-160
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	16	16	16	16
Axial Stress Summary	-110	119	238	-174

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	11	11

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 178 of 264
 Nozzle Calcs.: N4-Shell Inlet Nozl: 19 9:47am Apr 20,2026

Two * Max Shear Stress 143 120 239 218

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 323.85 + 2 \cdot 1.65 \cdot \sqrt{1023.0(20.0 - 2.0)}$
 $= 771.654 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2046.000 mm
 Nozzle Outside Diameter (d) = 543.850 mm
 Vessel Thickness used (T) = 18.000 mm
 Nozzle Thickness used (t) = 122.490 mm
 $T / t = 0.147$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.834$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.063
 Mr/P = 0.112
 M0/P = 0.033
 N0/P = 0.053
 MrD/Mc = 0.272
 NrDL/Mc = 0.102
 M0d/Mc = 0.082
 N0DL/Mc = 0.074
 MrD/Ml = 0.085
 NrDL/Ml = 0.079
 M0D/Ml = 0.025
 N0DL/Ml = 0.068

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	135	137	135	137
Outplane Stress Summary	137	133	168	121

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	46	-46	46	-46
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-36	36	36	-36
Inplane Pressure Stress	67	67	67	67
Inplane Stress Summary	76	56	160	-5



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N4-Shell Inlet

Nozl: 19 9:47am Apr 20,2026

Page 179 of 264

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			2	2	2	2
Shear Stress Summary			3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			137	133	168	127

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			3	3	3	3
Outplane Bending (P)			13	-13	13	-13
Outplane Membrane (Mc)			-5	-5	5	5
Outplane Bending (Mc)			-34	34	34	-34
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			67	67	67	67
Outplane Stress Summary			44	87	125	27

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			4	4	4	4
Inplane Bending (P)			46	-46	46	-46
Inplane Membrane (Mc)			-7	-7	7	7
Inplane Bending (Mc)			-116	116	116	-116
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			135	137	135	137
Inplane Stress Summary			62	204	310	-13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Torsional Shear (Mt)			2	2	2	2
Shear Stress Summary			1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 180 of 264
 Nozzle Calcs.: N4-Shell Inlet Nozl: 19 9:47am Apr 20,2026

Two * Max Shear Stress 62 204 310 42

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	65	63	65	63	65	63	65
Circ. Pl (SUS)	-5	-5	11	11	-2	-2	4	4
Circ. Q (SUS)	-14	14	30	-30	-19	19	35	-35
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-1	-1	3	3	0	0	6	6
Long. Q (SUS)	-13	13	29	-29	-22	22	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	63.0	65.0	63.0	65.0	63.0	65.0	63.0	65.0
Pm+Pl (SUS)	58.0	60.0	74.0	76.0	61.0	63.0	67.0	69.0
Pm+Pl+Q (Total)	44.1	74.1	104.1	46.1	42.1	82.1	102.1	35.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	138.00	Passed
Pm+Pl (SUS)	76.00	207.00	Passed
Pm+Pl+Q (TOTAL)	104.10	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	135	137	135	137	135	137	135	137
Circ. Pl (SUS)	-1	-1	7	7	-3	-3	11	11
Circ. Q (SUS)	3	-3	23	-23	-70	70	162	-162
Long. Pm (SUS)	67	67	67	67	67	67	67	67
Long. Pl (SUS)	-1	-1	9	9	-2	-2	8	8
Long. Q (SUS)	10	-10	82	-82	-21	21	47	-47
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	135.0	137.0	135.0	137.0	135.0	137.0	135.0	137.0
Pm+Pl (SUS)	134.0	136.0	142.0	144.0	132.0	134.0	146.0	148.0
Pm+Pl+Q (Total)	137.1	133.1	165.5	127.1	62.2	204.0	308.0	42.2



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 181 of 264
 Nozzle Calcs.: N4-Shell Inlet Nozl: 19 9:47am Apr 20,2026

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	137.00	138.00	Passed
Pm+Pl (SUS)	148.00	207.00	Passed
Pm+Pl+Q (TOTAL)	308.02	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	33	36	33	36	33	36	33	36
Circ. Pl (SUS)	-5	-5	11	11	0	0	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-12	-12	14	14	-12	-12	14	14
Long. Q (SUS)	-75	75	165	-165	-115	115	205	-205
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	33.0	36.0	33.0	36.0	33.0	36.0	33.0	36.0
Pm+Pl (SUS)	28.2	31.1	44.3	47.2	33.1	36.1	39.4	42.3
Pm+Pl+Q (Total)	101.0	81.0	195.2	182.4	144.5	119.4	235.5	217.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	36.00	118.00	Passed
Pm+Pl (SUS)	47.23	177.00	Passed
Pm+Pl+Q (TOTAL)	235.51	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 182 of 264
 Nozzle Calcs.: N2 Tube outlet Nozl: 20 9:47am Apr 20,2026

Input, Nozzle Desc: N2 Tube outlet

From: 50

Pressure for Reinforcement Calculations	P	5293.880	KPa.
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2024.00	mm
Shell Finished (Minimum) Thickness	t	46.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		18225.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

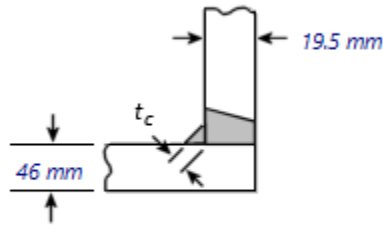
Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		36.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	19.5000	mm
Flange Material		SA-182 F304	
Hub Height of Integral Nozzle	h	70.0000	mm
Height of Beveled Transition	L'	82.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	500.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		44.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		a	
Flange Class		600	
Flange Grade		GR 2.1	
Flange Series		B	

The Pressure Design option was Overall MAWP.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 183 of 264
 Nozzle Calcs.: N2 Tube outlet Nozl: 20 9:47am Apr 20,2026



UW-16 (a)

Reinforcement CALCULATION, Description: N2 Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 36.000 in.
 Actual Thickness Used in Calculation 0.768 in.

Note:
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5294 \cdot 1014 / (138 \cdot 1 - 0.6 \cdot 5294)$
 $= 39.8180 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5294 \cdot 457 / (138 \cdot 1 + 0.4 \cdot 5294)$
 $= 17.2751 \text{ mm}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 160 / 0.5773$
 $= 277.9942 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1750.7999	mm
Parallel to Vessel Wall, opening length	d	875.4000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	110.0000	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$



PV Elite 28

Licensee: SL User

FileName : 106-C-Cladding on Both Side

Page 184 of 264

Nozzle Calcs.: N2 Tube outlet

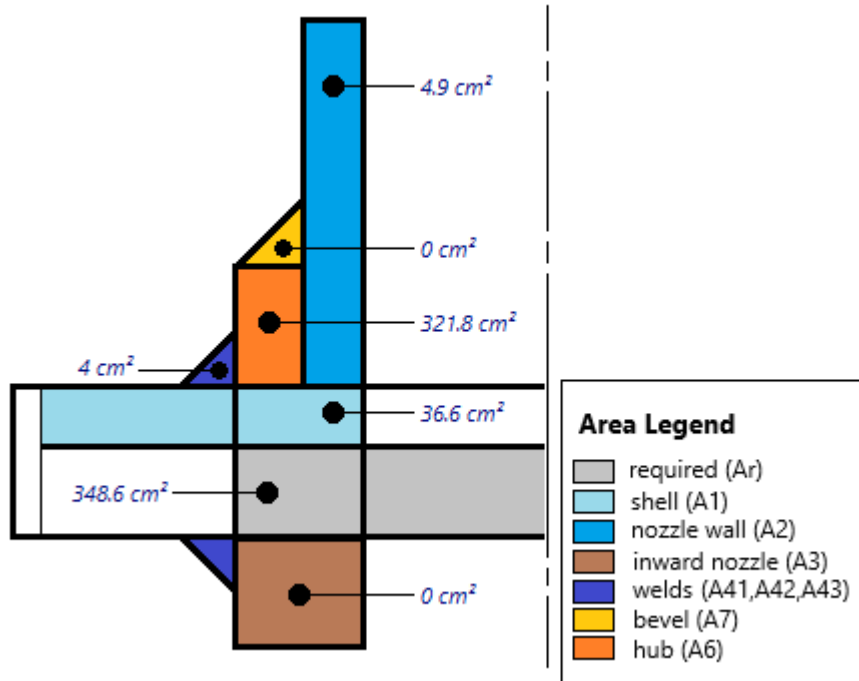
Nozl: 20 9:47am Apr 20,2026

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)

= min(1, 1)

= 1.000



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	348.567	NA	NA
Area in Shell	A1	36.609	NA	NA
Area in Nozzle Wall	A2	4.895	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		4.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub + Bevel	A6	321.783	NA	NA
TOTAL AREA AVAILABLE	Atot	367.287	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

= (d • tr•F + 2 • tn • tr•F • (1-fr1)) UG-37(c)

= (875•39.8•1+2•180•39.8•1•(1-1))

= 348.567 cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

= d(E1•t - F•tr) - 2 • tn(E1•t - F•tr) • (1 - fr1)

= 875 (1 • 44 - 1 • 39.8) - 2 • 180

(1 • 44 - 1 • 39.8) • (1 - 1)

= 36.609 cm²



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 185 of 264
 Nozzle Calcs.: N2 Tube outlet Nozl: 20 9:47am Apr 20,2026

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_r^2$$

$$= (2 \cdot 110) (19.5 - 17.3) 1$$

$$= 4.895 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_r^2 + (W_i - c_{an}/0.707)^2 \cdot f_r^2$$

$$= 20^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 4.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_r^2$$

$$= (2 \cdot \min(110, 500, 70)) \cdot (180 - 19.5) \cdot 1$$

$$= 321.783 \text{ cm}^2 \text{ Includes Bevel Area of } 97.083 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 17.2751 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 41.8180 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 41.8180 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[8.33, \max(41.8, 3.5)]$$

$$= 8.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(17.3, 8.33)$$

$$= 17.2751 \text{ mm}$$

Available Nozzle Neck Thickness = 19.5000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 75.7,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 269.3 N./mm ²	Passed
Occasional	: 58.1,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 11.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 19.5, tr = 17.3, c = 0 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.89, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-14 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 46, tr = 39.8, c = 2 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.9, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	15 °C
Min Metal Temp. at Required thickness (UCS 66.1)	9 °C



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 186 of 264
 Nozzle Calcs.: N2 Tube outlet Nozl: 20 9:47am Apr 20,2026

Gov. MDMT of the nozzle to shell joint welded assembly : 9 °C

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 160 / 0.5773$
 $= 277.9942 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1313.0999 mm
 Parallel to Vessel Wall, opening length d 656.5499 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 110.0000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		232.390	232.390
Area Available in Shell A1		18.304	18.304
Area Available in Nozzle Wall A2		4.895	4.895
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		4.000	4.000
Area Available in Pad A5		0.000	0.000
Area Available in Hub A6		321.783	321.783
TOTAL AREA AVAILABLE Atot		348.982	348.982

The area available without a pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N2 Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld 6.0000 = Min per Code 14.1400 = $0.7 * W_o$ mm

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5426 KPa.

The Drop for this Nozzle is : 109.1644 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 655.1644 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2 Tube outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	2.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2024.000	mm
Shell Thickness	46.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 187 of 264
 Nozzle Calcs.: N2 Tube outlet Nozl: 20 9:47am Apr 20,2026

Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	0.0000	mm	
Nozzle Diameter	1235.400	mm	
Nozzle Thickness	180.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-266 2		
Nozzle Cold S.I. Allowable	SNmc	138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:
External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5293.88	KPa.
Radial Load	P	-8260.24	Kgf
Circumferential Shear	Vc	5965.75	Kgf
Longitudinal Shear	Vl	5965.75	Kgf
Circumferential Moment	Mc	16514.19	Kg-m.
Longitudinal Moment	Ml	16514.19	Kg-m.
Torsional Moment	Mt	18991.26	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.584) is > 0.5
check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.863) is < 10
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 1235.4 + 2 \cdot 1.65 \cdot \sqrt{1036.0(46.0 - 2.0)}$
 $= 1939.963 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2072.000 mm
Nozzle Outside Diameter	(d) =	1235.400 mm
Vessel Thickness used	(T) =	44.000 mm
Nozzle Thickness used	(t) =	180.000 mm
T / t	=	0.244
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.092$$

Nr/P	=	0.050
Mr/P	=	0.079
M0/P	=	0.026
N0/P	=	0.057
MrD/Mc	=	0.255
NrDL/Mc	=	0.090
M0d/Mc	=	0.076
N0DL/Mc	=	0.084



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N2 Tube outlet

Nozl: 20 9:47am Apr 20,2026

MrD/M1 = 0.069
 NrDL/M1 = 0.046
 MØD/M1 = 0.021
 NØDL/M1 = 0.060

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	119	124	119	124
Outplane Stress Summary	115	125	140	116

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	59	59	59	59
Inplane Stress Summary	50	66	112	17

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	115	125	140	116

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-5	-5	5	5
Outplane Bending (Mc)	-30	30	30	-30
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	59	59	59	59



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N2 Tube outlet

Nozl: 20 9:47am Apr 20,2026

Page 189 of 264

Outplane Stress Summary	31	80	105	30
-------------------------	----	----	-----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-103	103	103	-103
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	119	124	119	124

Inplane Stress Summary	31	204	250	9
------------------------	----	-----	-----	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	32	204	250	30
------------------------	----	-----	-----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-4	-4	4	4
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	19	12	19

Hoop Stress Summary	11	17	19	25
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-1	-1	1	1
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	5	5	9	5
----------------------	---	---	---	---



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 Nozzle Calcs.: N2 Tube outlet

Nozl: 20 9:47am Apr 20,2026

Page 190 of 264

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (V1)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	11	17	19	25

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-5	-5	5	5
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	19	12	19
Hoop Stress Summary	9	15	20	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	D1
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	0	0	0	0
Axial Membrane (Mc)	-1	-1	1	1
Axial Bending (Mc)	-5	5	5	-5
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	1	9	13	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	D1
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (V1)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	D1
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	9	15	20	27

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 191 of 264
 Nozzle Calcs.: N2 Tube outlet Nozl: 20 9:47am Apr 20,2026

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	119	124	119	124	119	124	119	124
Circ. Pl (SUS)	-2	-2	6	6	-4	-4	8	8
Circ. Q (SUS)	-2	2	14	-14	-84	84	122	-122
Long. Pm (SUS)	59	59	59	59	59	59	59	59
Long. Pl (SUS)	-1	-1	5	5	-3	-3	7	7
Long. Q (SUS)	-8	8	46	-46	-24	24	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	119.0	124.0	119.0	124.0	119.0	124.0	119.0	124.0
Pm+Pl (SUS)	117.0	122.0	125.0	130.0	115.0	120.0	127.0	132.0
Pm+Pl+Q (Total)	115.0	124.0	139.0	116.0	32.6	204.0	249.0	30.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.00	138.00	Passed
Pm+Pl (SUS)	132.00	207.00	Passed
Pm+Pl+Q (TOTAL)	249.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	19	12	19	12	19	12	19
Circ. Pl (SUS)	-2	-2	6	6	-3	-3	7	7
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-1	-1	1	1	-1	-1	1	1
Long. Q (SUS)	-1	1	1	-1	-5	5	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	12.0	19.0	12.0	19.0	12.0	19.0	12.0	19.0
Pm+Pl (SUS)	10.0	17.0	18.0	25.0	9.0	16.0	19.0	26.0
Pm+Pl+Q (Total)	10.0	17.0	18.0	25.0	9.0	16.0	19.0	26.0

Nozzle Stress Summation Comparison (N./mm²):

Type of	Max. S.I.	S.I. Allowable	Result
---------	-----------	----------------	--------



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side Page 192 of 264
Nozzle Calcs.: N2 Tube outlet Nozl: 20 9:47am Apr 20,2026

Stress Int.			
Pm (SUS)	19.00	138.00	Passed
Pm+Pl (SUS)	26.00	207.00	Passed
Pm+Pl+Q (TOTAL)	26.00	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 193 of 264
 Nozzle Calcs.: V1 Nozl: 21 9:47am Apr 20,2026

Input, Nozzle Desc: V1

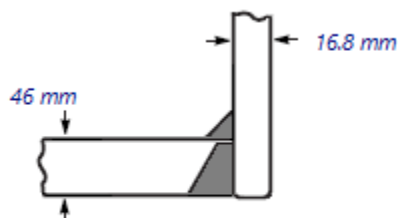
From: 50

Pressure for Reinforcement Calculations	P	5293.880	KPa.
Temperature for Internal Pressure	Temp	200	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2024.00	mm
Shell Finished (Minimum) Thickness	t	46.0000	mm
Shell Internal Corrosion Allowance	c	2.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		17725.53	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm
Flange Material		SA-182 F304	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 2.1	

The Pressure Design option was Overall MAWP.





PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 194 of 264
 Nozzle Calcs.: V1 Nozl: 21 9:47am Apr 20,2026

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5294 \cdot 1014 / (138 \cdot 1 - 0.6 \cdot 5294)$
 $= 39.8180 \text{ mm}$

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left(\frac{P}{S_n \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 25.4 \left(\exp \left(\frac{5294}{138 \cdot 1} \right) - 1 \right)$
 $= 0.9934 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	172.3280	mm
Parallel to Vessel Wall	Rn+tn+t	86.1640	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	41.9100	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: V1.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.9934	mm
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 41.8180	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 41.8180	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.8, \max(41.8, 3.5)]$
 $= 4.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(0.99, 4.8)$
 $= 4.8000 \text{ mm}$

Available Nozzle Neck Thickness = 16.7640 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 16.8, tr = 0.99, c = 0 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.059$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: V1



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side Page 195 of 264
Nozzle Calcs.: V1 Nozl: 21 9:47am Apr 20,2026

Intermediate Calc. for nozzle/shell Welds Tmin 16.7640 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.8787 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 346.8787 mm

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 196 of 264
 Case: 1 9:47a Apr 20,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell

Shell Maximum Design Pressure	Psd,max	920.00 KPa.
Shell Maximum Operating Pressure	Psox,max	920.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	2.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2024.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4950.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	2.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2024.000 mm
Channel Design Temperature	TEMPC	200.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5285
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13090.00 mm
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	144.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	986.713 mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 197 of 264
 Case: 1 9:47a Apr 20,2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	28995.0 cm ²
Perimeter of Tube Layout	Cp	6328.5 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	200.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	205.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2120.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

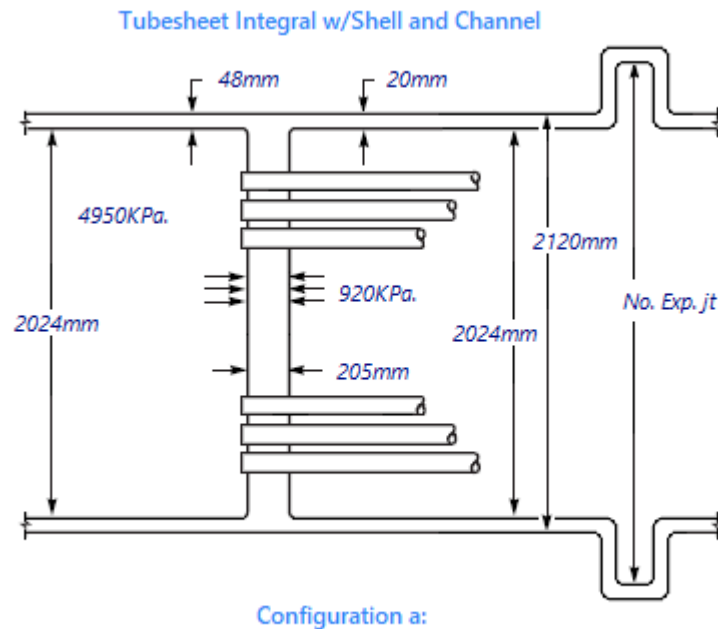
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	610.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	-102.0000 KPa.
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 198 of 264
 Case: 1 9:47a Apr 20,2026



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19120E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
 Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
 TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19120E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Note:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 199 of 264
 Case: 1 9:47a Apr 20,2026

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (5052 \cdot (19/2 - 0)) / (126 \cdot 1 + 0.4 \cdot 5052)$
 $= 0.3759 + 0.0000 = 0.3759 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-1 at 200.00 °C
 Elastic Modulus for Material 178957744.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	13090.00	11.55	50.0000	0.0082522	73.24

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 73.2) / (3 \cdot 11.5) = 8458 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3677	19.05	13090.00	51.81	50.0000	0.0004098	35.75

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 35.8) / (3 \cdot 51.8) = 920 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3759 mm
Allowable Internal Pressure at Corroded thickness	23450.25 KPa.
Required Internal Design Pressure	5052.00 KPa.
Allowable External Pressure at Corroded thickness	8457.97 KPa.
Required External Design Pressure	920.00 KPa.
Required Thickness due to Shell Side pressure	0.3677 mm

Detailed Results for load Case D1 un-corr. (Psd,min + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= l_{tx} / h$ (modified for corrosion if present)
 $= 0/205$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:
 $= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(19 - 2 \cdot 1.65 (182349680/191201920) \cdot (126/138) \cdot (0); 19 - 2 \cdot 1.65)$
 $= 19.0500 \text{ mm}$

Equivalent outer tube limit circle diameter $[Do]$:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 987 + 19$
 $= 1992.476 \text{ mm}$

Determine the basic ligament efficiency for shear $[\mu]$:
 $= (p - dt) / p$
 $= (25.4 - 19) / 25.4$
 $= 0.250$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 200 of 264
 Case: 1 9:47a Apr 20,2026

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 1992 / 2$$

$$= 996.2380 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 25.4 / \sqrt{1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1992 \cdot 25.4) / (\pi \cdot 1992^2) }$$

$$= 25.4000 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= as / ao$$

$$= 1012 / 996$$

$$= 1.0158$$

Ratio [pc]:

$$= ac / ao$$

$$= 1012 / 996$$

$$= 1.0158$$

Compute parameter [xt]:

$$= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 5285 / ((19 - 2 \cdot 1.65) / (2 \cdot 996))^2$$

$$= 0.6698$$

Determine parameter [xs]:

$$= 1 - Nt / (dt / (2 \cdot ao))^2$$

$$= 1 - 5285 / (19 / (2 \cdot 996))^2$$

$$= 0.5169$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts (D_s + ts) Es / L$$

$$= \pi \cdot 20 (2024 + 20) 194566320 / 13090$$

$$= 194663.8281 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt (Dt - tt) Et / L$$

$$= \pi \cdot 1.65 (19 - 1.65) 182349680 / 13090$$

$$= 128.1283 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= Ks / (Nt \cdot Kt)$$

$$= 194664 / (5285 \cdot 128)$$

$$= 0.28747$$

Rigidity Ratio [J]:

$$= 1 / (1 + Ks / Kj)$$

$$= 1 / (1 + 194664 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2024 + 20) 20)$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 201 of 264
 Case: 1 9:47a Apr 20, 2026

= 0.0090 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.009 \cdot 194566320 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 2563097600.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2024 + 20) \cdot 2.56E+09 / 205^3 \cdot (1 + 205 \cdot 0.009 + (205 \cdot 0.009)^2 / 2)$$

$$= 16571102.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2024^2 / (4 \cdot 194566320 \cdot 20) \cdot (1 - 2024 \cdot 0.3 / (2(2024 + 20)))$$

$$= 0.22408E+00 \text{ mm/N./mm}^2$$

Channel coefficient [βc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((D_c + t_c) t_c)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2024 + 48) 48)$$

$$= 0.0058 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0058 \cdot 191201920 \cdot 48^3 / (6(1 - 0.3^2))$$

$$= 22323562496.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2024 + 48 / \cos(0)) \cdot 22.32E+09 / 205^3 \cdot (1 + 205 \cdot 0.0058 \cdot (1 + 1) / 2 + (205 \cdot 0.0058)^2 / 2 \cdot 1)$$

$$= 92770432.0 \text{ KPa}$$

Channel coefficient [δc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2024^2 / (4 \cdot 191201920 \cdot 48) \cdot (1 - 2024 \cdot 0.3 / (2(2024 + 48)))$$

$$= 0.95234E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 8.070867$; $\mu^* = 0.250000$; $E^* = 41416348 \text{ KPa}$.
 $E^*/E = 0.216611$; $\nu^* = 0.396457$

Note: As h/p (8.071) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*) N t \cdot E t \cdot t t (d t - t t) a o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2) 5285 \cdot 182349680 \cdot 1.65 (19 - 1.65) 996^2) / (41416348 \cdot 13090 \cdot 205^3))^{1/4}$$

$$= 3.3023$$

Values from Table 4.18.3

$$Z_d = 0.041657 ; Z_v = 0.089090 ; Z_m = 0.446810$$

$$Z_a = 0.306418E+01 ; Z_w = 0.089090$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 202 of 264
 Case: 1 9:47a Apr 20, 2026

$$= A / D_o$$

$$= 2120/1992$$

$$= 1.0640$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda s + \lambda c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 41416348 \cdot (16571102 + 92770432 + 191201920 \cdot \ln(1.06))$$

$$= 1.7662$$

Compute parameter [ϕ]:

$$= (1 + v^*) F$$

$$= (1 + 0.4) 1.77$$

$$= 2.4665$$

Compute parameter [Q1]:

$$= (\psi s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.02 - 1 - 2.47 \cdot 0.089) / (1 + 2.47 \cdot 0.45)$$

$$= -0.097009$$

Compute Parameter [Qz1]:

$$= (Z_d + Q_1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.042 + -0.097 \cdot 0.089) / 2 \cdot 3.3^4$$

$$= 1.9632$$

Compute parameter [Qz2]:

$$= (Z_v + Q_1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.089 + -0.097 \cdot 0.45) / 2 \cdot 3.3^4$$

$$= 2.7202$$

Compute parameter [U]:

$$= (Z_w + (\psi s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.089 + (1.02 - 1) 0.45) 3.3^4 / (1 + 2.47 \cdot 0.45)$$

$$= 5.4404$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ωs]:

$$= \psi s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 2.56E+09 \cdot 0.009 \cdot 0.22408171 (1 + 205 \cdot 0.009)$$

$$= 14914.5410 \text{ mm}^2$$

Calculate parameter [$\omega^* s$]:

$$= a o^2 (\psi s^2 - 1) \cdot (\psi s - 1) / 4 - \omega s$$

$$= 996^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 14915$$

$$= -14789.3369 \text{ mm}^2$$

Parameter [ωc]:

$$= \psi c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.02 \cdot 22.32E+09 [0.095234253 (1 + 0.0058 \cdot 1 \cdot 205) - 0.000 (1/205 + 1 \cdot 0.0058/2)]$$

$$= 27159.8477 \text{ mm}^2$$

Parameter [$\omega^* c$]:

$$= a o^2 [(\psi c^2 + 1) \cdot (\psi c - 1) / 4 - (\psi s - 1) / 2] - \omega c$$

$$= 996^2 [(1.02^2 + 1) \cdot (1.02 - 1) / 4 - (1.02 - 1) / 2] - 27160$$

$$= -27034.6445 \text{ mm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 203 of 264
 Case: 1 9:47a Apr 20, 2026

Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= -102 \{ 0.52 + 2(1 - 0.52) 0.31 + [2/0.29 \cdot (2024/1992)^2] 0.3 - [(1.02^2 - 1)/(1 \cdot 0.29)] - [(1 - 1)/(2 \cdot 1 \cdot 0.29)] [(0^2 - (2024)^2)/1992^2] \}$$

 = -291.6384 KPa.

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.67 + 2(1 - 0.67) 0.31 + 1/(1 \cdot 0.29)] 4950$$

 = 21547.9102 KPa.

Pressure [P_γ]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 5285 \cdot 128 \cdot 0 / (3.14 \cdot 996^2)$$

 = 0.0000 KPa.

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 5.44 \cdot 0 / (2 \cdot \pi \cdot 996^2)$$

 = -0.0000 KPa.

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(5.44/996^2) \cdot (-14789 \cdot -102 - -27035 \cdot 4950)$$

 = -741.8189 KPa.

Calculate the pressure [P_ω]:

$$= (U/ao^2) \cdot (\omega_s \cdot P*s - \omega_c \cdot P*c)$$

$$= (5.44/996^2) \cdot (14915 \cdot 0 - 27160 \cdot 0)$$

 = 0.0000 KPa.

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + P_{\gamma} + P_{\omega} + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.29 / (1 + 1 \cdot 0.29(1.96 + (1.02 - 1)2.72)) \cdot (-292 - 21548 + 0 + 0 + -0 + -742)$$

 = -4117.0547 KPa.

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega_s \cdot P*s - \omega_c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-14789 \cdot -102 - -27035 \cdot 4950) - (14915 \cdot 0 - 27160 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 2.47 \cdot 0.45)$$

 = 6565.1943 Kgf

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.097 + 2 \cdot 6565 / (-4117 \cdot 996^2)$$

 = -0.12852

Fm Value from Table 4.18.3 = 0.064260



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 204 of 264
 Case: 1 9:47a Apr 20, 2026

Tubesheet Bending Stress - original thickness [σ]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.064 / 0.25) \cdot (2 \cdot 996 / (205 - 0))^2 \cdot -4117$
 $= -149.9623 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$

Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.061 / 0.25) \cdot (2 \cdot 996 / (167 - 0))^2 \cdot -3943$
 $= -206.9930 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 $= h + C_{ats} + C_{atc}$
 $= 167 + 0 + 0$
 $= 166.8918 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:
 $= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$
 $= (2 \cdot 0.25 \cdot 205 / 996) \cdot 110$
 $= 11.3585 \text{ N./mm}^2$

Shear Stress is not required to be computed; [P_e] $\leq$ τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:
 $= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$
 $= (28995 \cdot -3028) / (0.25 \cdot 51.2 \cdot 6328)$
 $= -108.3693 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:
 $= H + C_{ats} + C_{atc}$
 $= 51.2 + 0 + 0$
 $= 51.2044 \text{ mm}$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:
 $= \max(H_{reqB}; H_{reqS})$
 $= \max(167; 51.2)$
 $= 166.8918 \text{ mm}$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{tmin} = -0.014$, $F_{tmax} = 1.8$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$
 $= ((-102 \cdot 0.52 - 4950 \cdot 0.67) - (-4117) \cdot -0.014) / (0.67 - 0.52)$
 $= -22.4064 \text{ N./mm}^2$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:
 $= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$
 $= ((-102 \cdot 0.52 - 4950 \cdot 0.67) - (-4117) \cdot 1.8) / (0.67 - 0.52)$
 $= 26.3439 \text{ N./mm}^2$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:
 $= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 26.3 \text{ N./mm}^2$

Allowable Tube Stress [$\sigma_{t,A}$]:
 $= S_{ot} = 125.9991 \text{ N./mm}^2$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 205 of 264
 Case: 1 9:47a Apr 20,2026

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$
 $= \max((3.25 - 0.25(0.042 + -0.13 \cdot 0.089)3.3^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2 \text{)}$

Determine the factor [rt]:
 $= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$
 $= (\sqrt{ (19^2 + (19 - 2 \cdot 1.65)^2) }) / 4$
 $= 6.1794 \text{ mm}$

Determine the factor [Ct]:
 $= \sqrt{ (2 \cdot \pi^2 \cdot Et / S_{yt})}$
 $= \sqrt{ (2 \cdot \pi^2 \cdot 182349680 / 144)}$
 $= 158.1060$

Determine the factor [Ft]:
 $= k \cdot L / r$
 $= 1 \cdot 610 / 6.18$
 $= 98.7147$

Buckling Allowable Stress [Stb]:
 $= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$
 $= (144 / 2) \cdot (1 - 98.7 / (2 \cdot 158))$
 $= 49.523 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
 $= \sigma_{t,max} \cdot \text{Tube Area}$
 $= 26.3 \cdot 0.9$
 $= 242.29 \text{ Kg}$

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (19 - 1.65) 1.65$
 $= 0.902 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 1.65 (19 - 1.65) 126$
 $= 1158.836 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$
 $= 0.85 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 126$
 $= 1140.9987 \text{ Kg}$

Weld Strength Factor [fw]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 126 / 126$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$
 $= 0.55 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 126$
 $= 738.2932 \text{ Kg}$

Minimum Length of Weld [ar]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 206 of 264
 ASME TS Calc: D Case: 1 9:47a Apr 20,2026

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd)} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 1.65 \cdot (19 - 1.65) \cdot 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 2.075 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0374 mm
 Minimum Required Groove Weld Leg [agr]: 1.0374 mm

Tube-Tubesheet Jt allowable, 1159 is >= tube strength 1159 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σ_s, m]:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 996^2 / ((2024 + 20) 20) \cdot [-4117 + (1.02^2 - 1)(-102 - 4950)] + 1012^2 \cdot 4950 / ((2024 + 20) 20)$
 $= 20.1445 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 20 mm, the minimum Shell length adjacent to the tubesheet should be 362.154 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:
 $= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$
 $= 996^2 / ((2024 + 20) 20) \cdot [-4117 + (1.02^2 - 1)(-102 - 4950)] + 1012^2 \cdot 4950 / ((2024 + 20) 20)$
 $= 20.1445 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction [σ_s, b]:
 $= 6 \cdot k_s / t_s^2 \cdot \{ \beta_s [\delta_s \cdot P_s + a s^2 \cdot P \cdot s / (E \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) \cdot (a o / h)^3 \cdot (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 2.56E+09 / 20^2 \cdot \{ 0.009 [0.22408171 \cdot (-102 + 1012^2 \cdot 0 / (3.89E+09))] + 6(1 - 0.4^2) / (41416348) \cdot (996 / 20)^3 \cdot (1 + 205 \cdot 0.009 / 2) [-4117 (0.089 + 0.45 \cdot -0.097) + 2 / 996^2 \cdot 0.45 \cdot 6565] \}$
 $= -142.8769 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$
 $:: |20.1| + |-143| \leq 207 \text{ N./mm}^2$
 $:: 163, \text{ must be } \leq 207 \text{ N./mm}^2$

Note:
 For a given Channel thickness of 48 mm, the minimum Channel length adjacent to the tubesheet should be 561.046 mm

Channel Membrane Stress due to Joint Interaction [σ_c, m]:
 $= a c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$
 $= 1012^2 \cdot 4950 / ((2024 + 48) 48 \cdot \cos(0))$
 $= 50.975 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction [σ_c, b]:
 $= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6(1 - \nu^2) / (E \cdot h) \cdot (a o / h)^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) + [P_e (Z_v + Z_m \cdot Q_1) + 2 / a o^2 \cdot Z_m \cdot Q_2] \}$
 $= 6 \cdot 22.32E+09 / 48^2 \cdot \{ (0.095234253 \cdot 1 \cdot 0.0058 - 0.000 \cdot 1) \cdot 4950$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 207 of 264
 Case: 1 9:47a Apr 20,2026

$$\begin{aligned}
 & -6((1 - 0.4^2) / 41416348) \cdot \\
 & (996/205)^3 \cdot (1 + 205 \cdot 0.0058/2 \cdot 1) \cdot [-4117 (0.089 + 0.45 \\
 & \cdot -0.097) + 2/996^2 \cdot 0.45 \cdot 6565] \} \\
 & = 326.9534 \text{ N./mm}^2
 \end{aligned}$$

Channel Stress Summation vs. Allowable

:: $|\sigma_{c,m}| + |\sigma_{c,b}| \leq 1.5 \cdot S_c$
 :: $|51| + |327| \leq 207 \text{ N./mm}^2$
 :: 378 must be $\leq 207 \text{ N./mm}^2$ (See Elast-Plast iteration below)

Stress/Force Summary for Loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	150.0 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	36.8 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	26.3 <=	126.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-22.4 <=	-49.5 N./mm ²	Pass
Maximum Force on any one Tube	242.3 <=	1158.8 Kgf	Pass
Axial Membrane Stress in Shell	20.1 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	163.0 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	377.9 <=	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:
 = min(Sys, SPS,s/2)
 = min(230, 461/2) = 230 N./mm²

Compute Allowable [S*c]:
 = min(Syc, SPS,c/2)
 = min(225, 450/2) = 225 N./mm²

Compute Factor [facts]:
 = min([1.4 - 0.4 * abs(σs,b) / Sstars], 1)
 = min([1.4 - 0.4 * abs(-143) / 230], 1) = 1

Compute Factor [factc]:
 = min([1.4 - 0.4 * abs(σc,b) / Sstarc], 1)
 = min([1.4 - 0.4 * abs(327) / 225], 1) = 0.82

Compute Modified Modulus [Ec]:
 = Ec * factc
 = 191201920 * 0.82 = 156550720 KPa.

Tubesheet Bending Stress - final thickness [σf]:
 = (1.5 * Fm / μ*) * (2 * ao / (h - h'g))² * Pe
 = (1.5 * 0.058 / 0.25) * (2 * 996 / (162 - 0))² * -3891
 = -206.9923 N./mm²

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:
 = h + Cats + Catc
 = 162 + 0 + 0
 = 161.8151 mm

For the Elastic-Plastic iteration, Shell Allowable = SPS,s
 ShellAllow = 460.8179 N./mm²



ASME Section VIII-2 Part 4.18.8.4 Step 12:

For the Elastic-Plastic iteration, the channel allowable stress = SPS,c
 Challow = 450.0172 N./mm²

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D1 un-corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	142.9 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	36.8 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	26.3 <=	126.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-22.4 <=	-49.5 N./mm ²	Pass
Maximum Force on any one Tube	242.3 <=	1158.8 Kgf	Pass
Axial Membrane Stress in Shell	20.1 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	163.0 <=	460.8 N./mm ²	Pass
Channel Stress (jt. inter.)	377.9 <=	450.0 N./mm ²	Pass

Detailed Results for load Case O1 un-corr. (Psox,min + Ptox,max + Th)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:
 = ltx / h (modified for corrosion if present)
 = 0/205
 = 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d*]:
 = $\max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$
 = $\max(19 - 2 \cdot 1.65 (182349680/191201920) \cdot (126/138) \cdot (0); 19 - 2 \cdot 1.65)$
 = 19.0500 mm

Equivalent outer tube limit circle diameter [Do]:
 = $2 \cdot ro + dt$
 = $2 \cdot 987 + 19$
 = 1992.476 mm

Determine the basic ligament efficiency for shear [μ]:
 = $(p - dt) / p$
 = $(25.4 - 19) / 25.4$
 = 0.250

Equivalent outer tube limit radius [ao]:
 = $Do / 2$
 = $1992/2$
 = 996.2380 mm

Effective tube pitch [p*]:
 = $p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$
 = $25.4 / \sqrt{ (1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1992 \cdot 25.4) / (\pi \cdot 1992^2)) }$
 = 25.4000 mm

Effective ligament efficiency for bending [μ*]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 209 of 264
 Case: 1 9:47a Apr 20, 2026

$$= (p^* - d^*) / p^*$$

$$= (25.4 - 19) / 25.4$$

$$= 0.2500$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 1012 / 996$$

$$= 1.0158$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 1012 / 996$$

$$= 1.0158$$

Compute parameter [xt]:

$$= 1 - Nt ((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 5285 ((19 - 2 \cdot 1.65) / (2 \cdot 996))^2$$

$$= 0.6698$$

Determine parameter [xs]:

$$= 1 - Nt (dt / (2 \cdot a_o))^2$$

$$= 1 - 5285 (19 / (2 \cdot 996))^2$$

$$= 0.5169$$

Determine the value [h'g]:

$$= 0 \text{ (For pressure plus thermal and thermal only cases)}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 20 (2024 + 20) 194566320 / 13090$$

$$= 194663.8281 \text{ Kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 1.65 (19 - 1.65) 182349680 / 13090$$

$$= 128.1283 \text{ Kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 194664 / (5285 \cdot 128)$$

$$= 0.28747$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 194664 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2024 + 20) 20)$$

$$= 0.0090 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.009 \cdot 194566320 \cdot 20^3 / (6(1 - 0.3^2))$$

$$= 2563097600.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2024 + 20) \cdot 2.56E+09 / 205^3 \cdot (1 + 205 \cdot 0.009 + (205 \cdot 0.009)^2 / 2)$$

$$= 16571102.0000 \text{ KPa}$$

Shell coefficient [δ_s]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 210 of 264
 Case: 1 9:47a Apr 20, 2026

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 2024^2 / (4 \cdot 194566320 \cdot 20) \cdot (1 - 2024 \cdot 0.3 / (2(2024 + 20)))$$

$$= 0.22408E+00 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2024 + 48)48)$$

$$= 0.0058 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_c^2))$$

$$= 0.0058 \cdot 191201920 \cdot 48^3 / (6(1 - 0.3^2))$$

$$= 22323562496.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(2024 + 48 / \cos(0)) \cdot 22.32E+09 / 205^3 \cdot (1 + 205 \cdot 0.0058 \cdot (1 + 1) / 2 + (205 \cdot 0.0058)^2 / 2 \cdot 1)$$

$$= 92770432.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot T_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 2024^2 / (4 \cdot 191201920 \cdot 48) \cdot (1 - 2024 \cdot 0.3 / (2(2024 + 48)))$$

$$= 0.95234E-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 8.070867 ; \mu^* = 0.250000 ; E^* = 41416348 \text{ KPa}$$

$$E^*/E = 0.216611 ; v^* = 0.396457$$

Note: As h/p (8.071) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - v^{*2})Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2)5285 \cdot 182349680 \cdot 1.65(19 - 1.65)996^2) / (41416348 \cdot 13090 \cdot 205^3))^{1/4}$$

$$= 3.3023$$

Values from Table 4.18.3

$$Z_d = 0.041657 ; Z_v = 0.089090 ; Z_m = 0.446810$$

$$Z_a = 0.306418E+01 ; Z_w = 0.089090$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2120 / 1992$$

$$= 1.0640$$

Compute coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 41416348 \cdot (16571102 + 92770432 + 191201920 \cdot \ln(1.06))$$

$$= 1.7662$$

Compute parameter [ϕ]:

$$= (1 + v^*)^F$$

$$= (1 + 0.4)^{1.77}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 211 of 264
 Case: 1 9:47a Apr 20,2026

= 2.4665

Compute parameter [Q1]:

= ($\rho_s - 1 - \phi \cdot Z_v$) / ($1 + \phi \cdot Z_m$)
 = (1.02 - 1 - 2.47 • 0.089) / (1 + 2.47 • 0.45)
 = -0.097009

Compute Parameter [Qz1]:

= ($Z_d + Q1 \cdot Z_w$) / 2 • X_a^4
 = (0.042 + -0.097 • 0.089) / 2 • 3.3⁴
 = 1.9632

Compute parameter [Qz2]:

= ($Z_v + Q1 \cdot Z_m$) / 2 • X_a^4
 = (0.089 + -0.097 • 0.45) / 2 • 3.3⁴
 = 2.7202

Compute parameter [U]:

= ($Z_w + (\rho_s - 1) Z_m$) X_a^4 / ($1 + \phi \cdot Z_m$)
 = (0.089 + (1.02 - 1) 0.45) 3.3⁴ / (1 + 2.47 • 0.45)
 = 5.4404

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

= 0 (config a)

Compute Parameter [y]:

= ($\alpha_{tm} (T_{tm} - T_a) - \alpha_{sm} (T_{sm} - T_a)$) L
 = (0.15299E-04 (0 - 21.1) - 0.11215E-04 (0 - 21.1)) 13090
 = -1.1286 mm

Calculate parameter [ws]:

= $\rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$
 = 1.02 • 2.56E+09 • 0.009 • 0.22408171 (1 + 205 • 0.009)
 = 14914.5410 mm²

Calculate parameter [$\omega \cdot s$]:

= $a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega s$
 = 996² (1.02² - 1) • (1.02 - 1) / 4 - 14915
 = -14789.3369 mm²

Parameter [ωc]:

= $\rho_c \cdot k_c [\delta C (C_m \Theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \Theta + h \cdot C_q \Theta \cdot h \cdot \beta C / 2)]$
 = 1.02 • 22.32E+09 [0.095234253 (1 + 0.0058 • 1 • 205) - 0.000 (1/205 + 1 • 0.0058/2)]
 = 27159.8477 mm²

Parameter [$\omega \cdot c$]:

= $a o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\rho_s - 1) / 2] - \omega c$
 = 996² [(1.02² + 1) • (1.02 - 1) / 4 - (1.02 - 1) / 2] - 27160
 = -27034.6445 mm²

Average temperature of the unperforated rim [Tr]:

= ($T' + T's + T'c$) / 3 = (21.1 + 21.1 + 21.1) / 3 = 21.1 °C

Average temperature of the shell at the tubesheet [T*s]:

= ($T's + Tr$) / 2 = (21.1 + 21.1) / 2 = 21.1 °C

Average temperature of the channel at the tubesheet [T*c]:

= ($T'c + Tr$) / 2 = (21.1 + 21.1) / 2 = 21.1 °C

Pressure [P*S]:

= $E s \cdot t s / a s \cdot (\alpha s' \cdot (T \cdot s - T_a) - \alpha' \cdot (Tr - T_a))$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 212 of 264
 Case: 1 9:47a Apr 20, 2026

$$= 194566320 \cdot 20/1012 \cdot [0.11519E-04 \cdot (21.1 - 21.1) - 0.11519E-04 \cdot (21.1 - 21.1)]$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P*C]:

$$= Ec \cdot tc / ac \cdot (\alpha c (T \cdot c - Ta) - \alpha' (Tr - Ta))$$

$$= 191201920 \cdot 48/1012 \cdot [0.11519E-04 (21.1 - 21.1) - 0.11519E-04 (21.1 - 21.1)]$$

$$= 0.0000 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 0 \{ 0.52 + 2(1 - 0.52)0.31 + [2/0.29 \cdot (2024/1992)^2]0.3 - [(1.02^2 - 1)/(1 \cdot 0.29)] - [(1 - 1)/(2 \cdot 1 \cdot 0.29)] [(0^2 - (2024)^2)/1992^2] \}$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] P_t$$

$$= [0.67 + 2(1 - 0.67)0.31 + 1/(1 \cdot 0.29)] 4950$$

$$= 21547.9102 \text{ KPa.}$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 5285 \cdot 128 \cdot -1.13 / (3.14 \cdot 996^2)$$

$$= -2403.5955 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 5.44 \cdot 0 / (2 \cdot \pi \cdot 996^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt)$$

$$= -(5.44/996^2) \cdot (-14789 \cdot 0 - 27035 \cdot 4950)$$

$$= -733.5499 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/ao^2) \cdot (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)$$

$$= (5.44/996^2) \cdot (14915 \cdot 0 - 27160 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.29 / (1 + 1 \cdot 0.29 (1.96 + (1.02 - 1)2.72)) \cdot (0 - 21548 + -2404 + 0 + -0 + -734)$$

$$= -4500.6006 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-14789 \cdot 0 - 27035 \cdot 4950) - (14915 \cdot 0 - 27160 \cdot 0)) \cdot 0.0001 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 2.47 \cdot 0.45)$$

$$= 6492.0132 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.097 + 2 \cdot 6492 / (-4501 \cdot 996^2)$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 213 of 264
 Case: 1 9:47a Apr 20, 2026

$$= -0.12551$$

Fm Value from Table 4.18.3 = 0.062757

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.063 / 0.25) \cdot (2 \cdot 996 / (205 - 0))^2 \cdot -4501$$

$$= -160.0976 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [Sigma allowed]:
 SPS (ASME Sec VIII Div 1, UG-23(e)) = 425.9965 N./mm²

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.052 / 0.25) \cdot (2 \cdot 996 / (107 - 0))^2 \cdot -3947$$

$$= -425.9952 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 107 + 0 + 0$$

$$= 106.9756 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.25 \cdot 205 / 996) \cdot 110$$

$$= 11.3585 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (28995 \cdot -3422) / (0.25 \cdot 57.6 \cdot 6328)$$

$$= -108.9247 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 57.6 + 0 + 0$$

$$= 57.5719 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(107 ; 57.6)$$

$$= 106.9756 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:
 $F_{tmin} = -0.025$, $F_{tmax} = 1.81$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((0 \cdot 0.52 - 4950 \cdot 0.67) - (-4501) \cdot -0.025) /$$

$$(0.67 - 0.52)$$

$$= -22.4148 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((0 \cdot 0.52 - 4950 \cdot 0.67) - (-4501) \cdot 1.81) /$$

$$(0.67 - 0.52)$$

$$= 31.6645 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(| \sigma_{t,1} | ; | \sigma_{t,2} |) = 31.7 \text{ N./mm}^2$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 214 of 264
 Case: 1 9:47a Apr 20, 2026

Allowable Tube Stress [σ_t, A]:
 $= 2 \cdot S_{ot} = 251.9982 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_{a^4}); 1.25)$
 $= \max((3.25 - 0.25(0.042 + -0.13 \cdot 0.089) 3.3^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2 \text{)}$

Determine the factor [rt]:
 $= (\sqrt{ \frac{dt^2 + (dt - 2 \cdot tt)^2}{4} }) / 4$
 $= (\sqrt{ \frac{19^2 + (19 - 2 \cdot 1.65)^2}{4} }) / 4$
 $= 6.1794 \text{ mm}$

Determine the factor [Ct]:
 $= \sqrt{ \frac{2 \cdot \pi^2 \cdot E_t / S_{yt}}{2 \cdot \pi^2 \cdot 182349680 / 144} })$
 $= 158.1060$

Determine the factor [Ft]:
 $= k \cdot L / r$
 $= 1 \cdot 610 / 6.18$
 $= 98.7147$

Buckling Allowable Stress [Stb]:
 $= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$
 $= (144 / 2) \cdot (1 - 98.7 / (2 \cdot 158))$
 $= 49.523 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
 $= \sigma_{t,max} \cdot \text{Tube Area}$
 $= 31.7 \cdot 0.9$
 $= 291.22 \text{ Kg}$

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (19 - 1.65) 1.65$
 $= 0.902 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 1.65 (19 - 1.65) 126$
 $= 1158.836 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$
 $= 0.85 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 126$
 $= 1140.9987 \text{ Kg}$

Weld Strength Factor [fw]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 126 / 126$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 215 of 264
 Case: 1 9:47a Apr 20, 2026

$$= 0.55 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 126$$

$$= 738.2932 \text{ Kgf}$$

Maximum Allowed Tube-Tubesheet Joint load [Lmax]:

$$= 2 \cdot F_t$$

$$= 2 \cdot 1159$$

$$= 2317.671 \text{ Kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 1.65 \cdot (19 - 1.65) 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 2.075 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.0374 mm
 Minimum Required Groove Weld Leg [agr]: 1.0374 mm

Tube-Tubesheet Jt allowable, 2318 is >= tube strength 2318 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 = SPS,s = 461 N./mm²

Axial Membrane Stress in Shell [σs,m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1) (P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 996^2 / ((2024 + 20) 20) \cdot [-4501 + (1.02^2 - 1) (0 - 4950)] + 1012^2 \cdot 4950 / ((2024 + 20) 20)$$

$$= 10.9112 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 20 mm, the minimum Shell length adjacent to the tubesheet should be 362.154 mm

Shell Membrane Stress due to Joint Interaction [σs,m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1) (P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 996^2 / ((2024 + 20) 20) [-4501 + (1.02^2 - 1) (0 - 4950)] + 1012^2 \cdot 4950 / ((2024 + 20) 20)$$

$$= 10.9112 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σs,b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E_s \cdot t_s)] + 6 (1 - \nu^2) / (E_s) (a_o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 2.56 \text{E}+09 / 20^2 \{ 0.009 [0.22408171 \cdot 0 + 1012^2 \cdot 0 / (3.89 \text{E}+09)] + 6 (1 - 0.4^2) / (41416348) (996 / 205)^3 (1 + 205 \cdot 0.009 / 2) [-4501 (0.089 + 0.45 \cdot -0.097) + 2 / 996^2 \cdot 0.45 \cdot 6492] \}$$

$$= -153.8096 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_{s,m}| + |\sigma_{s,b}| \leq \text{SPS},s$$

$$\therefore |10.9| + |-154| \leq 461 \text{ N./mm}^2$$

$$\therefore 165, \text{ must be } \leq 461 \text{ N./mm}^2$$

Note:

For a given Channel thickness of 48 mm, the minimum Channel length adjacent to the tubesheet should be 561.046 mm

Channel Membrane Stress due to Joint Interaction [σc,m]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 216 of 264
 Case: 1 9:47a Apr 20,2026

$$= ac^2 \cdot Pt / ((Dc + tc) tc \cdot \cos(\theta))$$

$$= 1012^2 \cdot 4950 / ((2024 + 48) 48 \cdot \cos(0))$$

$$= 50.975 \text{ N./mm}^2$$

Channel Bending Stress due to Joint Interaction [σc,b]:

$$= 6 \cdot kc / tc^2 \cdot \{ (\delta c \cdot Cm \Delta \cdot \beta c - \zeta c \cdot Cm \Theta) \cdot Pt$$

$$- 6 ((1 - \nu^2) / E^*) \cdot (ao / h)^3 \cdot (Cm \Theta +$$

$$h \cdot \beta c / 2 \cdot Cm \Delta) \cdot [Pe (Zv + Zm \cdot Q1) + 2 / ao^2 \cdot Zm \cdot Q2] \}$$

$$= 6 \cdot 22.32E+09 / 48^2 \cdot \{ (0.095234253 \cdot 1 \cdot 0.0058 - 0.000 \cdot 1) \cdot 4950$$

$$- 6 ((1 - 0.4^2) / 41416348) \cdot$$

$$(996 / 205)^3 \cdot (1 + 205 \cdot 0.0058 / 2 \cdot 1) \cdot [-4501 (0.089 + 0.45$$

$$\cdot -0.097) + 2 / 996^2 \cdot 0.45 \cdot 6492] \}$$

$$= 350.5308 \text{ N./mm}^2$$

Channel Stress Summation vs. Allowable

$$:: | \sigma_{c,m} | + | \sigma_{c,b} | \leq \text{SPS}_{c,c}$$

$$:: | 51 | + | 351 | \leq 450 \text{ N./mm}^2$$

$$:: 402 \text{ must be } \leq 450 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase O1 un-corr. (Psox,min + Ptox,max + Th):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	160.1 <=	426.0 N./mm ²	Pass
Tubesheet Shear Stress	40.2 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	31.7 <=	252.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-22.4 <=	-49.5 N./mm ²	Pass
Maximum Force on any one Tube	291.2 <=	2317.7 Kgf	Pass
Axial Membrane Stress in Shell	10.9 <=	460.8 N./mm ²	Pass
Shell Stress (jt. inter.)	164.7 <=	460.8 N./mm ²	Pass
Channel Stress (jt. inter.)	401.5 <=	450.0 N./mm ²	Pass

Thickness Results for Loadcase O1 un-corr. (Psox,min + Ptox,max + Th):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	106.976	205.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.037	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.037	1.650	Ok

Detailed Results for load Case D1 corr. (Psd,min + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/179$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt (Et / E) \cdot (St / S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 1.65 (182349680 / 191201920) \cdot$$

$$(126 / 138) \cdot (0); 19 - 2 \cdot 1.65)$$

$$= 19.0500 \text{ mm}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 217 of 264
 Case: 1 9:47a Apr 20,2026

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 987 + 19 \\ &= 1992.476 \text{ mm} \end{aligned}$$

Determine the basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (25.4 - 19) / 25.4 \\ &= 0.250 \end{aligned}$$

Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= Do / 2 \\ &= 1992 / 2 \\ &= 996.2380 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) } \\ &= 25.4 / \sqrt{1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1992 \cdot 25.4) / (\pi \cdot 1992^2) } \\ &= 25.4000 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (25.4 - 19) / 25.4 \\ &= 0.2500 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= as / ao \\ &= 1014 / 996 \\ &= 1.0178 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= ac / ao \\ &= 1014 / 996 \\ &= 1.0178 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\ &= 1 - 5285 ((19 - 2 \cdot 1.65) / (2 \cdot 996))^2 \\ &= 0.6698 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt(dt / (2 \cdot ao))^2 \\ &= 1 - 5285 (19 / (2 \cdot 996))^2 \\ &= 0.5169 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((hg - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((0 - 13); 0) \\ &= 0.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot ts(D_s + ts) E_s / L \\ &= \pi \cdot 18 (2028 + 18) 194566320 / 13116 \\ &= 175021.2500 \text{ Kg/mm} \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned} &= \pi \cdot tt(D_t - tt) E_t / L \\ &= \pi \cdot 1.65 (19 - 1.65) 182349680 / 13116 \\ &= 127.8743 \text{ Kg/mm} \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned} &= K_s / (Nt \cdot K_t) \\ &= 175021 / (5285 \cdot 128) \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 218 of 264
 Case: 1 9:47a Apr 20,2026

$$= 0.25898$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s/K_j)$$

$$= 1 / (1 + 175021/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((Ds + ts)ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2028 + 18)18)$$

$$= 0.0095 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.0095 \cdot 194566320 \cdot 18^3 / (6(1 - 0.3^2))$$

$$= 1968607104.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(Ds + ts) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2028 + 18) \cdot 1968607104 / 179^3 \cdot (1 + 179 \cdot 0.0095 + (179 \cdot 0.0095)^2 / 2)$$

$$= 17415332.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= Ds^2 / (4 \cdot E_s \cdot ts) \cdot (1 - Ds \cdot \nu_s / (2(Ds + ts)))$$

$$= 2028^2 / (4 \cdot 194566320 \cdot 18) \cdot (1 - 2028 \cdot 0.3 / (2(2028 + 18)))$$

$$= 0.24992\text{E}+00 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / ((Dc + tc)tc)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((2028 + 46)46)$$

$$= 0.0059 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_c^2))$$

$$= 0.0059 \cdot 191201920 \cdot 46^3 / (6(1 - 0.3^2))$$

$$= 2006066952.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(Dc + tc / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2028 + 46 / \cos(0)) \cdot 2006066952 / 179^3 \cdot (1 + 179 \cdot 0.0059 \cdot (1 + 1) / 2 + (179 \cdot 0.0059)^2 / 2 \cdot 1)$$

$$= 113531960.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= Dc^2 / (4 \cdot E_c \cdot Tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc)))$$

$$= 2028^2 / (4 \cdot 191201920 \cdot 46) \cdot (1 - 2028 \cdot 0.3 / (2(2028 + 46)))$$

$$= 0.99751\text{E}-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 7.047244 ; \mu^* = 0.250000 ; E^* = 41416348 \text{ KPa}$$

$$E^*/E = 0.216611 ; \nu^* = 0.396457$$

Note: As h/p (7.047) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2)5285 \cdot 182349680 \cdot 1.65$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 219 of 264
 Case: 1 9:47a Apr 20, 2026

$$\begin{aligned} & (19 - 1.65)996^2 / (41416348 \cdot 13116 \cdot 179^3)^{1/4} \\ & = 3.6541 \end{aligned}$$

Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.031117 ; Z_v = 0.074231 ; Z_m = 0.400947 \\ Z_a &= 0.459497E+01 ; Z_w = 0.074231 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 2120/1992 \\ &= 1.0640 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.4) / 41416348 \cdot (17415332 + 113531960 + \\ &\quad 191201920 \cdot \ln(1.06)) \\ &= 2.0811 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + \nu^*) F \\ &= (1 + 0.4) 2.08 \\ &= 2.9062 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.02 - 1 - 2.91 \cdot 0.074) / (1 + 2.91 \cdot 0.4) \\ &= -0.091399 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ &= (0.031 + -0.091 \cdot 0.074) / 2 \cdot 3.65^4 \\ &= 2.1691 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ &= (0.074 + -0.091 \cdot 0.4) / 2 \cdot 3.65^4 \\ &= 3.3505 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\rho_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ &= (0.074 + (1.02 - 1) 0.4) 3.65^4 / (1 + 2.91 \cdot 0.4) \\ &= 6.7010 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \rho_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 1.02 \cdot 1968607104 \cdot 0.0095 \cdot 0.24992131 (1 + 179 \cdot 0.0095) \\ &= 12787.3945 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= a o^2 (\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s \\ &= 996^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 12787 \\ &= -12628.2441 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) -$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 220 of 264
 Case: 1 9:47a Apr 20, 2026

$$\begin{aligned} & \zeta c(Cm\theta + h \cdot Cq\theta \cdot h \cdot \beta C / 2)) \\ & = 1.02 \cdot 20.06E+09 [0.099750735 (1 + 0.0059 \cdot 1 \cdot 179) - \\ & \quad 0.000 (1/179 + 1 \cdot 0.0059/2)) \\ & = 24616.3613 \text{ mm}^2 \end{aligned}$$

Parameter [$\omega \cdot c$]:

$$\begin{aligned} & = ao^2 [(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\rho s - 1) / 2] - \omega c \\ & = 996^2 [(1.02^2 + 1) \cdot (1.02 - 1) / 4 - (1.02 - 1) / 2] - 24616 \\ & = -24457.2129 \text{ mm}^2 \end{aligned}$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$\begin{aligned} & = Ps \{ xs + 2(1 - xs) vt + [2 / Kst \cdot (Ds/Do)^2] vs - \\ & \quad [(\rho s^2 - 1) / (J \cdot Kst)] - [(1 - J) / (2J \cdot Kst)] [(Dj^2 - (Ds)^2) / Do^2] \} \\ & = -102 \{ 0.52 + 2(1 - 0.52) 0.31 + [2 / 0.26 \cdot \\ & \quad (2028 / 1992)^2] 0.3 - [(1.02^2 - 1) / (1 \cdot 0.26)] - \\ & \quad [(1 - 1) / (2 \cdot 1 \cdot 0.26)] [(0^2 - (2028)^2) / 1992^2] \} \\ & = -313.9204 \text{ KPa.} \end{aligned}$$

Pressure [$P't$]:

$$\begin{aligned} & = [xt + 2(1 - xt) vt + 1 / (J \cdot Kst)] Pt \\ & = [0.67 + 2(1 - 0.67) 0.31 + 1 / (1 \cdot 0.26)] 4950 \\ & = 23442.4355 \text{ KPa.} \end{aligned}$$

Pressure [P_y]:

$$\begin{aligned} & = Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\ & = 5285 \cdot 128 \cdot 0 / (3.14 \cdot 996^2) \\ & = 0.0000 \text{ KPa.} \end{aligned}$$

Pressure [P_w]:

$$\begin{aligned} & = -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot ao^2) \\ & = -0 \cdot 6.7 \cdot 0 / (2 \cdot \pi \cdot 996^2) \\ & = -0.0000 \text{ KPa.} \end{aligned}$$

Calculate the pressure [$Prim$]:

$$\begin{aligned} & = - (U / ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) \\ & = - (6.7 / 996^2) \cdot (-12628 \cdot -102 - -24457 \cdot 4950) \\ & = -826.0808 \text{ KPa.} \end{aligned}$$

Calculate the pressure [$P\omega$]:

$$\begin{aligned} & = (U / ao^2) \cdot (\omega s \cdot P \cdot s - \omega c \cdot P \cdot c) \\ & = (6.7 / 996^2) \cdot (12787 \cdot 0 - 24616 \cdot 0) \\ & = 0.0000 \text{ KPa.} \end{aligned}$$

Determine the effective pressure [Pe]:

$$\begin{aligned} & = J \cdot Kst / (1 + J \cdot Kst(Qz1 + (\rho s - 1) Qz2)) \cdot \\ & \quad (P's - P't + P_y + P_w + P_w + Prim) \\ & = 0.1000E+01 \cdot 0.26 / (1 + 1 \cdot 0.26 (2.17 + (1.02 - \\ & \quad 1) 3.35)) \cdot (-314 - 23442 + 0 + 0 + -0 + -826) \\ & = -4036.3914 \text{ KPa.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$\begin{aligned} & = [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega s \cdot P \cdot s - \omega c \cdot P \cdot c)) + \\ & \quad W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\ & = [((-12628 \cdot -102 - -24457 \cdot 4950) - \\ & \quad (12787 \cdot 0 - 24616 \cdot 0)) \cdot 0.0001 + \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 221 of 264
 Case: 1 9:47a Apr 20, 2026

$$0 \cdot 0 / (2 \cdot \pi) / (1 + 2.91 \cdot 0.4) \\ = 5762.4009 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\ = -0.091 + 2 \cdot 5762 / (-4036 \cdot 996^2) \\ = -0.11961$$

Fm Value from Table 4.18.3 = 0.059805

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.06 / 0.25) \cdot (2 \cdot 996 / (179 - 0))^2 \cdot -4036 \\ = -179.4687 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138 = 207 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\ = (1.5 \cdot 0.058 / 0.25) \cdot (2 \cdot 996 / (163 - 0))^2 \cdot -3962 \\ = -206.9947 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc \\ = 163 + 13 + 13 \\ = 189.2279 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\ = (2 \cdot 0.25 \cdot 179 / 996) \cdot 110 \\ = 9.9179 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress - final thickness [τ_f]:

$$= (Ap \cdot Pe) / (\mu \cdot h \cdot Cp) \\ = (28995 \cdot -3114) / (0.25 \cdot 51.8 \cdot 6328) \\ = -110.2120 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + Cats + Catc \\ = 51.8 + 13 + 13 \\ = 77.7830 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(HreqB; HreqS) \\ = \max(189; 77.8) \\ = 189.2279 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:
 Ftmin = -0.19, Ftmax = 1.98

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs) \\ = ((-102 \cdot 0.52 - 4950 \cdot 0.67) - (-4036) \cdot -0.19) / \\ (0.67 - 0.52) \\ = -27.0388 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs)$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 222 of 264
 Case: 1 9:47a Apr 20, 2026

$$= ((-102 \cdot 0.52 - 4950 \cdot 0.67) - (-4036)) \cdot 1.98 / (0.67 - 0.52)$$

$$= 30.3122 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_t, \max$]:
 $= \max(|\sigma_t, 1|; |\sigma_t, 2|) = 30.3 \text{ N./mm}^2$

Allowable Tube Stress [σ_t, A]:
 $= S_{ot} = 125.9991 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_{a^4}); 1.25)$
 $= \max((3.25 - 0.25(0.031 + -0.12 \cdot 0.074) 3.65^4); 1.25)$
 $= 2.0000 \text{ (Should be } \leq 2 \text{)}$

Determine the factor [rt]:
 $= (\sqrt{dt^2 + (dt - 2 \cdot tt)^2}) / 4$
 $= (\sqrt{19^2 + (19 - 2 \cdot 1.65)^2}) / 4$
 $= 6.1794 \text{ mm}$

Determine the factor [Ct]:
 $= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 182349680 / 144)}$
 $= 158.1060$

Determine the factor [Ft]:
 $= k \cdot L / r$
 $= 1 \cdot 610 / 6.18$
 $= 98.7147$

Buckling Allowable Stress [Stb]:
 $= (S_y / F_s) \cdot (1 - F_t / (2 \cdot C_t))$
 $= (144 / 2) \cdot (1 - 98.7 / (2 \cdot 158))$
 $= 49.523 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
 $= \sigma_{t, \max} \cdot \text{Tube Area}$
 $= 30.3 \cdot 0.9$
 $= 278.79 \text{ Kg}$

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (19 - 1.65) 1.65$
 $= 0.902 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 1.65 (19 - 1.65) 126$
 $= 1158.836 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 a_g) S_w \text{ (but not } > F_t \text{)}$
 $= 0.85 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 126$
 $= 1140.9987 \text{ Kg}$

Weld Strength Factor [fw]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 126 / 126$
 $= 1.000$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 223 of 264
 Case: 1 9:47a Apr 20, 2026

Design Strength Ratio [fd]:
 = 1 for full strength welds

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)
 $= 0.55 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 126$
 $= 738.2932 \text{ Kgf}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) f_w \cdot fd)} - 0.75 \cdot do \right)$
 $= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 1.65 \cdot (19 - 1.65) 1 \cdot 1)} - 0.75 \cdot 19 \right)$
 $= 2.075 \text{ mm}$

Minimum Required Fillet Weld Leg [afr]: 1.0374 mm
 Minimum Required Groove Weld Leg [agr]: 1.0374 mm

Tube-Tubesheet Jt allowable, 1159 is >= tube strength 1159 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 $= S_s \cdot E_{sw} = 138 \cdot 1 = 138 \text{ N./mm}^2$

Axial Membrane Stress in Shell [σs,m]:
 $= ao^2 / ((Ds + ts) ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) ts)$
 $= 996^2 / ((2028 + 18) 18) \cdot [-4036 + (1.02^2 - 1) (-102 - 4950)] + 1014^2 \cdot 4950 / ((2028 + 18) 18)$
 $= 24.5237 \text{ N./mm}^2$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 18 mm, the minimum Shell length adjacent to the tubesheet should be 343.908 mm

Shell Membrane Stress due to Joint Interaction [σs,m]:
 $= ao^2 / ((Ds + ts) ts) [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) ts)$
 $= 996^2 / ((2028 + 18) 18) [-4036 + (1.02^2 - 1) (-102 - 4950)] + 1014^2 \cdot 4950 / ((2028 + 18) 18)$
 $= 24.5237 \text{ N./mm}^2$

Shell Bending Stress due to Joint Interaction [σs,b]:
 $= 6 \cdot ks / ts^2 \{ \beta_s [\delta_s \cdot Ps + as^2 \cdot P \cdot s / (Es \cdot ts)] + 6(1 - \nu^2) / (E \cdot h) (ao/h)^3 (1 + h \cdot \beta_s / 2) [Pe (Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$
 $= 6 \cdot 1968607104 / 18^2 \{ 0.0095 [0.24992131 \cdot -102 + 1014^2 \cdot 0 / (3.50E+09)] + 6(1 - 0.4^2) / (41416348) (996/179)^3 (1 + 179 \cdot 0.0095/2) [-4036 (0.074 + 0.4 \cdot -0.091) + 2/996^2 \cdot 0.4 \cdot 5762] \}$
 $= -159.1908 \text{ N./mm}^2$

Shell Stress Summation vs. Allowable
 $:: |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$
 $:: |24.5| + |-159| \leq 207 \text{ N./mm}^2$
 $:: 184, \text{ must be } \leq 207 \text{ N./mm}^2$

Note:
 For a given Channel thickness of 46 mm, the minimum Channel length adjacent to the tubesheet should be 549.776 mm



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 224 of 264
 Case: 1 9:47a Apr 20,2026

Channel Membrane Stress due to Joint Interaction [σ_c, m]:

$$= ac^2 \cdot Pt / ((Dc + tc) tc \cdot \cos(\theta))$$

$$= 1014^2 \cdot 4950 / ((2028 + 46) 46 \cdot \cos(0))$$

$$= 53.351 \text{ N./mm}^2$$

Channel Bending Stress due to Joint Interaction [σ_c, b]:

$$= 6 \cdot kc / tc^2 \cdot \{ (\delta_c \cdot Cm \Delta \cdot \beta_c - \zeta_c \cdot Cm \Theta) \cdot Pt$$

$$- 6 ((1 - \nu^2) / E^*) \cdot (ao / h)^3 \cdot (Cm \Theta +$$

$$h \cdot \beta_c / 2 \cdot Cm \Delta) \cdot [Pe (Z_v + Z_m \cdot Q_1) + 2 / ao^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 20.06E+09 / 46^2 \cdot \{ (0.099750735 \cdot 1 \cdot 0.0059 - 0.000 \cdot 1) \cdot 4950$$

$$- 6 ((1 - 0.4^2) / 41416348) \cdot$$

$$(996 / 179)^3 \cdot (1 + 179 \cdot 0.0059 / 2 \cdot 1) \cdot [-4036 (0.074 + 0.4$$

$$\cdot -0.091) + 2 / 996^2 \cdot 0.4 \cdot 5762] \}$$

$$= 359.2014 \text{ N./mm}^2$$

Channel Stress Summation vs. Allowable

$$:: | \sigma_{c,m} | + | \sigma_{c,b} | \leq 1.5 \cdot S_c$$

$$:: | 53.4 | + | 359 | \leq 207 \text{ N./mm}^2$$

$$:: 413 \text{ must be } \leq 207 \text{ N./mm}^2 \text{ (See Elast-Plast iteration below)}$$

Stress/Force Summary for Loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	179.5 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	41.3 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	30.3 <=	126.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-27.0 <=	-49.5 N./mm ²	Pass
Maximum Force on any one Tube	278.8 <=	1158.8 Kgf	Pass
Axial Membrane Stress in Shell	24.5 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	183.7 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	412.6 <=	207.0 N./mm ²	See below

Additional Results for the 2nd Elastic-Plastic Iteration.



These calculations are not intended to be used in the creep range. Check Part 4.18 in Section VIII Division 2.

Compute Allowable [Sstars]:

$$= \min(\text{Sys}, \text{SPS}, s/2)$$

$$= \min(230, 461/2) = 230 \text{ N./mm}^2$$

Compute Allowable [S*c]:

$$= \min(\text{Syc}, \text{SPS}, c/2)$$

$$= \min(225, 450/2) = 225 \text{ N./mm}^2$$

Compute Factor [facts]:

$$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_c, b) / \text{Sstars}], 1)$$

$$= \min([1.4 - 0.4 \cdot \text{abs}(-159) / 230], 1) = 1$$

Compute Factor [factc]:

$$= \min([1.4 - 0.4 \cdot \text{abs}(\sigma_c, b) / \text{Sstarc}], 1)$$

$$= \min([1.4 - 0.4 \cdot \text{abs}(359) / 225], 1) = 0.76$$

Compute Modified Modulus [Ec]:

$$= E_c \cdot \text{factc}$$

$$= 191201920 \cdot 0.76 = 145589584 \text{ KPa.}$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu^*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 225 of 264
 Case: 1 9:47a Apr 20,2026

$$= (1.5 \cdot 0.054 / 0.25) \cdot (2 \cdot 996 / (156 - 0))^2 \cdot -3889$$

$$= -206.9938 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + \text{Cats} + \text{Catc}$$

$$= 156 + 13 + 13$$

$$= 182.0375 \text{ mm}$$

For the Elastic-Plastic iteration, Shell Allowable = SPS,s
 ShellAllow = 460.8179 N./mm²

ASME Section VIII-2 Part 4.18.8.4 Step 12:

For the Elastic-Plastic iteration, the channel allowable stress = SPS,c
 Challow = 450.0172 N./mm²

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D1 corr. (Psd,min + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	167.2 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	41.3 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	30.3 <=	126.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-27.0 <=	-49.5 N./mm ²	Pass
Maximum Force on any one Tube	278.8 <=	1158.8 Kg _f	Pass
Axial Membrane Stress in Shell	24.5 <=	138.0 N./mm ²	Pass
Shell Stress (jt. inter.)	183.7 <=	460.8 N./mm ²	Pass
Channel Stress (jt. inter.)	412.6 <=	450.0 N./mm ²	Pass

Detailed Results for load Case O1 corr. (Psox,min + Ptox,max + Th)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/179$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(19 - 2 \cdot 1.65 (182349680/191201920) \cdot$$

$$(126/138) \cdot (0); 19 - 2 \cdot 1.65)$$

$$= 19.0500 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 987 + 19$$

$$= 1992.476 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (25.4 - 19) / 25.4$$

$$= 0.250$$

Equivalent outer tube limit radius [ao]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 226 of 264
 Case: 1 9:47a Apr 20, 2026

$$\begin{aligned}
 &= Do / 2 \\
 &= 1992/2 \\
 &= 996.2380 \text{ mm}
 \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned}
 &= p / \sqrt{(1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))} \\
 &= 25.4 / \sqrt{(1 - 4 \cdot \min(0 \cdot 100 ; 4 \cdot 1992 \cdot 25.4) / (\pi \cdot 1992^2))} \\
 &= 25.4000 \text{ mm}
 \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned}
 &= (p^* - d^*) / p^* \\
 &= (25.4 - 19) / 25.4 \\
 &= 0.2500
 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned}
 &= as / ao \\
 &= 1014/996 \\
 &= 1.0178
 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned}
 &= ac / ao \\
 &= 1014/996 \\
 &= 1.0178
 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned}
 &= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2 \\
 &= 1 - 5285 ((19 - 2 \cdot 1.65) / (2 \cdot 996))^2 \\
 &= 0.6698
 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned}
 &= 1 - Nt(dt / (2 \cdot ao))^2 \\
 &= 1 - 5285 (19 / (2 \cdot 996))^2 \\
 &= 0.5169
 \end{aligned}$$

Determine the value [h'g]:

$$= 0 \text{ (For pressure plus thermal and thermal only cases)}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned}
 &= \pi \cdot ts(D_s + ts) Es / L \\
 &= \pi \cdot 18 (2028 + 18) 194566320 / 13116 \\
 &= 175021.2500 \text{ Kg/mm}
 \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned}
 &= \pi \cdot tt(Dt - tt) Et / L \\
 &= \pi \cdot 1.65 (19 - 1.65) 182349680 / 13116 \\
 &= 127.8743 \text{ Kg/mm}
 \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned}
 &= K_s / (Nt \cdot K_t) \\
 &= 175021 / (5285 \cdot 128) \\
 &= 0.25898
 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned}
 &= 1 / (1 + K_s / K_j) \\
 &= 1 / (1 + 175021 / 0) \\
 &= 1.0000 (= 1 \text{ if No Exp. Jt.})
 \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned}
 &= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts) \\
 &= ((12(1 - 0.3^2))^{1/4}) / ((2028 + 18) 18) \\
 &= 0.0095 \text{ 1/mm}
 \end{aligned}$$

Shell coefficient [ks]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 227 of 264
 Case: 1 9:47a Apr 20, 2026

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.0095 \cdot 194566320 \cdot 18^3 / (6(1 - 0.3^2))$$

$$= 1968607104.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(2028 + 18) \cdot 1968607104 / 179^3 \cdot (1 + 179 \cdot 0.0095 + (179 \cdot 0.0095)^2 / 2)$$

$$= 17415332.0000 \text{ KPa}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 2028^2 / (4 \cdot 194566320 \cdot 18) \cdot (1 - 2028 \cdot 0.3 / (2(2028 + 18)))$$

$$= 0.24992\text{E}+00 \text{ mm/N./mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(2028 + 46)46}$$

$$= 0.0059 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.0059 \cdot 191201920 \cdot 46^3 / (6(1 - 0.3^2))$$

$$= 20060669952.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(2028 + 46 / \cos(0)) \cdot 20.06\text{E}+09 / 179^3 \cdot (1 + 179 \cdot 0.0059 \cdot (1 + 1) / 2 + (179 \cdot 0.0059)^2 / 2 \cdot 1)$$

$$= 113531960.0 \text{ KPa}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 2028^2 / (4 \cdot 191201920 \cdot 46) \cdot (1 - 2028 \cdot 0.3 / (2(2028 + 46)))$$

$$= 0.99751\text{E}-01 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 7.047244 ; \mu^* = 0.250000 ; E^* = 41416348 \text{ KPa}$$

$$E^*/E = 0.216611 ; \nu^* = 0.396457$$

Note: As h/p (7.047) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [X_a]:

$$= ((24(1 - \nu^{*2})Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.4^2)5285 \cdot 182349680 \cdot 1.65(19 - 1.65)996^2) / (41416348 \cdot 13116 \cdot 179^3))^{1/4}$$

$$= 3.6541$$

Values from Table 4.18.3

$$Z_d = 0.031117 ; Z_v = 0.074231 ; Z_m = 0.400947$$

$$Z_a = 0.459497\text{E}+01 ; Z_w = 0.074231$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 2120 / 1992$$

$$= 1.0640$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 228 of 264
 Case: 1 9:47a Apr 20,2026

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.4) / 41416348 \cdot (17415332 + 113531960 + 191201920 \cdot \ln(1.06))$$

$$= 2.0811$$

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.4) 2.08$$

$$= 2.9062$$

Compute parameter [Q1]:

$$= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.02 - 1 - 2.91 \cdot 0.074) / (1 + 2.91 \cdot 0.4)$$

$$= -0.091399$$

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.031 + -0.091 \cdot 0.074) / 2 \cdot 3.65^4$$

$$= 2.1691$$

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.074 + -0.091 \cdot 0.4) / 2 \cdot 3.65^4$$

$$= 3.3505$$

Compute parameter [U]:

$$= (Z_w + (\psi_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.074 + (1.02 - 1) 0.4) 3.65^4 / (1 + 2.91 \cdot 0.4)$$

$$= 6.7010$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$= 0 \text{ (config a)}$$

Compute Parameter [γ]:

$$= (\alpha_m (T_{tm} - T_a) - \alpha_{sm} (T_{sm} - T_a)) L$$

$$= (0.15299E-04 (0 - 21.1) - 0.11215E-04 (0 - 21.1)) 13116$$

$$= -1.1309 \text{ mm}$$

Calculate parameter [ω_s]:

$$= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.02 \cdot 1968607104 \cdot 0.0095 \cdot 0.24992131 (1 + 179 \cdot 0.0095)$$

$$= 12787.3945 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= a_o^2 (\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s$$

$$= 996^2 (1.02^2 - 1) \cdot (1.02 - 1) / 4 - 12787$$

$$= -12628.2441 \text{ mm}^2$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta_c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.02 \cdot 20.06E+09 [0.099750735 (1 + 0.0059 \cdot 1 \cdot 179) - 0.000 (1/179 + 1 \cdot 0.0059/2)]$$

$$= 24616.3613 \text{ mm}^2$$

Parameter [ω^*c]:

$$= a_o^2 [(\rho_c^2 + 1) \cdot (\rho_c - 1) / 4 - (\psi_s - 1) / 2] - \omega_c$$

$$= 996^2 [(1.02^2 + 1) \cdot (1.02 - 1) / 4 - (1.02 - 1) / 2] - 24616$$

$$= -24457.2129 \text{ mm}^2$$

Average temperature of the unperforated rim [Tr]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 229 of 264
 Case: 1 9:47a Apr 20, 2026

$$= (T' + T's + T'c) / 3 = (21.1 + 21.1 + 21.1) / 3 = 21.1 \text{ } ^\circ\text{C}$$

Average temperature of the shell at the tubesheet [T*s]:
 $= (T's + Tr) / 2 = (21.1 + 21.1) / 2 = 21.1 \text{ } ^\circ\text{C}$

Average temperature of the channel at the tubesheet [T*c]:
 $= (T'c + Tr) / 2 = (21.1 + 21.1) / 2 = 21.1 \text{ } ^\circ\text{C}$

Pressure [P*S]:
 $= Es \cdot ts / as \cdot (\alpha s' \cdot (T's - Ta) - \alpha' \cdot (Tr - Ta))$
 $= 194566320 \cdot 18 / 1014 \cdot [0.11519E-04 \cdot (21.1 - 21.1) - 0.11519E-04 \cdot (21.1 - 21.1)]$
 $= 0.0000 \text{ KPa.}$

Pressure [P*C]:
 $= Ec \cdot tc / ac \cdot (\alpha c' \cdot (T'c - Ta) - \alpha' \cdot (Tr - Ta))$
 $= 191201920 \cdot 46 / 1014 \cdot [0.11519E-04 \cdot (21.1 - 21.1) - 0.11519E-04 \cdot (21.1 - 21.1)]$
 $= 0.0000 \text{ KPa.}$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:
 $= Ps \{ xs + 2(1 - xs) \cdot vt + [2 / Kst \cdot (Ds / Do)^2] \cdot vs - [(ps^2 - 1) / (J \cdot Kst)] \cdot [(1 - J) / (2J \cdot Kst)] \cdot [Dj^2 - (Ds)^2 / Do^2] \}$
 $= 0 \{ 0.52 + 2(1 - 0.52) \cdot 0.31 + [2 / 0.26 \cdot (2028 / 1992)^2] \cdot 0.3 - [(1.02^2 - 1) / (1 \cdot 0.26)] \cdot [(1 - 1) / (2 \cdot 1 \cdot 0.26)] \cdot [(0^2 - (2028)^2) / 1992^2] \}$
 $= 0.0000 \text{ KPa.}$

Pressure [P't]:
 $= [xt + 2(1 - xt) \cdot vt + 1 / (J \cdot Kst)] \cdot Pt$
 $= [0.67 + 2(1 - 0.67) \cdot 0.31 + 1 / (1 \cdot 0.26)] \cdot 4950$
 $= 23442.4355 \text{ KPa.}$

Pressure [P_y]:
 $= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$
 $= 5285 \cdot 128 \cdot -1.13 / (3.14 \cdot 996^2)$
 $= -2403.5955 \text{ KPa.}$

Pressure [P_w]:
 $= -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot ao^2)$
 $= -0 \cdot 6.7 \cdot 0 / (2 \cdot \pi \cdot 996^2)$
 $= -0.0000 \text{ KPa.}$

Calculate the pressure [Prim]:
 $= -(U / ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt)$
 $= -(6.7 / 996^2) \cdot (-12628 \cdot 0 - 24457 \cdot 4950)$
 $= -817.3841 \text{ KPa.}$

Calculate the pressure [P_w]:
 $= (U / ao^2) \cdot (\omega \cdot s \cdot P's - \omega \cdot c \cdot P'c)$
 $= (6.7 / 996^2) \cdot (12787 \cdot 0 - 24616 \cdot 0)$
 $= 0.0000 \text{ KPa.}$

Determine the effective pressure [Pe]:
 $= J \cdot Kst / (1 + J \cdot Kst \cdot (Qz1 + (ps - 1) \cdot Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$
 $= 0.1000E+01 \cdot 0.26 / (1 + 1 \cdot 0.26 \cdot (2.17 + (1.02 - 1) \cdot 3.35)) \cdot (0 - 23442 + -2404 + 0 + -0 + -817)$
 $= -4378.0840 \text{ KPa.}$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 230 of 264
 Case: 1 9:47a Apr 20, 2026

$$\begin{aligned}
 &= [((\omega * s \cdot P_s - \omega * c \cdot P_t) - (\omega s \cdot P * s - \omega c \cdot P * c)) + \\
 &\quad W * \cdot y_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m) \\
 &= [((-12628 \cdot 0 - -24457 \cdot 4950) - \\
 &\quad (12787 \cdot 0 - 24616 \cdot 0)) \cdot 0.0001 + \\
 &\quad 0 \cdot 0 / (2 \cdot \pi)] / (1 + 2.91 \cdot 0.4) \\
 &= 5701.7358 \text{ Kgf}
 \end{aligned}$$

Calculate factor [Q3]:

$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot a_o^2) \\
 &= -0.091 + 2 \cdot 5702 / (-4378 \cdot 996^2) \\
 &= -0.11713
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.058567

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.059 / 0.25) \cdot (2 \cdot 996 / (179 - 0))^2 \cdot -4378 \\
 &= -190.6323 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [Sigma allowed]:

$$\text{SPS (ASME Sec VIII Div 1, UG-23(e))} = 425.9965 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned}
 &= (1.5 \cdot F_m / \mu *) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.05 / 0.25) \cdot (2 \cdot 996 / (105 - 0))^2 \cdot -3955 \\
 &= -425.9954 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + C_{ats} + C_{atc} \\
 &= 105 + 13 + 13 \\
 &= 130.9829 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$\begin{aligned}
 &= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y) \\
 &= (2 \cdot 0.25 \cdot 179 / 996) \cdot 110 \\
 &= 9.9179 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Taulimit

Tubesheet Shear Stress - final thickness [τ_f]:

$$\begin{aligned}
 &= (A_p \cdot Pe) / (\mu \cdot h \cdot C_p) \\
 &= (28995 \cdot -3527) / (0.25 \cdot 61.3 \cdot 6328) \\
 &= -105.5439 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$\begin{aligned}
 &= H + C_{ats} + C_{atc} \\
 &= 61.3 + 13 + 13 \\
 &= 87.2529 \text{ mm}
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd] :

$$\begin{aligned}
 &= \max(H_{reqB}; H_{reqS}) \\
 &= \max(131 ; 87.3) \\
 &= 130.9829 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.2, F_{tmax} = 2$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$\begin{aligned}
 &= ((P_s \cdot x_s - P_t \cdot x_t) - Pe \cdot F_{tmin}) / (X_t - X_s) \\
 &= ((0 \cdot 0.52 - 4950 \cdot 0.67) - (-4378) \cdot -0.2) / \\
 &\quad (0.67 - 0.52)
 \end{aligned}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 231 of 264
 Case: 1 9:47a Apr 20, 2026

$$= -27.3841 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s) \\ = ((0 \cdot 0.52 - 4950 \cdot 0.67) - (-4378) \cdot 2) / \\ (0.67 - 0.52) \\ = 35.5573 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 35.6 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= 2 \cdot S_{ot} = 251.9982 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25) \\ = \max((3.25 - 0.25(0.031 + -0.12 \cdot 0.074)3.65^4); 1.25) \\ = 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4 \\ = (\sqrt{ 19^2 + (19 - 2 \cdot 1.65)^2 }) / 4 \\ = 6.1794 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot E_t / S_{yt}) } \\ = \sqrt{ (2 \cdot \pi^2 \cdot 182349680 / 144) } \\ = 158.1060$$

Determine the factor [Ft]:

$$= k \cdot L / r \\ = 1 \cdot 610 / 6.18 \\ = 98.7147$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t)) \\ = (144 / 2) \cdot (1 - 98.7 / (2 \cdot 158)) \\ = 49.523 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area} \\ = 35.6 \cdot 0.9 \\ = 327.03 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t \\ = \pi (19 - 1.65) 1.65 \\ = 0.902 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S \\ = \pi \cdot 1.65 (19 - 1.65) 126 \\ = 1158.836 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)} \\ = 0.85 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 126 \\ = 1140.9987 \text{ Kg}$$



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 232 of 264
 Case: 1 9:47a Apr 20, 2026

Weld Strength Factor [fw]:
 = S / Sw where Sw = min(S and St)
 = 126/126
 = 1.000

Design Strength Ratio [fd]:
 = 1 for full strength welds

Fillet Weld Strength [Ff]:
 = $0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw$ (but not > Ft)
 = $0.55 \cdot \pi \cdot 1.65 (19 + 0.67 \cdot 1.65) 126$
 = 738.2932 Kgf

Maximum Allowed Tube-Tubesheet Joint load [Lmax]:
 = $2 \cdot Ft$
 = $2 \cdot 1159$
 = 2317.671 Kgf

Minimum Length of Weld [ar]:
 = $2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) fw \cdot fd} - 0.75 \cdot do \right)$
 = $2 \left(\sqrt{(0.75 \cdot 19)^2 + 1.07 \cdot 1.65 \cdot (19 - 1.65) 1 \cdot 1} - 0.75 \cdot 19 \right)$
 = 2.075 mm

Minimum Required Fillet Weld Leg [afr]: 1.0374 mm
 Minimum Required Groove Weld Leg [agr]: 1.0374 mm

Tube-Tubesheet Jt allowable, 2318 is >= tube strength 2318 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:
 = SPS,s = 461 N./mm²

Axial Membrane Stress in Shell [σs,m]:
 = $ao^2 / ((Ds+ts)ts) \cdot [Pe + (ps^2-1)(Ps-Pt)] + as^2 \cdot Pt / ((Ds+ts)ts)$
 = $996^2 / ((2028 + 18)18) \cdot [-4378 + (1.02^2-1)(0-4950)] + 1014^2 \cdot 4950 / ((2028 + 18)18)$
 = 15.4136 N./mm²

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:
 For a given Shell thickness of 18 mm, the minimum Shell length adjacent to the tubesheet should be 343.908 mm

Shell Membrane Stress due to Joint Interaction [σs,m]:
 = $ao^2 / ((Ds+ts)ts) [Pe + (ps^2-1)(Ps-Pt)] + as^2 \cdot Pt / ((Ds+ts)ts)$
 = $996^2 / ((2028 + 18)18) [-4378 + (1.02^2-1)(0-4950)] + 1014^2 \cdot 4950 / ((2028 + 18)18)$
 = 15.4136 N./mm²

Shell Bending Stress due to Joint Interaction [σs,b]:
 = $6 \cdot ks / ts^2 \{ \beta s [\delta s \cdot Ps + as^2 \cdot P \cdot s / (Es \cdot ts)] + 6(1 - \nu^2) / (E \cdot h) (ao/h)^3 (1 + h \cdot \beta s / 2) [Pe (Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$
 = $6 \cdot 1968607104 / 18^2 \{ 0.0095 [0.24992131 \cdot 0 + 1014^2 \cdot 0 / (3.50E+09)] + 6(1 - 0.4^2) / (41416348) (996/179)^3 (1 + 179 \cdot 0.0095/2) [-4378 (0.074 + 0.4 \cdot -0.091) + 2/996^2 \cdot 0.4 \cdot 5702] \}$
 = -169.2795 N./mm²



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 233 of 264
 Case: 1 9:47a Apr 20,2026

Shell Stress Summation vs. Allowable

$||\sigma_{s,m}| + |\sigma_{s,b}| \leq \text{SPS}_s$
 $||15.4| + |-169| \leq 461 \text{ N./mm}^2$
 $185, \text{ must be } \leq 461 \text{ N./mm}^2$

Note:

For a given Channel thickness of 46 mm, the minimum Channel length adjacent to the tubesheet should be 549.776 mm

Channel Membrane Stress due to Joint Interaction $[\sigma_{c,m}]$:

$= ac^2 \cdot Pt / ((Dc + tc)tc \cdot \cos(\theta))$
 $= 1014^2 \cdot 4950 / ((2028 + 46)46 \cdot \cos(0))$
 $= 53.351 \text{ N./mm}^2$

Channel Bending Stress due to Joint Interaction $[\sigma_{c,b}]$:

$= 6 \cdot kc/tc^2 \cdot \{ (\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt - 6((1 - \nu^2)/E) \cdot (ao/h)^3 \cdot (Cm\Theta + h \cdot \beta_c/2 \cdot Cm\Delta) \cdot [Pe(Zv + Zm \cdot Q1) + 2/ao^2 \cdot Zm \cdot Q2] \}$
 $= 6 \cdot 20.06E+09/46^2 \cdot \{ (0.099750735 \cdot 1 \cdot 0.0059 - 0.000 \cdot 1) \cdot 4950 - 6((1 - 0.4^2)/41416348) \cdot (996/179)^3 \cdot (1 + 179 \cdot 0.0059/2 \cdot 1) \cdot [-4378(0.074 + 0.4 \cdot -0.091) + 2/996^2 \cdot 0.4 \cdot 5702] \}$
 $= 383.5585 \text{ N./mm}^2$

Channel Stress Summation vs. Allowable

$||\sigma_{c,m}| + |\sigma_{c,b}| \leq \text{SPS}_c$
 $||53.4| + |384| \leq 450 \text{ N./mm}^2$
 $437 \text{ must be } \leq 450 \text{ N./mm}^2$

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase O1 corr. (Psox,min + Ptox,max + Th):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	190.6 <=	426.0 N./mm ²	Pass
Tubesheet Shear Stress	44.8 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	35.6 <=	252.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-27.4 <=	-49.5 N./mm ²	Pass
Maximum Force on any one Tube	327.0 <=	2317.7 Kg	Pass
Axial Membrane Stress in Shell	15.4 <=	460.8 N./mm ²	Pass
Shell Stress (jt. inter.)	184.7 <=	460.8 N./mm ²	Pass
Channel Stress (jt. inter.)	436.9 <=	450.0 N./mm ²	Pass

Thickness Results for Loadcase O1 corr. (Psox,min + Ptox,max + Th):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	130.983	205.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.037	1.650	Ok
Tube-Tubesheet Groove Weld Leg :	1.037	1.650	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/Fail
D1uc	166.892	...	143 207 37 110	Ps+Pt-Th	D1 Ok
D2uc	29.742	...	17 207 4 110	Ps+Pt-Th	D2 Ok
D3uc	152.387	...	126 207 32 110	Ps+Pt-Th	D3 Ok
D4uc	4.500	...	2 207 ... 110	Ps+Pt-Th	D4 Ok
O1uc	106.976	...	160 426 40 110	Ps+Pt+Th	O1 Ok



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 234 of 264
 Case: 1 9:47a Apr 20,2026

O2uc	6.198	...	5	426	...	110	Ps+Pt+Th	O2	Ok
O3uc	98.620	...	143	426	36	110	Ps+Pt+Th	O3	Ok
O4uc	10.891	...	14	426	4	110	Ps+Pt+Th	O4	Ok
D1c	189.228	...	167	207	41	110	Ps+Pt-Th-c	D1	Ok
D2c	56.222	...	20	207	5	110	Ps+Pt-Th-c	D2	Ok
D3c	175.483	...	149	207	36	110	Ps+Pt-Th-c	D3	Ok
D4c	30.554	...	2	207	1	110	Ps+Pt-Th-c	D4	Ok
O1c	130.983	...	191	426	45	110	Ps+Pt+Th-c	O1	Ok
O2c	30.253	...	7	426	1	110	Ps+Pt+Th-c	O2	Ok
O3c	123.058	...	170	426	40	110	Ps+Pt+Th-c	O3	Ok
O4c	36.471	...	13	426	4	110	Ps+Pt+Th-c	O4	Ok

Max: 189.2279 ... mm 0.808 0.406 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Pass
D1uc	20	137	...	...	:	...	...	...	...	:	Ok
D2uc	13	137	...	...	:	...	...	...	...	:	Ok
D3uc	34	137	...	...	:	...	...	...	...	:	Ok
D4uc	1	137	-1	-105	:	...	...	...	...	:	Ok
O1uc	11	460	...	...	:	...	...	...	...	:	Ok
O2uc	2	460	...	...	:	...	...	...	...	:	Ok
O3uc	24	460	...	...	:	...	...	...	...	:	Ok
O4uc	11	460	-11	-105	:	...	...	...	...	:	Ok
D1c	25	137	...	...	:	...	...	...	...	:	Ok
D2c	14	137	...	...	:	...	...	...	...	:	Ok
D3c	40	137	...	...	:	...	...	...	...	:	Ok
D4c	2	137	-2	-103	:	...	...	...	...	:	Ok
O1c	15	460	...	...	:	...	...	...	...	:	Ok
O2c	3	460	...	...	:	...	...	...	...	:	Ok
O3c	29	460	...	...	:	...	...	...	...	:	Ok
O4c	11	460	-11	-103	:	...	...	...	...	:	Ok
Max	RATIO	0.289		0.103	:	...			...	:	

Tube, Shell and Channel Stress Summary:

Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Shell Stress	Channel Stress	Pass
							Stress Allwd	Stress Allwd	Fail



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 235 of 264
 Case: 1 9:47a Apr 20,2026

D1uc	26	125	-22	-49	242	1158	163	460	378	450	Ok
D2uc	3	125	-3	-49	30	1158	101	206	22	206	Ok
D3uc	23	125	-19	-49	214	1158	79	460	354	450	Ok
D4uc	...	125	...	-49	3	1158	11	206	2	206	Ok
O1uc	32	251	-22	-49	291	2317	165	460	402	450	Ok
O2uc	3	251	...	...	26	2317	70	460	4	450	Ok
O3uc	29	251	-19	-49	266	2317	89	460	380	450	Ok
O4uc	6	251	...	-49	52	2317	31	460	26	450	Ok
D1c	30	125	-27	-49	279	1158	184	460	413	450	Ok
D2c	4	125	-3	-49	35	1158	112	206	25	206	Ok
D3c	27	125	-23	-49	246	1158	89	460	385	450	Ok
D4c	...	125	...	-49	4	1158	12	206	3	206	Ok
O1c	36	251	-27	-49	327	2317	185	460	437	450	Ok
O2c	3	251	...	...	28	2317	81	460	2	450	Ok
O3c	32	251	-24	-49	298	2317	100	460	412	450	Ok
O4c	6	251	-1	-49	52	2317	32	460	27	450	Ok

Max RATIO 0.241 0.553 0.241 0.543 0.971

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	189.228	205.000	Ok
Tube thickness :	0.376	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 362.154 mm

Min Channel length of thk, (48.000) adj. to tubesheet: 561.046 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.131 mm
 Due to Pressure Shell Extends by : +.216 mm
 Due to Pressure + Thermal Shell Compresses by : -1.225 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	6773.16	0.999	9579.78	1.000
Tubesheet Shear Stress	12907.99	1.000	20761.03	1.000
Tube Tensile Stress	20819.47	1.000	30756.82	1.000
Tube Compressive Stress	9066.62	1.000	14234.16	1.000
Tube-Tubesheet Joint load	20819.45	1.000	30756.82	1.000
Shell Stress (Axial, Junction)	5589.44	0.450	3768.58	1.000
Tubesheet-Channel Junction	5088.22	0.996	3768.58	0.227
Tube Pressure Stress	23450.24	1.000	8457.96	1.000
Minimum MAWP	5088.22		3768.58	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	7773.46	0.999	11080.71	1.000
Tubesheet Shear Stress	15013.27	1.000	23033.23	1.000
Tube Tensile Stress	25910.90	1.000	39180.37	1.000



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: D

Page 236 of 264
 Case: 1 9:47a Apr 20,2026

Tube Compressive Stress	14579.48	1.000	21567.13	1.000
Tube-Tubesheet Joint load	25910.89	1.000	39180.37	1.000
Shell Stress (Axial, Junction)	7773.46	0.999	11080.71	2.340
Tubesheet-Channel Junction	6859.40	1.000	11080.71	1.000
Tube Pressure Stress	25683.31	1.000	11364.51	1.000
Minimum MAPnc	6859.40		11080.71	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.2159	0.2164	...	...	...	...
2	-0.0943	-0.0945	...	...	...	...
3	0.1111	0.1114	...	...	...	...
4	0.0105	0.0105	...	...	...	...
5	0.2055	0.2059	-1.1286	-1.1309	...	...
6	-0.0943	-0.0945	-1.1286	-1.1309	...	...
7	0.1111	0.1114	-1.1286	-1.1309	...	...
8	...	...	-1.1286	-1.1309	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(205/4, \min(205, 20))$$

$$= 51.250 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(920/11081, 1)$$

$$= 0.083$$

Thickness Ratio = 0.083, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 17 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(205/4, \min(205, 48))$$

$$= 51.250 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(4950/6859, 1)$$

$$= 0.722$$

Thickness Ratio = 0.72, Temperature Reduction per Fig. UCS 66.1 = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 17 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit 1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -15 °C



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-2 Q-7

Page 237 of 264
 Case: 2 9:47a Apr 20,2026

Input Echo, Tubesheet Number 2, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell

Shell Maximum Design Pressure	Psd,max	920.00 KPa.
Shell Maximum Operating Pressure	Psox,max	781.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	2.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2024.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4347.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	2.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2024.000 mm
Channel Design Temperature	TEMPC	200.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5285
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13090.00 mm
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Allowable Stress at Operating Temperature		133.04 N./mm ²
Tube Yield Stress At design Temperature	Syt	144.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-2 Q-7

Page 238 of 264
 Case: 2 9:47a Apr 20,2026

Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	986.713 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	28995.0 cm ²
Perimeter of Tube Layout	Cp	6328.5 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	200.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	205.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2120.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

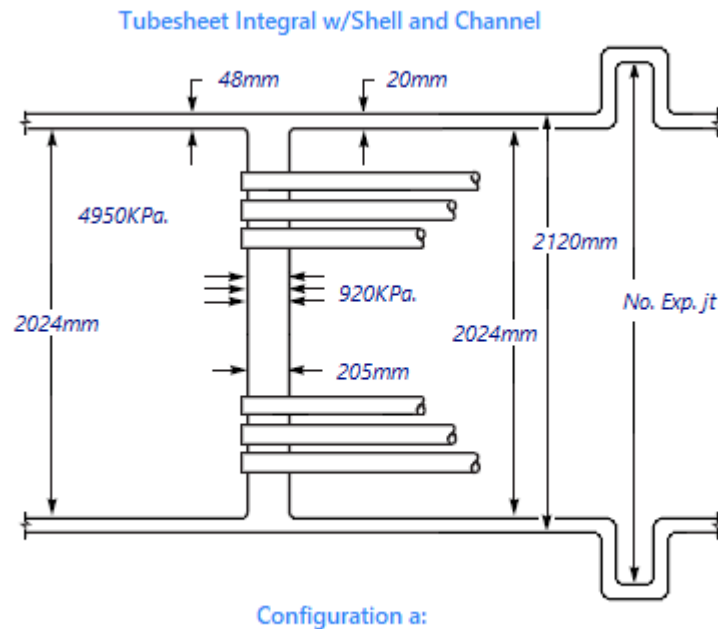
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	610.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	123.40 °C
Channel Operating Temperature	Tcop	131.00 °C
Tube Operating Temperature	Ttop	131.00 °C
Tubesheet Operating Temperature	Tshop	131.00 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side
ASME TS Calc: CASE-2 Q-7

Page 239 of 264
Case: 2 9:47a Apr 20,2026



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	123.4 °C	0.19670E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19120E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Operating Temperature	131.0 °C	0.19485E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Operating Temperature	131.0 °C	0.18726E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	131.0 °C	0.18726E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
------------------------------------	----------	------------------



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 240 of 264
 ASME TS Calc: CASE-2 Q-7 Case: 2 9:47a Apr 20,2026

Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.
 Elastic Mod. at Operating Temperature 131.0 °C 0.19623E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (4950 \cdot (19/2 - 0)) / (126 \cdot 1 + 0.4 \cdot 4950)$
 $= 0.3684 + 0.0000 = 0.3684 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-1 at 200.00 °C
 Elastic Modulus for Material 178957744.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	13090.00	11.55	50.0000	0.0082522	73.24

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 73.2) / (3 \cdot 11.5) = 8458 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3677	19.05	13090.00	51.81	50.0000	0.0004098	35.75

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 35.8) / (3 \cdot 51.8) = 920 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3684 mm
Allowable Internal Pressure at Corroded thickness	23450.25 KPa.
Required Internal Design Pressure	4950.00 KPa.
Allowable External Pressure at Corroded thickness	8457.97 KPa.
Required External Design Pressure	920.00 KPa.
Required Thickness due to Shell Side pressure	0.3677 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Req'd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	165.526	...	141	207	36	110	Ps+Pt-Th	D1 Ok
D2uc	29.821	...	17	207	4	110	Ps+Pt-Th	D2 Ok
D3uc	152.439	...	126	207	32	110	Ps+Pt-Th	D3 Ok
O1uc	95.551	...	142	444	36	110	Ps+Pt+Th	O1 Ok
O2uc	5.016	...	3	444	...	110	Ps+Pt+Th	O2 Ok
O3uc	88.273	...	127	444	32	110	Ps+Pt+Th	O3 Ok
O4uc	10.632	...	14	444	4	110	Ps+Pt+Th	O4 Ok
D1c	187.924	...	165	207	41	110	Ps+Pt-Th-c	D1 Ok
D2c	56.296	...	20	207	5	110	Ps+Pt-Th-c	D2 Ok
D3c	175.524	...	149	207	36	110	Ps+Pt-Th-c	D3 Ok
O1c	119.950	...	169	444	40	110	Ps+Pt+Th-c	O1 Ok
O2c	67.778	...	4	444	...	110	Ps+Pt+Th-c	O2 Ok
O3c	113.032	...	152	444	36	110	Ps+Pt+Th-c	O3 Ok
O4c	36.221	...	14	444	4	110	Ps+Pt+Th-c	O4 Ok
Max:	187.9240	... mm		0.799		0.370	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-2 Q-7

Page 241 of 264
 Case: 2 9:47a Apr 20,2026

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	22	137	...	...		...	...	...	...	Ok
D2uc	13	137	...	...		...	...	...	...	Ok
D3uc	34	137	...	...		...	...	...	...	Ok
O1uc	8	470	...	...		...	...	...	...	Ok
O2uc	...	470	...	-105		...	...	...	...	Ok
O3uc	19	470	...	...		...	...	...	...	Ok
O4uc	11	470	-11	-105		...	...	...	...	Ok
D1c	26	137	...	...		...	...	...	...	Ok
D2c	14	137	...	...		...	...	...	...	Ok
D3c	40	137	...	...		...	...	...	...	Ok
O1c	12	470	...	...		...	...	...	...	Ok
O2c	1	470	...	...		...	...	...	...	Ok
O3c	24	470	...	...		...	...	...	...	Ok
O4c	11	470	-11	-103		...	...	...	...	Ok
Max RATIO	0.288			0.105		...			...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel	Stress	Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	26	125	-22	-49	239	1158	154	460	375	450	Ok
D2uc	3	125	-3	-49	29	1158	101	206	22	206	Ok
D3uc	23	125	-19	-49	214	1158	79	460	354	450	Ok
O1uc	29	266	-20	-53	264	2447	145	470	356	468	Ok
O2uc	3	266	...	...	31	2447	54	470	8	468	Ok
O3uc	26	266	-17	-53	242	2447	80	470	338	468	Ok
O4uc	6	266	...	-53	53	2447	32	470	26	468	Ok
D1c	30	125	-27	-49	275	1158	174	460	410	450	Ok
D2c	4	125	-3	-49	35	1158	112	206	25	206	Ok
D3c	27	125	-23	-49	246	1158	89	460	385	450	Ok
O1c	32	266	-24	-53	296	2447	162	470	387	468	Ok
O2c	3	266	...	...	27	2447	63	470	6	468	Ok
O3c	29	266	-21	-53	271	2447	90	470	366	468	Ok
O4c	6	266	-1	-53	52	2447	32	470	27	468	Ok
Max RATIO	0.237			0.536		0.237		0.543		0.910	

Summary of Thickness Comparisons for 14 Load Cases:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-2 Q-7

Page 242 of 264
 Case: 2 9:47a Apr 20,2026

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	187.924	205.000	Ok
Tube thickness :	0.368	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 362.154 mm

Min Channel length of thk, (48.000) adj. to tubesheet: 561.046 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.131 mm
 Due to Pressure Shell Extends by : +.206 mm
 Due to Pressure + Thermal Shell Compresses by : -1.209 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6773.14	0.999	9579.34	1.000
Tubesheet Shear Stress	12873.75	1.000	20753.20	1.000
Tube Tensile Stress	20844.87	1.000	30808.05	1.000
Tube Compressive Stress	9227.14	1.000	14254.08	1.000
Tube-Tubesheet Joint load	20844.89	1.000	30808.05	1.000
Shell Stress (Axial, Junction)	5831.60	0.455	3768.76	1.000
Tubesheet-Channel Junction	5307.54	0.996	3768.76	0.227
Tube Pressure Stress	23450.24	1.000	8457.96	1.000
Minimum MAWP	5307.54		3768.76	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7774.00	0.999	11081.93	1.000
Tubesheet Shear Stress	15007.69	1.000	23024.52	1.000
Tube Tensile Stress	25945.04	1.000	39274.05	1.000
Tube Compressive Stress	14621.68	1.000	21600.23	1.000
Tube-Tubesheet Joint load	25945.04	1.000	39274.05	1.000
Shell Stress (Axial, Junction)	7774.00	0.999	11081.93	2.340
Tubesheet-Channel Junction	6861.48	1.000	11081.93	1.000
Tube Pressure Stress	25683.31	1.000	11364.51	1.000
Minimum MAPnc	6861.48		11081.93	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.2055	0.2059	...	...	...	...
2	-0.0943	-0.0945	...	...	...	...
3	0.1111	0.1114	...	...	...	...
4	...	...	...	...	...	...
5	0.1757	0.1761	-1.1286	-1.1309	...	...
6	-0.0780	-0.0781	-1.1286	-1.1309	...	...



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
ASME TS Calc: CASE-2 Q-7

Page 243 of 264
Case: 2 9:47a Apr 20,2026

7	0.0977	0.0979	-1.1286	-1.1309	...	...
8	...	...	-1.1286	-1.1309	...	...

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE 3- START U

Page 244 of 264
 Case: 3 9:47a Apr 20,2026

Input Echo, Tubesheet Number 3, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell

Shell Maximum Design Pressure	Psd,max	920.00 KPa.
Shell Maximum Operating Pressure	Psox,max	600.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	2.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2024.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4950.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	2.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2024.000 mm
Channel Design Temperature	TEMPC	200.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5285
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13090.00 mm
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Allowable Stress at Operating Temperature		138.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	144.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE 3- START U

Page 245 of 264
 Case: 3 9:47a Apr 20,2026

Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	986.713 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	28995.0 cm ²
Perimeter of Tube Layout	Cp	6328.5 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	200.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	205.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2120.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

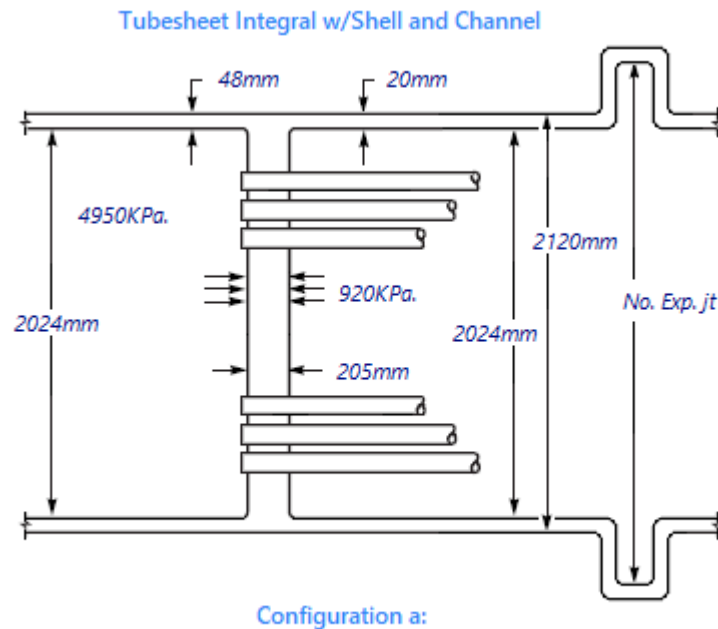
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	610.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	52.00 °C
Channel Operating Temperature	Tcop	28.00 °C
Tube Operating Temperature	Ttop	52.00 °C
Tubesheet Operating Temperature	Tshop	52.00 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side
ASME TS Calc: CASE 3- START U

Page 246 of 264
Case: 3 9:47a Apr 20,2026



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	52.0 °C	0.20093E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19120E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Operating Temperature	28.0 °C	0.20093E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Operating Temperature	52.0 °C	0.19276E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	52.0 °C	0.19276E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
------------------------------------	----------	------------------



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE 3- START U

Page 247 of 264

Case: 3 9:47a Apr 20,2026

Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.
 Elastic Mod. at Operating Temperature 52.0 °C 0.20093E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (4950 \cdot (19/2 - 0)) / (126 \cdot 1 + 0.4 \cdot 4950)$
 $= 0.3684 + 0.0000 = 0.3684 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-1 at 200.00 °C
 Elastic Modulus for Material 178957744.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	13090.00	11.55	50.0000	0.0082522	73.24

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 73.2) / (3 \cdot 11.5) = 8458 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3677	19.05	13090.00	51.81	50.0000	0.0004098	35.75

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 35.8) / (3 \cdot 51.8) = 920 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3684 mm
Allowable Internal Pressure at Corroded thickness	23450.25 KPa.
Required Internal Design Pressure	4950.00 KPa.
Allowable External Pressure at Corroded thickness	8457.97 KPa.
Required External Design Pressure	920.00 KPa.
Required Thickness due to Shell Side pressure	0.3677 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Req'd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	165.526	...	141	207	36	110	Ps+Pt-Th	D1 Ok
D2uc	29.821	...	17	207	4	110	Ps+Pt-Th	D2 Ok
D3uc	152.439	...	126	207	32	110	Ps+Pt-Th	D3 Ok
O1uc	98.340	...	161	482	40	110	Ps+Pt+Th	O1 Ok
O2uc	4.027	...	5	482	1	110	Ps+Pt+Th	O2 Ok
O3uc	93.313	...	149	482	38	110	Ps+Pt+Th	O3 Ok
O4uc	9.904	...	15	482	4	110	Ps+Pt+Th	O4 Ok
D1c	187.924	...	165	207	41	110	Ps+Pt-Th-c	D1 Ok
D2c	56.296	...	20	207	5	110	Ps+Pt-Th-c	D2 Ok
D3c	175.524	...	149	207	36	110	Ps+Pt-Th-c	D3 Ok
O1c	122.682	...	191	482	45	110	Ps+Pt+Th-c	O1 Ok
O2c	30.458	...	5	482	1	110	Ps+Pt+Th-c	O2 Ok
O3c	117.909	...	178	482	42	110	Ps+Pt+Th-c	O3 Ok
O4c	35.529	...	14	482	4	110	Ps+Pt+Th-c	O4 Ok
Max:	187.9240	... mm		0.799		0.409	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 248 of 264
 ASME TS Calc: CASE 3- START U Case: 3 9:47a Apr 20,2026

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	22	137	...	...		...	...	...	...	Ok
D2uc	13	137	...	...		...	...	...	...	Ok
D3uc	34	137	...	...		...	...	...	...	Ok
O1uc	10	508	...	...		...	...	...	...	Ok
O2uc	3	508	-3	-105		...	...	...	...	Ok
O3uc	19	508	...	...		...	...	...	...	Ok
O4uc	11	508	-11	-105		...	...	...	...	Ok
D1c	26	137	...	...		...	...	...	...	Ok
D2c	14	137	...	...		...	...	...	...	Ok
D3c	40	137	...	...		...	...	...	...	Ok
O1c	15	508	...	...		...	...	...	...	Ok
O2c	2	508	-2	-103		...	...	...	...	Ok
O3c	24	508	...	...		...	...	...	...	Ok
O4c	11	508	-11	-103		...	...	...	...	Ok
Max RATIO	0.288			0.107		...			...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel	Stress	Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	26	125	-22	-49	239	1158	154	460	375	450	Ok
D2uc	3	125	-3	-49	29	1158	101	206	22	206	Ok
D3uc	23	125	-19	-49	214	1158	79	460	354	450	Ok
O1uc	32	275	-23	-63	295	2538	163	508	403	524	Ok
O2uc	4	275	...	...	37	2538	39	508	13	524	Ok
O3uc	30	275	-20	-63	279	2538	113	508	389	524	Ok
O4uc	6	275	...	-63	54	2538	32	508	27	524	Ok
D1c	30	125	-27	-49	275	1158	174	460	410	450	Ok
D2c	4	125	-3	-49	35	1158	112	206	25	206	Ok
D3c	27	125	-23	-49	246	1158	89	460	385	450	Ok
O1c	36	275	-28	-63	332	2538	182	508	439	524	Ok
O2c	4	275	...	...	34	2538	45	508	12	524	Ok
O3c	34	275	-25	-63	312	2538	127	508	422	524	Ok
O4c	6	275	-1	-63	54	2538	33	508	28	524	Ok
Max RATIO	0.237			0.536		0.237		0.543		0.910	

Summary of Thickness Comparisons for 14 Load Cases:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE 3- START U

Page 249 of 264
 Case: 3 9:47a Apr 20,2026

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	187.924	205.000	Ok
Tube thickness :	0.368	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 362.154 mm

Min Channel length of thk, (48.000) adj. to tubesheet: 561.046 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.131 mm
 Due to Pressure Shell Extends by : +.206 mm
 Due to Pressure + Thermal Shell Compresses by : -1.189 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6773.14	0.999	9579.34	1.000
Tubesheet Shear Stress	12849.90	1.000	20753.20	1.000
Tube Tensile Stress	20844.87	1.000	30808.05	1.000
Tube Compressive Stress	9227.14	1.000	14254.08	1.000
Tube-Tubesheet Joint load	20844.89	1.000	30808.05	1.000
Shell Stress (Axial, Junction)	5933.82	0.915	3768.76	1.000
Tubesheet-Channel Junction	5436.80	1.000	3768.76	0.227
Tube Pressure Stress	23450.24	1.000	8457.96	1.000
Minimum MAWP	5436.80		3768.76	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7774.00	0.999	11081.93	1.000
Tubesheet Shear Stress	15007.69	1.000	23024.52	1.000
Tube Tensile Stress	25945.04	1.000	39274.05	1.000
Tube Compressive Stress	14621.68	1.000	21600.23	1.000
Tube-Tubesheet Joint load	25945.04	1.000	39274.05	1.000
Shell Stress (Axial, Junction)	7774.00	0.999	11081.93	2.340
Tubesheet-Channel Junction	6861.48	1.000	11081.93	1.000
Tube Pressure Stress	25683.31	1.000	11364.51	1.000
Minimum MAPnc	6861.48		11081.93	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.2055	0.2059	...	...	...	...
2	-0.0943	-0.0945	...	...	...	...
3	0.1111	0.1114	...	...	...	...
4	...	...	...	...	...	...
5	0.1944	0.1948	-1.1286	-1.1309	...	...
6	-0.0582	-0.0583	-1.1286	-1.1309	...	...



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
ASME TS Calc: CASE 3- START U

Page 250 of 264
Case: 3 9:47a Apr 20,2026

7	0.1362	0.1365	-1.1286	-1.1309	...	...
8	...	...	-1.1286	-1.1309	...	...

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-4 Q-7

Page 251 of 264
 Case: 4 9:47a Apr 20,2026

Input Echo, Tubesheet Number 4, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell

Shell Maximum Design Pressure	Psd,max	920.00 KPa.
Shell Maximum Operating Pressure	Psox,max	781.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	2.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2024.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4347.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	2.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2024.000 mm
Channel Design Temperature	TEMPC	200.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5285
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13090.00 mm
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Allowable Stress at Operating Temperature		133.04 N./mm ²
Tube Yield Stress At design Temperature	Syt	144.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-4 Q-7

Page 252 of 264
 Case: 4 9:47a Apr 20,2026

Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	986.713 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	28995.0 cm ²
Perimeter of Tube Layout	Cp	6328.5 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	200.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	205.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2120.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

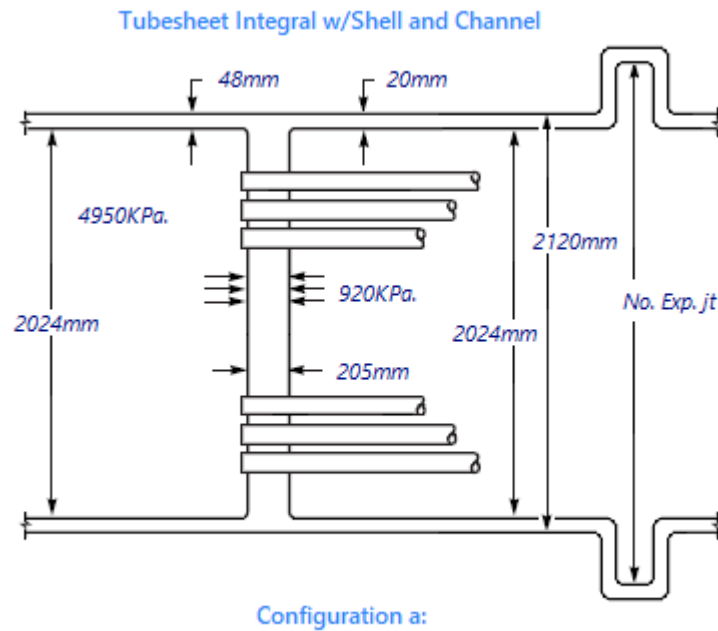
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	610.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	64.50 °C
Channel Operating Temperature	Tcop	131.00 °C
Tube Operating Temperature	Ttop	131.00 °C
Tubesheet Operating Temperature	Tshop	131.00 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side
ASME TS Calc: CASE-4 Q-7

Page 253 of 264
Case: 4 9:47a Apr 20,2026



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	64.5 °C	0.20022E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19120E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Operating Temperature	131.0 °C	0.19485E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Operating Temperature	131.0 °C	0.18726E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	131.0 °C	0.18726E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
------------------------------------	----------	------------------



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-4 Q-7

Page 254 of 264
 Case: 4 9:47a Apr 20,2026

Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.
 Elastic Mod. at Operating Temperature 131.0 °C 0.19623E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (4950 \cdot (19/2 - 0)) / (126 \cdot 1 + 0.4 \cdot 4950)$
 $= 0.3684 + 0.0000 = 0.3684 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-1 at 200.00 °C
 Elastic Modulus for Material 178957744.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	13090.00	11.55	50.0000	0.0082522	73.24

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 73.2) / (3 \cdot 11.5) = 8458 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3677	19.05	13090.00	51.81	50.0000	0.0004098	35.75

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 35.8) / (3 \cdot 51.8) = 920 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3684 mm
Allowable Internal Pressure at Corroded thickness	23450.25 KPa.
Required Internal Design Pressure	4950.00 KPa.
Allowable External Pressure at Corroded thickness	8457.97 KPa.
Required External Design Pressure	920.00 KPa.
Required Thickness due to Shell Side pressure	0.3677 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Req'd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	165.526	...	141	207	36	110	Ps+Pt-Th	D1 Ok
D2uc	29.821	...	17	207	4	110	Ps+Pt-Th	D2 Ok
D3uc	152.439	...	126	207	32	110	Ps+Pt-Th	D3 Ok
O1uc	95.428	...	142	444	36	110	Ps+Pt+Th	O1 Ok
O2uc	3.922	...	3	444	...	110	Ps+Pt+Th	O2 Ok
O3uc	88.145	...	127	444	32	110	Ps+Pt+Th	O3 Ok
O4uc	10.691	...	14	444	4	110	Ps+Pt+Th	O4 Ok
D1c	187.924	...	165	207	41	110	Ps+Pt-Th-c	D1 Ok
D2c	56.296	...	20	207	5	110	Ps+Pt-Th-c	D2 Ok
D3c	175.524	...	149	207	36	110	Ps+Pt-Th-c	D3 Ok
O1c	119.830	...	169	444	40	110	Ps+Pt+Th-c	O1 Ok
O2c	30.342	...	3	444	...	110	Ps+Pt+Th-c	O2 Ok
O3c	112.907	...	152	444	36	110	Ps+Pt+Th-c	O3 Ok
O4c	36.281	...	14	444	4	110	Ps+Pt+Th-c	O4 Ok
Max:	187.9240	... mm		0.799		0.370	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-4 Q-7

Page 255 of 264
 Case: 4 9:47a Apr 20,2026

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	22	137	...	...		...	...	...	...	Ok
D2uc	13	137	...	...		...	...	...	...	Ok
D3uc	34	137	...	...		...	...	...	...	Ok
O1uc	8	492	...	...		...	...	...	...	Ok
O2uc	...	492	...	-105		...	...	...	...	Ok
O3uc	19	492	...	...		...	...	...	...	Ok
O4uc	11	492	-11	-105		...	...	...	...	Ok
D1c	26	137	...	...		...	...	...	...	Ok
D2c	14	137	...	...		...	...	...	...	Ok
D3c	40	137	...	...		...	...	...	...	Ok
O1c	12	492	...	...		...	...	...	...	Ok
O2c	1	492	...	...		...	...	...	...	Ok
O3c	24	492	...	...		...	...	...	...	Ok
O4c	11	492	-11	-103		...	...	...	...	Ok
Max RATIO	0.288			0.106		...			...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel	Stress	Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	26	125	-22	-49	239	1158	154	460	375	450	Ok
D2uc	3	125	-3	-49	29	1158	101	206	22	206	Ok
D3uc	23	125	-19	-49	214	1158	79	460	354	450	Ok
O1uc	29	266	-20	-53	264	2447	147	492	356	468	Ok
O2uc	3	266	...	...	32	2447	54	492	8	468	Ok
O3uc	26	266	-17	-53	242	2447	82	492	337	468	Ok
O4uc	6	266	...	-53	53	2447	32	492	27	468	Ok
D1c	30	125	-27	-49	275	1158	174	460	410	450	Ok
D2c	4	125	-3	-49	35	1158	112	206	25	206	Ok
D3c	27	125	-23	-49	246	1158	89	460	385	450	Ok
O1c	32	266	-24	-53	295	2447	164	492	387	468	Ok
O2c	3	266	...	...	28	2447	63	492	7	468	Ok
O3c	29	266	-21	-53	270	2447	92	492	366	468	Ok
O4c	6	266	-1	-53	53	2447	33	492	28	468	Ok
Max RATIO	0.237			0.536		0.237		0.543		0.910	

Summary of Thickness Comparisons for 14 Load Cases:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE-4 Q-7

Page 256 of 264
 Case: 4 9:47a Apr 20,2026

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	187.924	205.000	Ok
Tube thickness :	0.368	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 362.154 mm

Min Channel length of thk, (48.000) adj. to tubesheet: 561.046 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.131 mm
 Due to Pressure Shell Extends by : +.206 mm
 Due to Pressure + Thermal Shell Compresses by : -1.209 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6773.14	0.999	9579.34	1.000
Tubesheet Shear Stress	12901.95	1.000	20753.20	1.000
Tube Tensile Stress	20844.87	1.000	30808.05	1.000
Tube Compressive Stress	9227.14	1.000	14254.08	1.000
Tube-Tubesheet Joint load	20844.89	1.000	30808.05	1.000
Shell Stress (Axial, Junction)	5840.14	0.441	3768.76	1.000
Tubesheet-Channel Junction	5316.54	0.996	3768.76	0.227
Tube Pressure Stress	23450.24	1.000	8457.96	1.000
Minimum MAWP	5316.54		3768.76	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7774.00	0.999	11081.93	1.000
Tubesheet Shear Stress	15007.69	1.000	23024.52	1.000
Tube Tensile Stress	25945.04	1.000	39274.05	1.000
Tube Compressive Stress	14621.68	1.000	21600.23	1.000
Tube-Tubesheet Joint load	25945.04	1.000	39274.05	1.000
Shell Stress (Axial, Junction)	7774.00	0.999	11081.93	2.340
Tubesheet-Channel Junction	6861.48	1.000	11081.93	1.000
Tube Pressure Stress	25683.31	1.000	11364.51	1.000
Minimum MAPnc	6861.48		11081.93	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.2055	0.2059	...	...	...	...
2	-0.0943	-0.0945	...	...	...	...
3	0.1111	0.1114	...	...	...	...
4	...	...	...	...	...	...
5	0.1757	0.1761	-1.1286	-1.1309	...	...
6	-0.0780	-0.0781	-1.1286	-1.1309	...	...



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
ASME TS Calc: CASE-4 Q-7

Page 257 of 264
Case: 4 9:47a Apr 20,2026

7	0.0977	0.0979	-1.1286	-1.1309	...	...
8	...	...	-1.1286	-1.1309	...	...

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE 5- Q-7

Page 258 of 264
 Case: 5 9:47a Apr 20,2026

Input Echo, Tubesheet Number 5, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: main shell

Shell Maximum Design Pressure	Psd,max	920.00 KPa.
Shell Maximum Operating Pressure	Psox,max	781.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	20.0000 mm
Shell Internal Corrosion Allowance	cas	2.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2024.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²
Shell Allowable Stress at Operating Temperature		138.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	4950.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	4347.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	48.0000 mm
Channel Internal Corrosion Allowance	cac	2.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2024.000 mm
Channel Design Temperature	TEMPC	200.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²
Channel Allowable Stress at Operating Temperature		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5285
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	13500.00 mm
Straight Tube Length (bet. inner tubsht faces) L		13090.00 mm
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-213 TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		126.00 N./mm ²
Tube Allowable Stress At Ambient		138.00 N./mm ²
Tube Allowable Stress at Operating Temperature		134.19 N./mm ²
Tube Yield Stress At design Temperature	Syt	144.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6500 mm
Groove Weld Leg	ag	1.6500 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE 5- Q-7

Page 259 of 264
 Case: 5 9:47a Apr 20,2026

Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	986.713 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	28995.0 cm ²
Perimeter of Tube Layout	Cp	6328.5 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	200.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Tubesheet Allowable Stress at Operating Temper.		138.00 N./mm ²
Thickness of Tubesheet	h	205.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	13.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	13.0000 mm
Tubesheet Outside Diameter	A	2120.000 mm

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

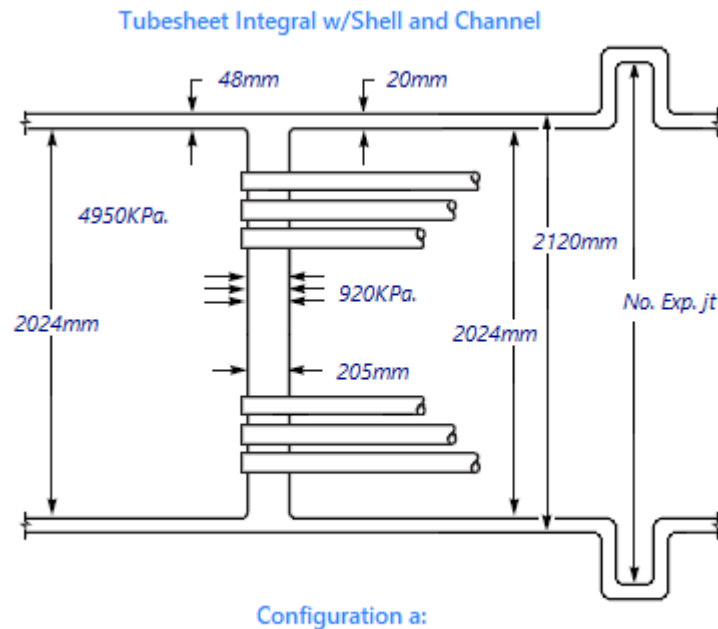
Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	610.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Use Operating Temperatures for Thermal cases		Yes
Shell Operating Temperature	Tsop	123.40 °C
Channel Operating Temperature	Tcop	70.00 °C
Tube Operating Temperature	Ttop	123.40 °C
Tubesheet Operating Temperature	Tshop	123.40 °C
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



PV Elite 28 Licensee: SL User
FileName : 106-C-Cladding on Both Side
ASME TS Calc: CASE 5- Q-7

Page 260 of 264
Case: 5 9:47a Apr 20,2026



ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Operating Temperature	123.4 °C	0.19670E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C> 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.19120E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Operating Temperature	70.0 °C	0.19852E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	200.0 °C	0.18235E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Operating Temperature	123.4 °C	0.18774E+09 KPa.
Elastic Mod. at Tubsht. Operat. Temp.	123.4 °C	0.18774E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	200.0 °C	0.19258E+09 KPa.
------------------------------------	----------	------------------



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side Page 261 of 264
 ASME TS Calc: CASE 5- Q-7 Case: 5 9:47a Apr 20,2026

Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.
 Elastic Mod. at Operating Temperature 123.4 °C 0.19670E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (4950 \cdot (19/2 - 0)) / (126 \cdot 1 + 0.4 \cdot 4950)$
 $= 0.3684 + 0.0000 = 0.3684 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-1 at 200.00 °C
 Elastic Modulus for Material 178957744.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
1.6500	19.05	13090.00	11.55	50.0000	0.0082522	73.24

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 73.2) / (3 \cdot 11.5) = 8458 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3677	19.05	13090.00	51.81	50.0000	0.0004098	35.75

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 35.8) / (3 \cdot 51.8) = 920 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3684 mm
Allowable Internal Pressure at Corroded thickness	23450.25 KPa.
Required Internal Design Pressure	4950.00 KPa.
Allowable External Pressure at Corroded thickness	8457.97 KPa.
Required External Design Pressure	920.00 KPa.
Required Thickness due to Shell Side pressure	0.3677 mm

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

Case#	--Req'd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	165.526	...	141	207	36	110	Ps+Pt-Th	D1 Ok
D2uc	29.821	...	17	207	4	110	Ps+Pt-Th	D2 Ok
D3uc	152.439	...	126	207	32	110	Ps+Pt-Th	D3 Ok
O1uc	95.398	...	143	447	36	110	Ps+Pt+Th	O1 Ok
O2uc	4.985	...	3	447	...	110	Ps+Pt+Th	O2 Ok
O3uc	88.115	...	128	447	32	110	Ps+Pt+Th	O3 Ok
O4uc	10.587	...	14	447	4	110	Ps+Pt+Th	O4 Ok
D1c	187.924	...	165	207	41	110	Ps+Pt-Th-c	D1 Ok
D2c	56.296	...	20	207	5	110	Ps+Pt-Th-c	D2 Ok
D3c	175.524	...	149	207	36	110	Ps+Pt-Th-c	D3 Ok
O1c	119.823	...	170	447	40	110	Ps+Pt+Th-c	O1 Ok
O2c	67.778	...	4	447	...	110	Ps+Pt+Th-c	O2 Ok
O3c	112.898	...	153	447	36	110	Ps+Pt+Th-c	O3 Ok
O4c	36.179	...	14	447	4	110	Ps+Pt+Th-c	O4 Ok
Max:	187.9240	... mm		0.799		0.370	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Cladding on Both Side
 ASME TS Calc: CASE 5- Q-7

Page 262 of 264
 Case: 5 9:47a Apr 20,2026

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	22	137	...	...		...	...	...	...	Ok
D2uc	13	137	...	...		...	...	...	...	Ok
D3uc	34	137	...	...		...	...	...	...	Ok
O1uc	8	470	...	...		...	...	...	...	Ok
O2uc	...	470	...	-105		...	...	...	...	Ok
O3uc	19	470	...	...		...	...	...	...	Ok
O4uc	11	470	-11	-105		...	...	...	...	Ok
D1c	26	137	...	...		...	...	...	...	Ok
D2c	14	137	...	...		...	...	...	...	Ok
D3c	40	137	...	...		...	...	...	...	Ok
O1c	12	470	...	...		...	...	...	...	Ok
O2c	1	470	...	...		...	...	...	...	Ok
O3c	23	470	...	...		...	...	...	...	Ok
O4c	11	470	-11	-103		...	...	...	...	Ok
Max RATIO	0.288			0.105		...			...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel	Stress	Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	26	125	-22	-49	239	1158	154	460	375	450	Ok
D2uc	3	125	-3	-49	29	1158	101	206	22	206	Ok
D3uc	23	125	-19	-49	214	1158	79	460	354	450	Ok
O1uc	29	268	-20	-54	264	2468	144	470	358	490	Ok
O2uc	3	268	...	...	31	2468	54	470	8	490	Ok
O3uc	26	268	-17	-54	242	2468	79	470	339	490	Ok
O4uc	6	268	...	-54	53	2468	32	470	27	490	Ok
D1c	30	125	-27	-49	275	1158	174	460	410	450	Ok
D2c	4	125	-3	-49	35	1158	112	206	25	206	Ok
D3c	27	125	-23	-49	246	1158	89	460	385	450	Ok
O1c	32	268	-24	-54	296	2468	161	470	389	490	Ok
O2c	3	268	...	...	27	2468	63	470	6	490	Ok
O3c	29	268	-21	-54	270	2468	89	470	368	490	Ok
O4c	6	268	-1	-54	52	2468	32	470	28	490	Ok
Max RATIO	0.237			0.536		0.237		0.543		0.910	

Summary of Thickness Comparisons for 14 Load Cases:



PV Elite 28 Licensee: SL User
 FileName : 106-C-Clading on Both Side
 ASME TS Calc: CASE 5- Q-7

Page 263 of 264
 Case: 5 9:47a Apr 20,2026

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	187.924	205.000	Ok
Tube thickness :	0.368	1.650	Ok
Tube-tubesheet fillet weld leg :	1.037	1.650	Ok
Tube-tubesheet groove weld leg :	1.037	1.650	Ok

Min Shell length of thk, (20.000) adj. to tubesheet: 362.154 mm

Min Channel length of thk, (48.000) adj. to tubesheet: 561.046 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.131 mm
 Due to Pressure Shell Extends by : +.206 mm
 Due to Pressure + Thermal Shell Compresses by : -1.209 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	6773.14	0.999	9579.34	1.000
Tubesheet Shear Stress	12861.33	1.000	20753.20	1.000
Tube Tensile Stress	20844.87	1.000	30808.05	1.000
Tube Compressive Stress	9227.14	1.000	14254.08	1.000
Tube-Tubesheet Joint load	20844.89	1.000	30808.05	1.000
Shell Stress (Axial, Junction)	5933.82	0.915	3768.76	1.000
Tubesheet-Channel Junction	5436.80	1.000	3768.76	0.227
Tube Pressure Stress	23450.24	1.000	8457.96	1.000
Minimum MAWP	5436.80		3768.76	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7774.00	0.999	11081.93	1.000
Tubesheet Shear Stress	15007.69	1.000	23024.52	1.000
Tube Tensile Stress	25945.04	1.000	39274.05	1.000
Tube Compressive Stress	14621.68	1.000	21600.23	1.000
Tube-Tubesheet Joint load	25945.04	1.000	39274.05	1.000
Shell Stress (Axial, Junction)	7774.00	0.999	11081.93	2.340
Tubesheet-Channel Junction	6861.48	1.000	11081.93	1.000
Tube Pressure Stress	25683.31	1.000	11364.51	1.000
Minimum MAPnc	6861.48		11081.93	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.2055	0.2059	...	...	...	...
2	-0.0943	-0.0945	...	...	...	...
3	0.1111	0.1114	...	...	...	...
4	...	...	...	...	...	...
5	0.1753	0.1756	-1.1286	-1.1309	...	...
6	-0.0778	-0.0779	-1.1286	-1.1309	...	...



PV Elite 28 Licensee: SL User
FileName : 106-C-Clading on Both Side
ASME TS Calc: CASE 5- Q-7

Page 264 of 264
Case: 5 9:47a Apr 20,2026

7	0.0975	0.0977	-1.1286	-1.1309	...	...
8	...	...	-1.1286	-1.1309	...	...

PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.